



Lidl US, LLC
3500 S. Clark St.
Arlington , Virginia 22202
P: (571) 398-5435

Project: 1517 Norcross GA - Jimmy Carter Blvd
4990 Jimmy Carter Blvd
Norcross, Georgia 30093

Submittal #22-1.0 - Plumbing 22 - PLUMBING

Distribution Summary

Distributed by Luis Marcelino (Place Services Inc.) on Oct 1, 2026

To Jens Byer (28101155 Lidl US, LLC), Amy Schoch (21118060 Progressive Companies), Shelley Chovan (21118060 Progressive Companies), Luis Marcelino (Place Services Inc.), Tucker Teague (Place Services Inc.), Bobby McTeague (Place Services Inc.)

Message None

Attachments

Name	Response	Attachments	Comments
Amy Schoch (21118060 Progressive Companies)	Approved as Noted	Lidl-Norcross - Submittal 22-1.0 - Plumbing Package_PC Reviewed.pdf	Please see reviewed submittal attached, thank you.

Revision	0	Submittal Manager	Amy Schoch (21118060 Progressive Companies)
Status	Closed	Date Created	Sep 3, 2026
Issue Date	Sep 3, 2026	Spec Section	22 - PLUMBING
Responsible Contractor	Place Services Inc.	Received From	Tucker Teague (Place Services Inc.)
Received Date	Sep 11, 2026	Submit By	Sep 3, 2026
Final Due Date	Oct 14, 2026	Lead Time	
		Cost Code	
Location		Type	Cutsheet

Submittal Package

Approvers Amy Schoch (21118060 Progressive Companies)

Ball in Court

Distribution Amy Schoch (21118060 Progressive Companies), Bobby McTeague (Place Services Inc.), Jens Byer (28101155 Lidl US, LLC), Luis Marcelino (Place Services Inc.), Shelley Chovan (21118060 Progressive Companies)

Description

Submittal Workflow

Name	Sent Date	Due Date	Returned Date	Response	Attachments
General Information Attachments					Lidl-1517-Norcross-GA-Ground-Up-Submittal-PSI-0003-Plumbing-Package.pdf
Amy Schoch		Oct 14, 2026	Sep 25, 2026	Approved as Noted	Lidl-Norcross - Submittal 22-1.0 - Plumbing Package_PC Reviewed.pdf (Current)
Comment	Please see reviewed submittal attached, thank you.				



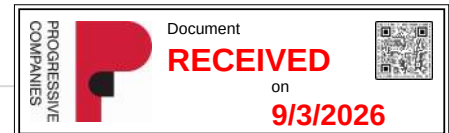
Lidl US, LLC
3500 S. Clark St.
Arlington , Virginia 22202
P: (571) 398-5435

Project: 1517 Norcross GA - Jimmy Carter Blvd
4990 Jimmy Carter Blvd
Norcross, Georgia 30093

Submittal #22-1.0 - Plumbing Package 22 - PLUMBING

Revision	0	Submittal Manager	Amy Schoch (21118060 Progressive Companies)
Status	Open	Date Created	Sep 3, 2026
Issue Date	Sep 3, 2026	Spec Section	22 - PLUMBING
Responsible Contractor	Place Services Inc.	Received From	Tucker Teague (Place Services Inc.)
Received Date	Sep 11, 2026	Submit By	Sep 3, 2026
Final Due Date	Sep 11, 2026	Lead Time	
		Cost Code	
Location		Type	Cutsheet
Submittal Package	#2: Plumbing		
Approvers	Amy Schoch (21118060 Progressive Companies)		
Ball in Court	Amy Schoch (21118060 Progressive Companies)		
Distribution	Amy Schoch (21118060 Progressive Companies), Bobby McTeague (Place Services Inc.), Jens Byer (28101155 Lidl US, LLC), Luis Marcelino (Place Services Inc.), Shelley Chovan (21118060 Progressive Companies)		

Description



Submittal Workflow

Name	Sent Date	Due Date	Returned Date	Response	Attachments
General Information					Lidl-1517-Norcross-GA-Ground-Up-Submittal-PSI-0003-Plumbing-Package.pdf
Attachments					
Amy Schoch	Sep 3, 2026	Sep 11, 2026		Pending	

- ☐ NO EXCEPTIONS TAKEN ☒ MAKE CORRECTIONS NOTED
☐ REVISE & RESUBMIT ☐ REJECTED
☐ FOR RECORD PURPOSES ONLY

bizons / Sep 25, 2026

This submittal is reviewed for general conformance with the design concept of the project and with information given in the Contract Documents. This review shall not constitute approval of safety precautions and is not conducted for substantiating instructions for installation or performance of equipment or systems. This review does not relieve the contractor of responsibility for conformance with the Contract Documents and applicable codes, all of which have priority over this submittal. The Architect/Engineer does not warrant or represent that the information within the submittal is either accurate or complete. The contractor is responsible for coordinating and verifying all quantities, dimensions, tolerances, clearances, fabrication processes, techniques, sequences, means, methods of construction and compatibility of materials. It is also understood that the contractor has reviewed and coordinated all related trades and components of an assembly prior to issuing a submittal for review.

PROGRESSIVE COMPANIES
Progressive Architecture Engineering & Planning III, Inc.

Lidl 1517 Norcross, GA - Ground Up

4990 Jimmy Carter Blvd. Norcorss, GA 30093

Report generated by Tucker Teague (Place Services, Inc.)



Submittal #PSI-0003 - Tucker Teague

Creator	Tucker Teague (Place Services, Inc.)	Submission due	-
Assignee	Tucker Teague (Place Services, Inc.)	Response due	09/17/2026
Reviewers	-	Closed on	-
Status	Pending approval	Reference number	-
Type	Product data	Specification	22 11 19 DOMESTIC WATER PIPING SPECIALTIES
Description	-		

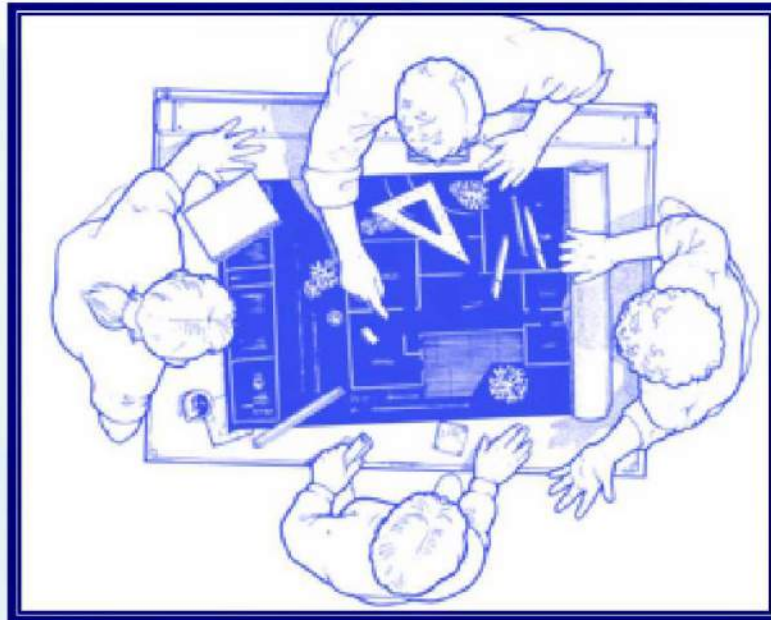
Approval (rev. 0)

Responded by	-	Responded on	-
Response	-		
Notes	-		
Attachments	-		

Submission (rev. 0)

Submitted by	Tucker Teague (Place Services, Inc.)	Submitted on	09/03/2026
Notes	-		
Attachments	PDF attachments are included in this export. ↓ #1517 Lidl Norcross, GA - Plumbing Submittals.pdf (Page 2)		

Plumbing Submittal



LIDL NORCROSS

Submitted by

GOLDEN HANDS PLUMBING

8/26/2026

COWAN SUPPLY

3032A, BANKERS IND DR, ST A
ATLANTA, GA 30360-2715
Phone: 404 351-6351
Fax: 404 351-1259

SHOP DRAWING / SUBMITTAL REVIEW

- ☒ NO EXCEPTION TAKEN
- ☐ REVISE AND RESUBMIT
- ☐ MAKE CORRECTIONS NOTED
- ☐ REJECTED

SUBMITTAL WAS REVIEWED FOR DESIGN CONFORMITY AND GENERAL CONFORMANCE TO CONTRACT DOCUMENTS ONLY. THE CONTRACTOR IS RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AT JOBSITE FOR TOLERANCE, CLEARANCE, QUANTITIES, FABRICATION PROCESSES AND TECHNIQUES OF CONSTRUCTION, COORDINATION OF HIS WORK WITH OTHER TRADES AND FULL COMPLIANCE WITH CONTRACT DOCUMENTS.

By: Tucker Teague
Date: 9/3/2026

Place Services, Inc
Canton, GA 30115

Spec#**Table of Contents**

1		ALL MODELS ARE SUBJECT TO APPROVAL
2		DF-1
3	630619	ELKAY VRCTLDDSC SS ADA WM BI-LEVEL VP PUSHBUTTON VP BUBBLER DRINKING FTN
4		HB-1
5	65231	WOODFORD B65C-4 3/4X4 SWT CHROME AUTO DRAIN FRZLS WALL HYDRANT W/ BOX & ANTI-SIPHON VAC BREAKER
6		HB-2
7	1624299	TSBRASS MV-0771-12N-BVT BRASS 49.93 GPM 4H 6 IN. CENTERSET WM DBL LEVER HDL MANUAL FCT
8		L-1
9	342134	KOHLER 2084-0 SOHO WHT 20 X 18 X 7IN VC 1H WM COM LAV SINK
10	834804	KOHLER 13460-CP SCULPTED POL CHROME 0.5 GPM 1H DM TOUCHLESS SENSOR FCT
11	258659	ZURN Z8743-PC 11/4X5 CP 17 GA BRASS COMM LAV GRID DRAIN W/ PO PLUG & OVERFLOW HOLES
12	974288	MAINLINE ML27001 11/4 CP 22GA BRASS P-TRAP L/ CLEANOUT
13	1166059	ZURN Z8804-XL-LR-PC 1/2X3/8 COMPCOMP CP BRASS XL ANGLE STOP & SUPPLY KIT W/ HDL EXT LEAD FREE
14	1017492	MAINLINE ML102EZ WHT LAV-GUARD TRAP & SUPPLY COVER F/ TUBULAR P-TRAP 2-ANGLE STOP & 2- SUPPLY ADA APPROVED
15	815216	JRS 0700-M32 LAV CARRIER
16		RD-1
17	1380508	JRSMITH 1015Y04
18		RH-1

Disclaimer:

The specification sheets provided herein were written and provided by the product's manufacturer and not by Hajoca Corporation. Consequently, Hajoca Corporation cannot assume and disclaims any responsibility for errors, omissions or any misstatements in the product specifications provided herein.

Spec#**Table of Contents**

19	2693001	PRIER P-RH4 SANITARY ROOF HYDRANT ASSEMBLY- ROOF MOUNT BRACKET & SANITARY HYDRANT
20		SK-1
21	1822786	KROWNE HS-5 HAND SINK SS 12 1/2 X 9 3/4 X 5 5/8IN 20 GA SS 2H 4 IN. CENTERSET 1 STATION WM HAND WASH SINK INCL WALL BRKT
22	1225542	KROWNE 10-400L POL CHROME 0.5 GPM 2H 4 IN. CENTERSET WM DBL LEVER HDL BAR FCT
23		SK-2
24	970812	ELKAY ELUHAD281655 LUSTERTONE CLASSIC LUSTROUS SATIN 30 1/2 X 18 1/2IN SGL BASIN UM SS KITCHEN SINK
25	2102151	ELKAY LKAV3032CR ELKAY AVADO 1 HOLE BAR FAUCET PULL-DOWN SPRAY LEVER HANDLE CHROME
26	2715468	ISE 80926 BADGER 5 1/2 HP CONT DISPOSER (OLD PART# 79883-ISE)
27		SK-3
28	2081398	ELKAY SS8354LR4 STURDIBILT BUFFED SATIN 102 X 27 1/2 X 44IN 14 GA SS 4H FM SCULLERY SINK INCL LEFT & RIGHT DRAINBOARD
29	1130255	KROWNE 17-109WL POL CHROME 1.15 GPM 2H 8 IN. WS WM TRIPLE LEVER HDL PRE-RINSE FCT W/ PULL-DOWN SPRAY
30		UB-1
31	1245652	GUYGRAY 88531 MIB1HAAB 1/2 SWT ICE MAKER BOX W/ ARRESTER LF
32		WC-1
33	2002188	KOHLER K-84325-0 WHITE KINGSTON ULTRA 11/2 TOP SPUD TOILET
34	1208094	ZURN ZTR-6200-WS1 1.6GPF FLUSHVALVE
35	793988	BEMIS 1655SSCT 000 WHT ELONG OF L/ COVER SSTA PLAS TOILET SEAT

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Spec#**Table of Contents**

36		WC-1 CARRIER OPTIONS
37	1886448	JRS 0211LY SGL LH CLOSET CARRIER
38	1466233	JR SMITH 0211DY BACK TO BACK DOUBLE CARRIER
39	1101626	JR SMITH 0230Y-M58/54 VERTICAL SINGLE CLOSET CARRIER
40		PP-1
41	997733	B&G 103401LF - NBF-36 3-SPEED WET ROTOR CIRC 115V/60HZ AB1953 LEAD FREE, 10GPM @ 15'TDH
42	532078	BG 113210 TC-1 TIMER
43	2174063	B&G 101501LF 3/4 SS FLG SET
44	684757	BG AQS 3/4 AQUASTAT CONTROL 113224
45		FD-1
46	2421064	MIFAB TS5-1 TOP SET STRAINER FOR 5" NB ROUND STRAINER
47	2726031	MIFAB A4-3-30-4 3" DRAIN BODY ONLY
48		FD-2
49	2514481	MIFAB TSS5-1 5" TOP SET SQUARE HEELPROOF STRAINER
50	2726031	MIFAB A4-3-30-4 3" DRAIN BODY ONLY
51		FD-3
52	2102082	ZURN ZN511-3NL-S-Y FLOOR DRAIN W/ SEDIMENT BUCKET
53		FS-1
54	1401756	ZURN Z1900-3NL-2 3X12 NEO-LOC SANI-FLOR ACID RESISTANT ENAMEL COAT CI RECEPT W/ 6 SUMP DEPTH & 1/2 GRATE
55		FS-2

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Spec#**Table of Contents**

56	64495	ELMUST 63M DURASTONE WHT 24 X 24 X 10IN MOP SVC BASIN
57		PRV
58	1240786	WATTS LFN223M1-S-B 0298617 3 TXT HIGH CAP IRON BODY 25-75PSI WTR PRESS RED VLV W/ STR SET AT 50PSI LEAD FREE
59		GREASE INTERCEPTOR
60	1460432	MISC NON-STOCK ITEM ***DO NOT USE TO BUY PRODUCT***
61		EWB-1
62	1918115	RHEEM ELD52-TB 683744 480V 6KW/6KW 50 GAL COM LT DUTY ELEC WTR HTR W/ TERM BLOCK 58-5/8H 20-1/4D
63		GAS METER
64	2496658	ON F-5500-3114-1150 INSERTION THERMAL MASS FLOW METER W/ 1.0 NPT ADAPTER, 24 VDC LOOP POWERED 4-20MA AND RS485 SERIAL COMM, INTEGRAL ENCLOSURE WITH LCD DISPLAY
65	2052116	ONICON INSTL0094-FMH INSTALLATION KIT

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ALL MODELS ARE SUBJECT TO APPROVAL



DF-1



PRODUCT SPECIFICATIONS

Elkay Wall Mount Vandal Resistant Bi-level ADA Cooler Non-filtered Non-refrigerated Stainless. Features shall include Vandal Resistant. Furnished with Vandal Resistant bubbler. Mechanical Front Bubbler Button activation. Product shall be Wall Mount (On Wall), for Indoor + Outdoor applications, serving 2 station(s). Unit shall be lead-free design which is certified to NSF/ANSI 61 & 372 (lead free) and meets Federal and State low-lead requirements.

Special Features:	Vandal Resistant
Finish:	Stainless Steel
Power:	No Electrical Required
Bubbler Style:	Vandal Resistant
Activation by:	Mechanical Front Bubbler Button
Mounting Type:	Wall Mount (On Wall)
Chilling Capacity:	Non-refrigerated
Dimensions (L x W x H):	36-5/16" x 18-5/8" x 27-1/8"
Approx. Shipping Weight:	71 lbs.
Installation Location:	Indoor + Outdoor
No. of Stations Served:	2

****When used in non-temperature controlled environments, unit(s) must be adequately winterized and/or protected from extreme heat to prevent damage where climates dictate.**

- Mechanically-Activated bubbler continues to supply water in event of service disruptions.



Included with Product: Water Cooler
Ships in one box.

A Century of Tradition and Quality. For more than 100 years, Elkay has been making innovative products and providing exceptional customer care. We take pride in offering plumbing products that make life easier, inspire change and leave the world a better place.



PRODUCT COMPLIANCE

ADA & ICC A117.1

ASME A112.19.3/CSA B45.4

Buy American Act

NSF/ANSI 61 & 372 (lead free)



Complies with ADA & ICC A117.1 accessibility requirements when installed according to the requirements outlined in these standards. Installation may require additional components and/or construction features to be fully compliant. Consult the local Authority Having Jurisdiction if necessary.

[Installation Instructions \(PDF\) - 98958C](#)

Water Systems are warranted for 12 months from date of installation.

Warranty pertains to drinking water applications only. Non-drinking water applications are not covered under warranty.

[Warranty \(PDF\)](#)

PART: _____ QTY: _____

PROJECT: _____

CONTACT: _____

DATE: _____

NOTES: _____

APPROVAL: _____

In keeping with our policy of continuing product improvement, Elkay reserves the right to change product specifications without notice. Please visit elkay the most current version of Elkay product specification sheets. This specification describes an Elkay product with design, quality, and functional benefits to When making a comparison of other producers' offerings, be certain these features are not overlooked.



630619

IMPORTANT! INSTALLER PLEASE NOTE:

This water cooler has been designed and built to provide water to the user which has not been altered by materials in the cooler waterways.

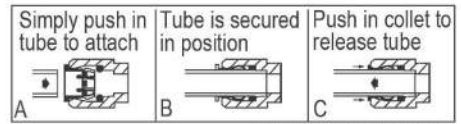
The grounding of electrical equipment such as telephone, computers, etc to water lines is a common procedure. This grounding may be in the building or occur away from the building. This grounding can cause electrical feedback into a water cooler creating an electrolysis which creates a metallic taste or causes an increase in the metal content of the water. This condition is avoidable by installing the cooler using the proper materials as shown below.

NOTICE

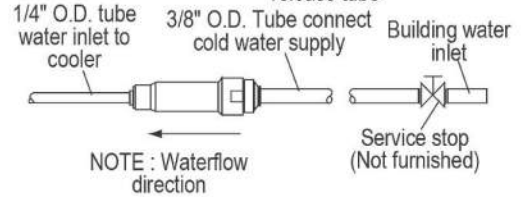
The water cooler must be connected to the water supply using a dielectric coupling. The cooler is furnished with a non-metallic strainer which meets this requirement. The drain trap which is provided by the installer should also be plastic to completely isolate the cooler from the building plumbing system.

These products are designed to operate on 20 psi to 105 psi supply line pressure. Simultaneous operation of both bubblers on a bi-level unit may not be possible depending on water supply pressure. If simultaneous operation is desired, please ensure a minimum of 50 psi supply.

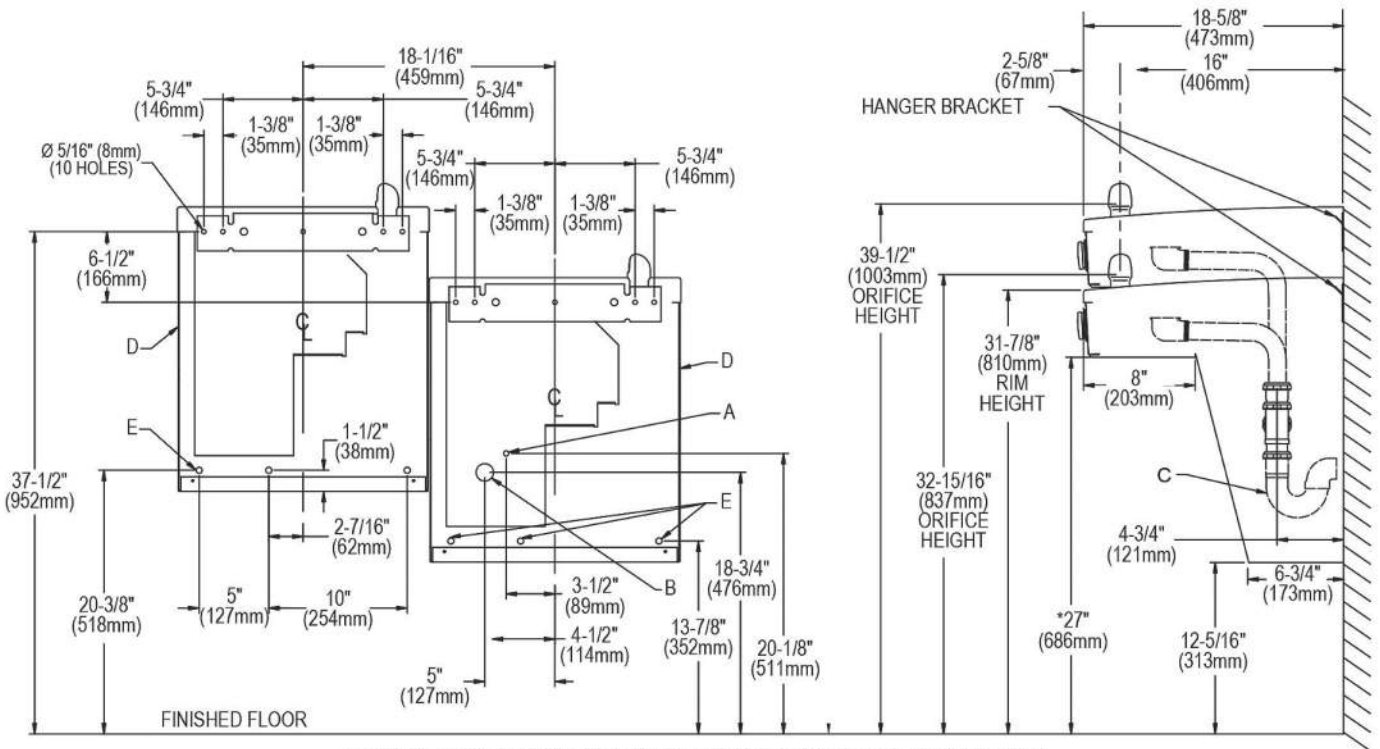
OPERATION OF QUICK CONNECT FITTINGS



Pushing tube in before pulling it out helps to release tube



NOTE : Waterflow direction



REDUCE HEIGHT BY 3" FOR INSTALLATION OF CHILDRENS ADA COOLER

LEGEND

- A = Recommended water supply location. Shut off valve (not furnished) to accept 3/8" O.D. unplated copper tube. 3" (76mm) maximum out from wall.
- B = Recommended location for waste outlet 1-1/2" O.D. drain. Drain stub 2" (51mm) out from wall.
- C = 1-1/2" Trap not furnished.
- D = Insure proper ventilation by maintaining 6" (152mm) (MIN.) clearance from cabinet louvers to wall.
- E = 7/16" Bolt holes for fastening unit to wall.

*ADA REQUIREMENT

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HB-1



The Model 65, B65, RB65 are automatic draining, freezeless wall hydrants with single check hose connection anti-siphon vacuum breakers. Hydrants drain as handle is shut off, *even if hose is attached*. All models are designed for irrigation purposes to blend in with modern architecture for installation on restaurants, schools, office buildings, churches, apartments, motels, stores, shopping centers and industrial buildings.

SPECIFICATIONS:

MODEL 65/B65/RB65

- Approved under ASSE Standard 1019-B
- Listed by IAPMO®
- Meets Government Specification WW-P-541b

VACUUM BREAKER - ANTI SIPHON -

- NIDEL® Model 34HA with 3/4 inch male hose thread
- Approved under ASSE 1011
- Listed by IAPMO®

VALVE SEAT – Permanent type brass valve body with hemispherical seating surface.

VALVE – One piece valve plunger accurately controls both flow and drainage with a minimum number of turns and without need for adjustments.

DRAIN – Under nozzle away from hands of operator and with a lip to divert water away from building.

CASING – Copper tubes.

NO LEAD SOLDER - All solder joints.

STEM – Hardened stainless steel stem resists damage.

TEE KEY – Loose key operates hydrant.

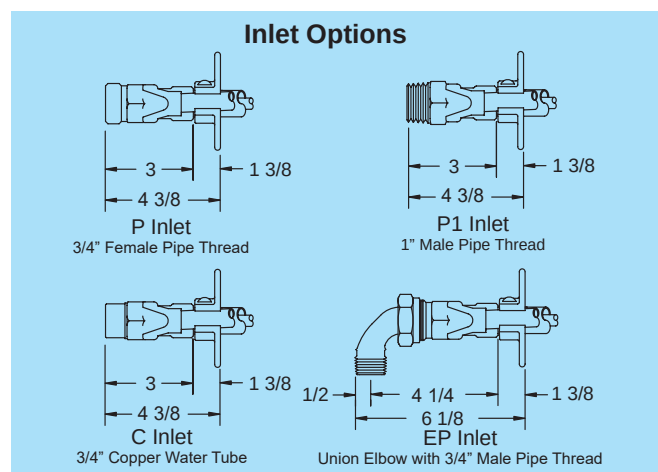
OPERATING ROD – 3/8" solid brass operating rod.

INLETS – As shown.

WALL CLAMP – Furnished on all 60 series except close coupled.

MAX PRESSURE – 125 p.s.i.

MAX TEMPERATURE - 120° F



Specify as follows:

Wall hydrant shall be Woodford Model 65 (exposed type), B65, RB65 or MB65 (concealed box type), automatic draining with anti-siphon vacuum breaker. ASSE Standard 1019-B approved. 3/4" inlet and outlet (specify type of inlet). Hardened stainless steel operating stem and one-piece valve plunger to control both flow and drain functions. Exterior finish to be Chrome Plated (options: Polished Brass or Rough Brass). Loose tee key to be furnished with each hydrant. Wall thickness to be _____ inches.

When ordering, specify model, inlet and wall thickness

Anti-Siphon Freezeless Wall Hydrants

Model 65/B65/RB65



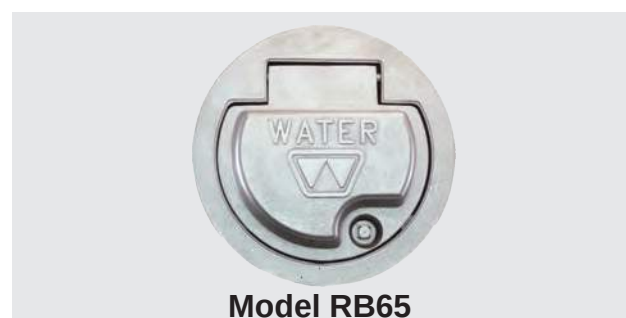
Model 65

Standard - Chrome (CH)
Optional - Brass (BR) Polished Brass (PB)
Fits one standard modular brick course.



Model B65

Exterior Finish: Box & Door
Standard - Chrome (CH)
Optional - Brass (BR) Polished Brass (PB)
Anodized Aluminum Box (AL)
Fits two std. modular brick course or one course of facing tile.

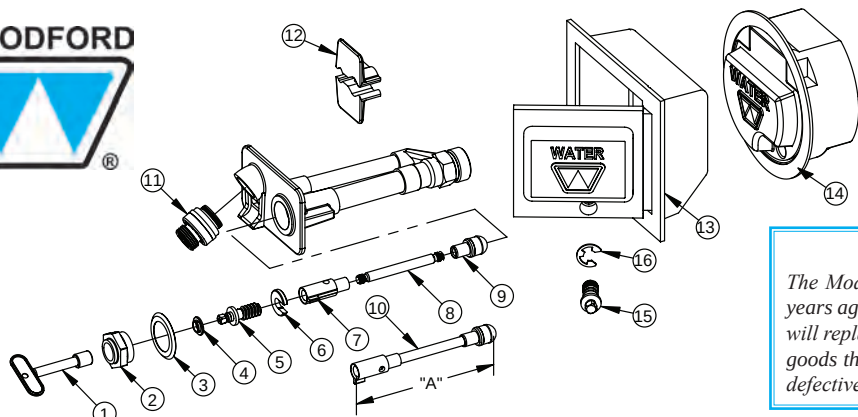


Model RB65

Exterior Finish: Box & Door - Chrome (CH) Only
Designed especially for tilt-up wall construction.
Install through 6" diameter hole.
Two 3/8"-16 tapped holes for all-thread anchors in back of box.
Lift and latch door stays open when hydrant is in use.

**SEE WOODFORDMFG.COM FOR INSTRUCTIONS
AND DETAILS OF APPLICABLE PATENTS AND
PATENTS PENDING OR CALL 1-800-621-6032**





LIMITED WARRANTY

The Model 65 is sold with a limited warranty for five years against defects in material and workmanship. We will replace or issue credit (at our option) for defective goods that are returned for inspection and found to be defective within five years of purchase from Woodford.

MODEL 65/B65/RB65 PARTS LIST

ITEM	PART #	DESCRIPTION
1	50009	Tee Key
	50010	Long Tee Key (Box Models)
2	50264	Head Nut – chrome
	50265	Head Nut – brass
3	50583	I.D. RING
4	50252	Stem Washer
5	55062	Stem Screw
6	50251	"C" washer
7	51014	Yoke Nut
8	Operating Rod (Sold With Item 10 Only)	
9	50250	Plunger
10	554XX	Operating Rod Assembly (Includes Items 7,8,9) (Select required assembled length from table below)
11	34HA-CH	34HA Vacuum Breaker – chrome
	34HA-BR	34HA Vacuum Breaker – brass
12	55063	Wall Clamp Assembly
13	67BX	Box/Door Assembly – Chrome
	65BXF	Box/Flat Door Assembly – Chrome
	67BX-BR	Box/Door Assembly - Brass
	65BXF-BR	Box/Flat Door Assembly - Brass
	67BX-PB	Box/Door Assembly – Polished Brass
	65BXF-PB	Box/Flat Door Assembly – Polished Brass
	67BX-AL	Box/Door Assembly – Aluminum
14	RB67BX	Round Box/Door Assembly - Chrome
	RK-65	Chrome Repair Kit (Includes Items 1, 2, 4-7 & 9)
15	51117	Door Lock Screw - Chrome
	51116	Door Lock Screw - Brass
16	51120	E-Ring

Operating Rod Assy. "A"

Wall Thickness	Length	Part No.
65	B65/RB65	
CC	CC	3 1/2"
N/A	4	4 1/2"
4	6	6 1/2"
6	8	8 1/2"
8	10	10 1/2"
10	12	12 1/2"
12	14	14 1/2"
14	16	16 1/2"
16	18	18 1/2"
18	20	20 1/2"
20	22	22 1/2"
22	24	24 1/2"
24	N/A	26 1/2"

Manufactured under one or more of the following patents:
 U.S. Patents:
 3,414,001
 4,178,956
 4,316,481
 4,532,954
 D216,790
 D216,791
 D277,365
 D277,366
 Canada Patents:
 852,529
 865,995
 1,146,438

Wall Thickness (Inch)	CC	4	6	8	10	12	14	16	18	20	22	24
Overall Length - Hydrant	5	8	10	12	14	16	18	20	22	24	26	28
Overall Length - Box Models	7 3/8	8 3/8	10 3/8	12 3/8	14 3/8	16 3/8	18 3/8	20 3/8	22 3/8	24 3/8	26 3/8	28 3/8
Shipping Wt. (Lbs)**	3.8	4.4	4.8	5.2	5.6	6	6.4	6.8	7.2	7.6	8	8.4

** Add 8.3 Lbs. for brass box models.

** Add 2.5 Lbs. for aluminum box models.

NOTE: Close Coupled Models are not recommended for use in freezing climates and cannot use wall clamp.

For more information contact...

WOODFORD MANUFACTURING COMPANY, LLC.

2121 Waynoka Road, Colorado Springs, Colorado 80915 • Phone: (800) 621-6032 • Fax: (800) 765-4115

To view our complete product line visit: www.woodfordmfg.com or email: sales@woodfordmfg.com



HB-2




T&S BRASS AND BRONZE WORKS, INC.

 2 Saddleback Cove / P.O. Box 1088
 Travelers Rest, SC 29690

Model No.

MV-0771-12N-BVT

Item No.

Travelers Rest, SC: 800-476-4103 • Simi Valley, CA: 800-423-0150 • Fax: 864-834-3518 • www.tsbrass.com

This Space for Architect/Engineer Approval

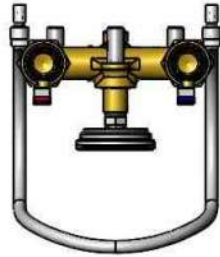
Job Name _____ Date _____

Model Specified _____ Quantity _____

Customer/Wholesaler _____

Contractor _____

Architect/Engineer _____


 3/4" Ball Valves
 w/ Stainless Steel,
 Color Coded
 Handles & Nuts

 3/4" Check
 Valves

3/4" Unions

 Bi-Metallic
 Thermometer
 w/ Stainless
 Steel Case

 5"
 [127mm]

 9 1/4"
 [235mm]

 6"
 [152mm]

Open

Open

Closed

Closed

 7 15/16"
 [202mm]

 2 3/4"
 [70mm]

 ϕ 7/16" (3)
 [11mm]

 ϕ 5/16" (2)
 [7mm]

Hose Swivel

 Stainless Steel
 Hose Rack w/
 Mounting Hardware

 8"
 [203mm]

 3 3/16"
 [81mm]

 3/4" NPT
 Female
 Thread Inlets

 1 1/2"
 [38mm]

 14 3/16"
 [361mm]

Wall

 9 3/8"
 [239mm]

Product Specifications:

 Natural Finish Washdown Station w/ 3/4" Mixing Valve, Ball Valves, Check
 Valves, Thermometer, Hose Swivel & Stainless Steel Hose Rack

Product Compliance:

NSF 61 Exempt (Non-Potable)

Drawn: DMH | Checked: JRM | Approved: JHB | Date: 07/17/20 | Scale: 1:8 | Sheet: 1

1624299





T&S BRASS AND BRONZE WORKS, INC.

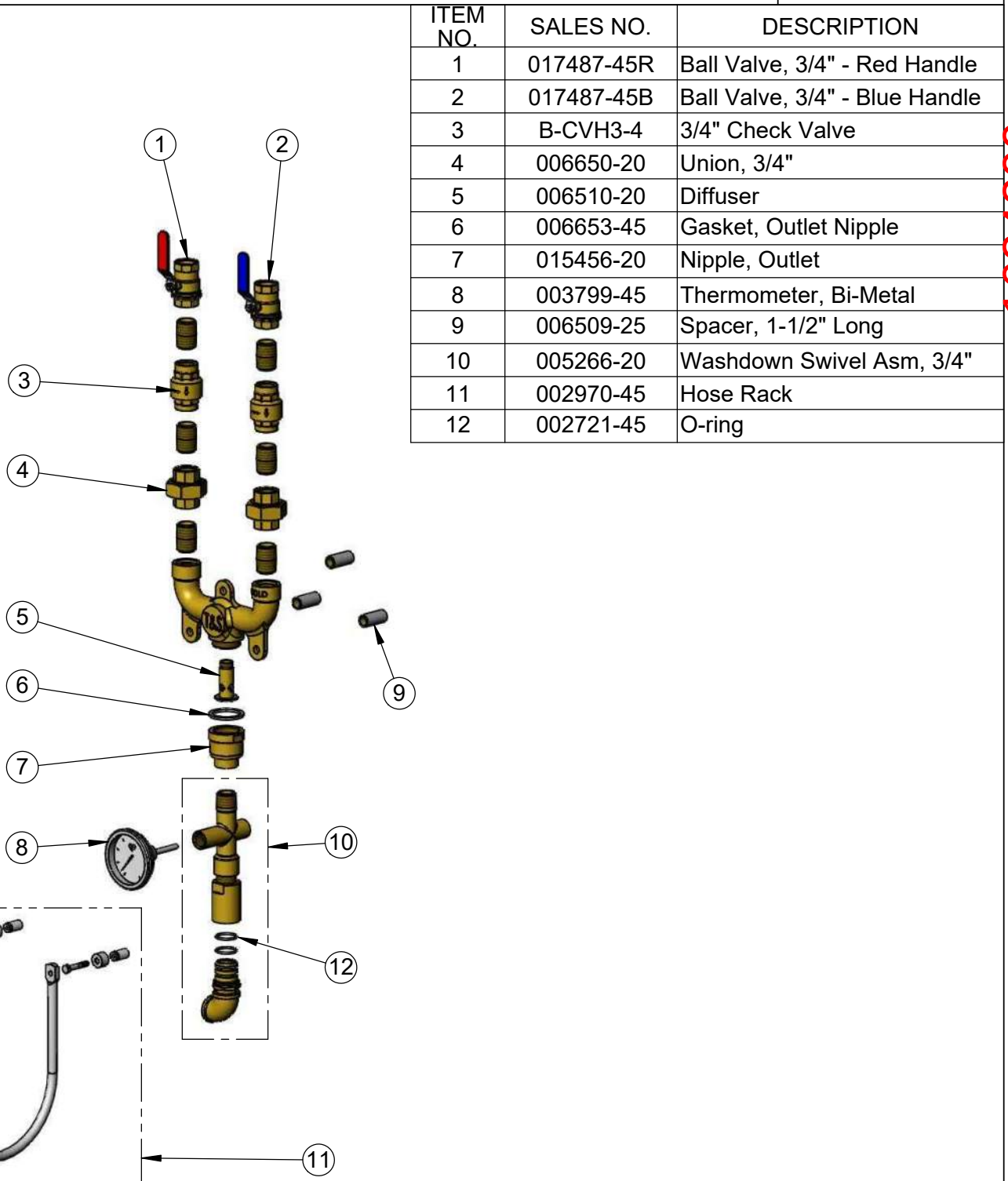
2 Saddleback Cove / P.O. Box 1088
Travelers Rest, SC 29690

Model No.

MV-0771-12N-BVT

Item No.

Travelers Rest, SC: 800-476-4103 • Simi Valley, CA: 800-423-0150 • Fax: 864-834-3518 • www.tsbrass.com



1624299

Product Specifications:

Natural Finish Washdown Station w/ 3/4" Mixing Valve, Ball Valves, Check Valves, Thermometer, Hose Swivel & Stainless Steel Hose Rack

Product Compliance:

NSF 61 Exempt (Non-Potable)



L-1





Soho®
Wall-Mount Bathroom Sink
K-2084

Features

- Made from premium materials that withstand high-volume usage
- 20" (508 mm) x 18" (457 mm)
- Faucet sold separately

Material

- Vitreous china

Installation

- Wall-mount
- Drilled for concealed arm carrier installation

Recommended Products/Accessories

K-8998 P-Trap
K-23726 Drain treatment
K-23725 Cast iron cleaner



342134

ADA **CSA B651** **OBC**

Codes/Standards

ASME A112.19.2/CSA B45.1
ADA
ICC/ANSI A117.1
CSA B651
OBC

KOHLER® Vitreous China Bathroom Sink Lifetime Limited Warranty

See website for detailed warranty information.

Available Colors/Finishes

Color tiles intended for reference only.

Color	Code	Description
	0	White
	96	Biscuit
	7	Black Black™

1-800-4KOHLER (1-800-456-4537)

Kohler Co. reserves the right to make revisions without notice to product specifications.

For the most current Specification Sheet, go to www.kohler.com USA or www.kohler.ca Canada

2-4-2026 21:47 - US/CA

THE BOLD LOOK
OF **KOHLER**



KOHLER Faucets

Sculpted

Touchless single-hole lavatory faucet with Insight™ sensor technology, DC-powered, 0.5 gpm
K-13460

Features

- 0.5 gpm (1.9 lpm) maximum flow rate
- Factory-set 30-second maximum continual run cycle
- Vandal-resistant aerator
- 24" (610 mm) flexible supply hose for easy installation
- Less drain
- Includes two AAA batteries

Material

- Brass

Technology

- Electronic Insight™ sensor technology for accurate and consistent activation

Installation

- Single-hole mounting
- Above-counter valve for easy installation and maintenance

Recommended Products/Accessories

K-23726 Drain treatment

Optional Products/Accessories

1160595 .35 gpm Aerator, Medium
 1075455 Medium Spray 0.5 gpm (1.9 l/min)
 1139284 Medium Aerated 1.0 gpm (3.8 l/min)
 1139285 Medium Laminar 1.0 gpm (3.8 l/min)
 1097180 Medium Aerated 1.5 gpm (5.7 l/min)
 1104546 Medium Laminar 1.5 gpm (5.7 l/min)
 79441 Sealing Washer

**ADA****CSA B651****OBC****Codes/Standards**

ASME A112.18.1/CSA B125.1
 NSF/ANSI/CAN 372
 DOE - Energy Policy Act 1992
 ADA
 ICC/ANSI A117.1
 CSA B651
 OBC
 NOM Standards
 NOM 012-CONAGUA

**KOHLER® One-Year Limited
 Warranty**

See website for detailed warranty information.

834804

1-800-4KOHLER (1-800-456-4537)

Kohler Co. reserves the right to make revisions without notice to product specifications.

For the most current Specification Sheet, go to www.kohler.com USA or www.kohler.ca Canada

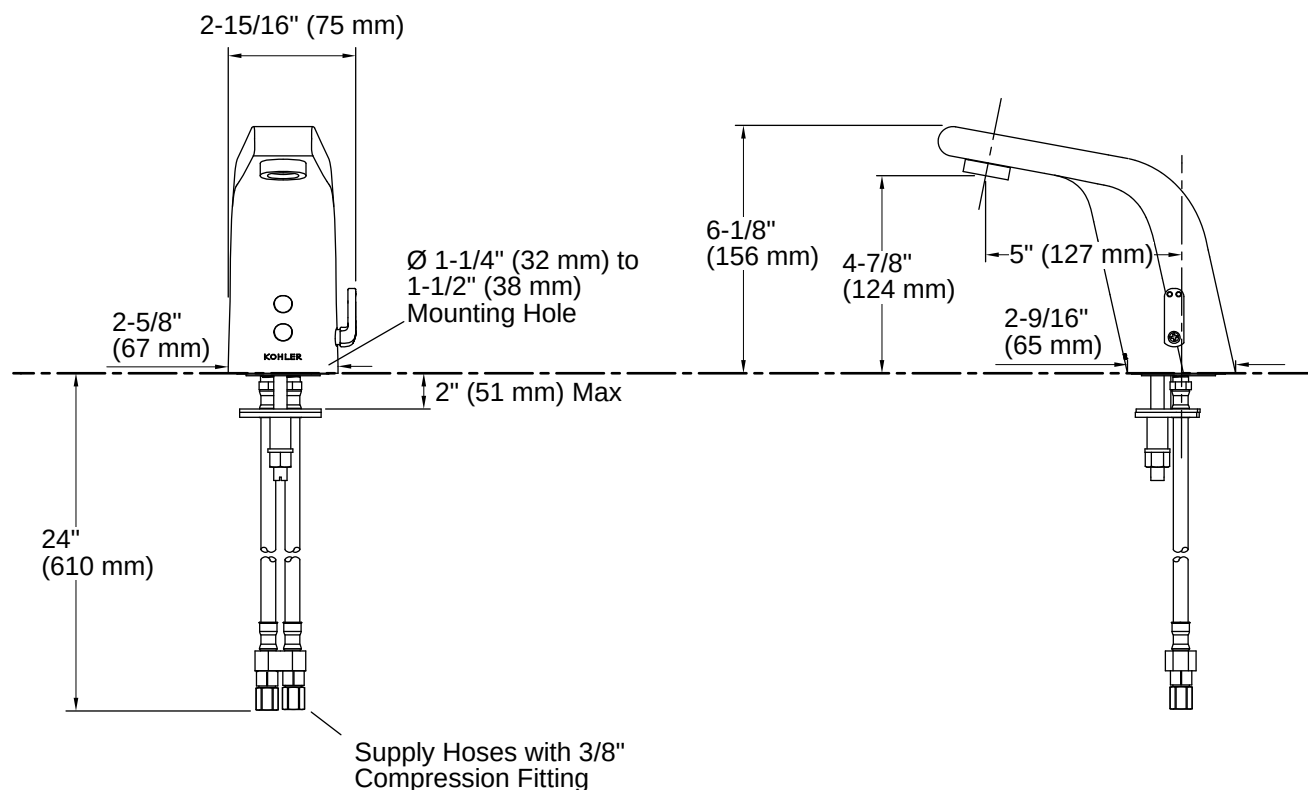
3-7-2026 19:31 - US/CA

THE BOLD LOOK
 OF **KOHLER**



KOHLER Faucets

Sculpted
 Touchless single-hole lavatory faucet with Insight™
 sensor technology, DC-powered, 0.5 gpm
K-13460



Technical Information

All product dimensions are nominal.

Power source: Battery - DC, included

Faucet flow rate: 0.5 gal/min (1.9 l/min)

Spout:

Spout reach: 5" (127 mm)

Notes

Install this product according to the installation guide.

A 1-1/4" (32 mm) minimum to 1-1/2" (38 mm) maximum diameter mounting hole is required.

The minimum distance between the back of the faucet spout and the wall must be 1-3/4" (44 mm). This will allow proper clearance for setscrew access.

The maximum distance between the bathroom sink basin edge to the base of the faucet spout must be 1-3/4" (44 mm).

The maximum height of the lip basin should be 3/4" (19 mm).

Product includes rosebud spray aerator. Other aerator styles are available.

ADA, OBC, CSA B651 compliant when installed to the specific requirements of these regulations.

1-800-4KOHLER (1-800-456-4537)

Kohler Co. reserves the right to make revisions without notice to product specifications.

For the most current Specification Sheet, go to www.kohler.com USA or www.kohler.ca Canada

3-7-2026 19:31 - US/CA

THE BOLD LOOK
 OF **KOHLER**



834804

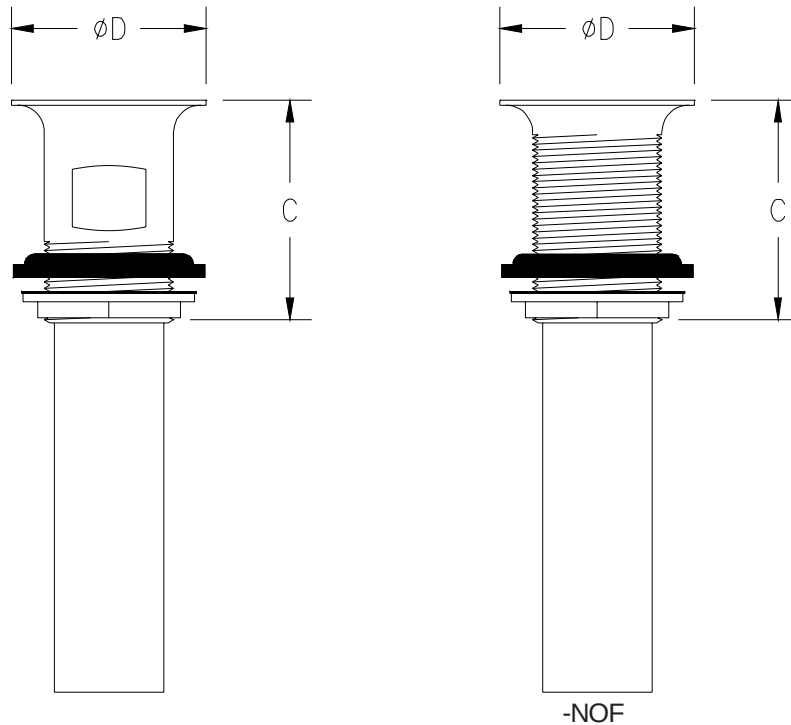


SOLID TOP, OPEN-GRID P.O. PLUGS

Z8743-PC

TAG _____

Engineering Specification: Zurn Z8743-PC – Chrome plated cast brass solid top, open grid, P.O. plug. Furnished with 5"[127mm] long, chrome plated brass 17 gauge tailpiece and chrome plated cast brass locknut for sink depth up to 2 1/4"[57mm]. 6"-, 8"-, and 12"- tailpieces available.



258659

Note: All dimensions are for reference only. Do not use for pre-plumbing

PRODUCT NUMBER	TAILPIECE in[mm]	DIMENSIONS		MASTER CARTON	WEIGHT CARTON
		Cin[mm]	ØDin[mm]		
Z8743-PC	1 1/4[32]	2 9/16[65]	2 1/4[57]	50	50
Z8743-1-PC	1 1/2[38]	2 9/16[65]	2 1/4[57]	30	30

OPTIONAL ACCESSORIES

Suffix	Description
_____ -NOF	No Overflow
_____ -6	w/6"[152mm] Tailpiece
_____ -8	w/8"[203mm] Tailpiece
_____ -12	w/12"[305mm] Tailpiece

ZURN INDUSTRIES, LLC ♦ COMMERCIAL BRASS OPERATION ♦ 2640 SOUTH WORK STREET ♦ FALCONER NY 14733

Phone: 1-716-665-1132 ♦ Fax: 1-716-665-1135 ♦ World Wide Web: www.zurn.com

In Canada: ZURN INDUSTRIES LIMITED ♦ 3544 Nashua Drive ♦ Mississauga, Ontario L4V1L2 ♦ Phone: 905/405-8272 Fax: 905/405-





1-1/4" TUBULAR P-TRAP

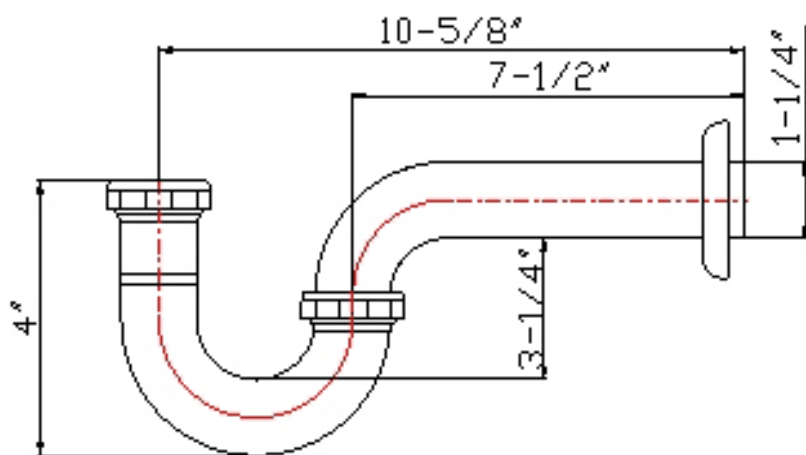
SPECIFICATIONS

1-1/4" TUBULAR LESS CLEANOUT

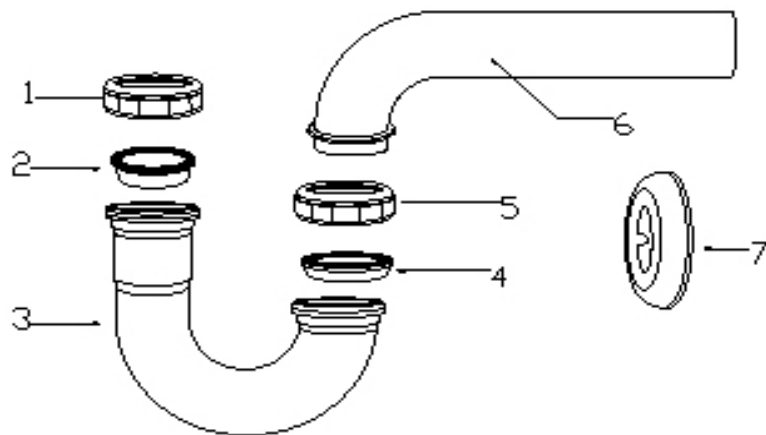
- 7-1/2" Brass Wall Tube
- 1-1/4" x 10-5/8"
- Zinc Nut Kit
- Shallow Escutcheon
- Chrome Plated



974288



Model #	ML7001	ML27001
Gauge	20	22



No.	Part Name	Material
1	Nut	Zinc
2	Washer	Rubber
3	J-Bend	Brass
4	Washer	Rubber
5	Nut	Zinc
6	Wall Tube	Brass
7	Escutcheon	Steel

CERTIFICATIONS

- cUPC



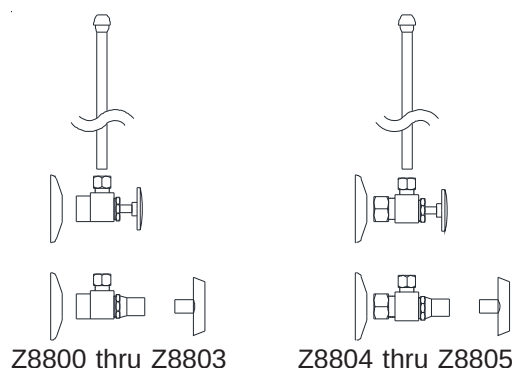


STANDARD STOP WITH FLEXIBLE LAVATORY SUPPLY

Z8800-XL-LR-PC TO Z8809-XL-LRLK-PC TAG _____

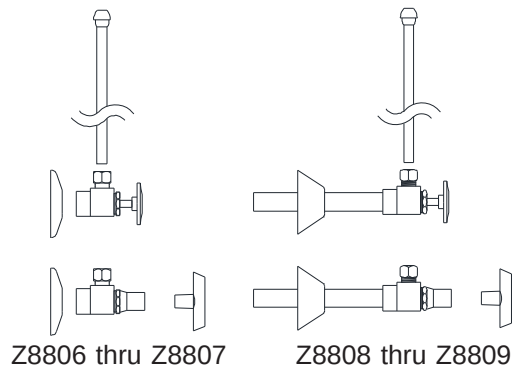
Engineering Specification: Zurn Z8800-XL-LR-PC to Z8809-XL-LRLK-PC Two Zurn chrome plated, solid brass angle stops with round wheel handles or loose key as specified, two 12"[305mm] flexible chrome plated copper lavatory risers complete with two chrome plated steel flanges. Z8808-XL-LR-PC to Z8809-XL-LRLK-PC include 5"[127mm] chrome plated copper extension tubes and deep bell steel flanges.

Zurn Lead Compliant* "XL" products are a line of durable, high quality brass faucets and fixtures that are designed and manufactured to comply with state laws and local codes that mandate lead content levels. Zurn "XL" products are manufactured with *not more than 0.25% lead content when used with respect to wetted surfaces of pipes and pipe fittings, plumbing fitting and fixtures: California Health & Safety Code § 116875; Vermont 9 VCA § 2470h.



Z8800 thru Z8803

Z8804 thru Z8805



Z8806 thru Z8807

Z8808 thru Z8809

PRODUCT NUMBER	CONNECTIONS	EXT.	MASTER CARTON	WEIGHT CARTON	
				LR	LRLK
Z8800-XL-LR-PC	3/8 [10] IPS x 3/8 [10] OD	-	25	24	
Z8800-XL-LRLK-PC	3/8 [10] IPS x 3/8 [10] OD	-	25		25
Z8802-XL-LR-PC	1/2 [13] IPS x 3/8 [10] OD	-	25	24	
Z8802-XL-LRLK-PC	1/2 [13] IPS x 3/8 [10] OD	-	25		25
Z8803-XL-LR-PC	1/2 [13] IPS x 1/2 [13] OD	-	25	24	
Z8803-XL-LRLK-PC	1/2 [13] IPS x 1/2 [13] OD	-	25		25
Z8804-XL-LR-PC	1/2 [13] NOM x 3/8 [10] OD	-	25	24	
Z8804-XL-LRLK-PC	1/2 [13] NOM x 3/8 [10] OD	-	25		25
Z8805-XL-LR-PC	1/2 [13] NOM x 1/2 [13] OD	-	25	24	
Z8805-XL-LRLK-PC	1/2 [13] NOM x 1/2 [13] OD	-	25		25
Z8806-XL-LR-PC	1/2 [13] SWT x 3/8 [10] OD	-	25	24	
Z8806-XL-LRLK-PC	1/2 [13] SWT x 3/8 [10] OD	-	25		25
Z8807-XL-LR-PC	1/2 [13] SWT x 1/2 [13] OD	-	25	24	
Z8807-XL-LRLK-PC	1/2 [13] SWT x 1/2 [13] OD	-	25		25
Z8808-XL-LR-PC	1/2 [13] SWT x 3/8 [10] OD	5	25	28	
Z8808-XL-LRLK-PC	1/2 [13] SWT x 3/8 [10] OD	5	25		33
Z8809-XL-LR-PC	1/2[13] SWT x1/2[13] OD	5	25	28	
Z8809-XL-LRLK-PC	1/2 [13] SWT x 1/2 [13] OD	5	25		33

SUFFIXES

- ☐ -A Single supply option
- ☐ -C1 Standard Line Lock Shield Cap
- ☐ -CE Chrome Plated, Cast Brass, Flange with setscrew
- ☐ -D Chrome Plated Steel, Deep Bell Flange (Bell Type)
- ☐ -XP Cross Handle

*REGULARLY FURNISHED UNLESS OTHERWISE SPECIFIED

- ☐ - 3 3[76] Nipple
- ☐ - 5 5[127] Nipple
- ☐ -15 15[381] Supply Tube
- ☐ -20 20[508] Supply Tube
- ☐ -30 30[762] Supply Tube
- ☐ -36 36[914] Supply Tube

ZURN INDUSTRIES, LLC. ♦ COMMERCIAL BRASS OPERATION ♦ 5900 ELWIN BUCHANAN DRIVE ♦ SANFORD NC 27330

Phone: 1-800-997-3876 ♦ Fax: 919-775-3541 ♦ World Wide Web: www.zurn.com

In Canada: ZURN INDUSTRIES LIMITED ♦ 3544 Nashua Drive ♦ Mississauga, Ontario L4V1L2 ♦ Phone: 905-405-8272 Fax: 905-405-1292



1166059



Molded Trap Covers

Product Specifications

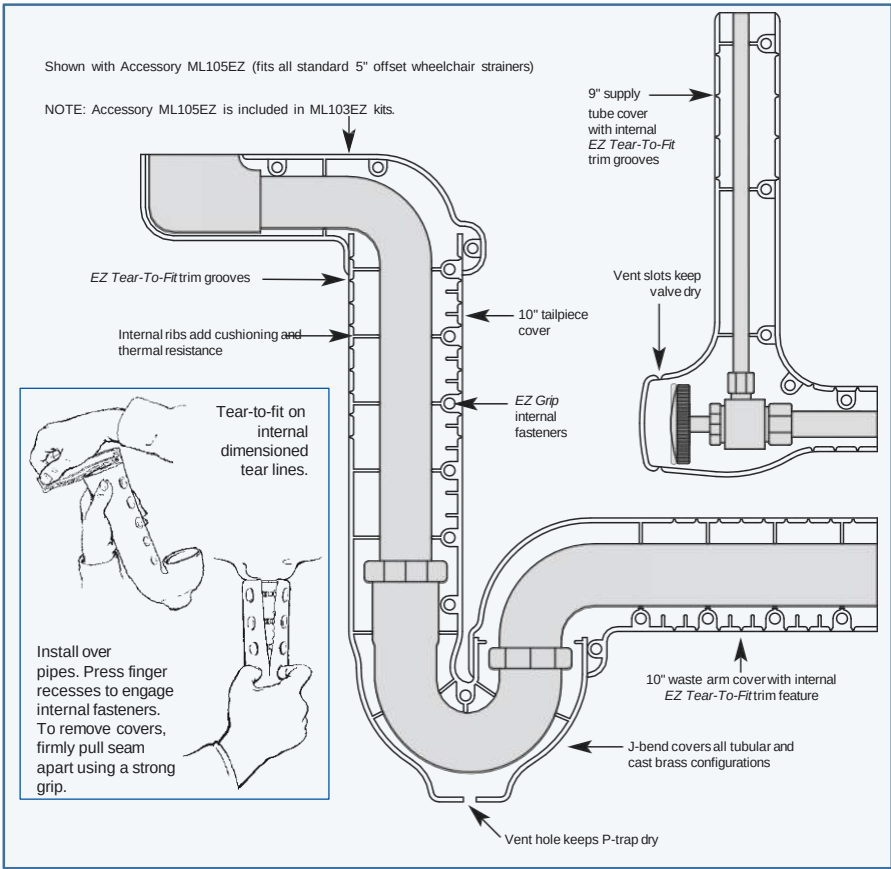
FEATURES

EZ Series waste and supply piping covers satisfy ADA compliance requirements. Built-in EZ Grip internal fasteners and internal EZ Tear-To-Fit trimming feature allow for fast installation without tools. Cover kits are designed to fit on all tubular and cast brass 1-1/4" or 1-1/2" P-trap assemblies. Valve cover completely encloses 3/8" or 1/2" angle valve and supply tube for both handled and keyed type water stops.

✓ To Submit	Part #	Description
	ML101EZ	One P-trap cover, one angle valve and supply covers
	ML102EZ	One P-trap cover, two angle valve and supply covers
	ML103EZ	One P-trap cover, two angle valve and supply covers, one 5" offset wheelchair strainer cover
	ML105EZ	One 5" offset wheelchair strainer cover



1017492



SPECIFICATIONS	
Material	Rigid high-impact, stain-resistant PVC
Nominal Wall	1/8" constant with internal ribs
Durometer	70-80 - Shore A
UV Protection	Will not fade or discolor
Durability	Virtually indestructible
Trimming (EZ Series)	EZ Tear-To-Fit trim feature (no tools needed)
Fasteners (EZ Series)	Internal EZ Grip fasteners, reusable
Color	China white
Paintability	Apply latex paint
Burning Characteristics ASTM D-635	Self extinguished 0 sec (ATB) 0 mm (AEB)
Bacteria/Fungus Resistance	ASTM G21 and G22 — Result: 0 growth
Maintenance	Wipe clean using common detergents

Warranty

See warranty information for more details.

All dimensions listed are nominal. MAINLINE® reserves the right to make product and material changes at any time without notice.

Contact your sales representative for more information or visit our website at mainlinecollection.com



815216



JAY R. SMITH MFG. CO.
 MEMBER OF MORRIS GROUP INTERNATIONAL
 POST OFFICE BOX 3237
 MONTGOMERY, ALABAMA 36109-0237 (USA)
 TEL: 334-277-8520 FAX: 888-377-7818 www.jrsmith.com



MEMBER OF:

LOCATION

LAVATORY SUPPORTS WITH CONCEALED ARMS

FLOOR MOUNTED WITH "PRO-SET" UPRIGHTS

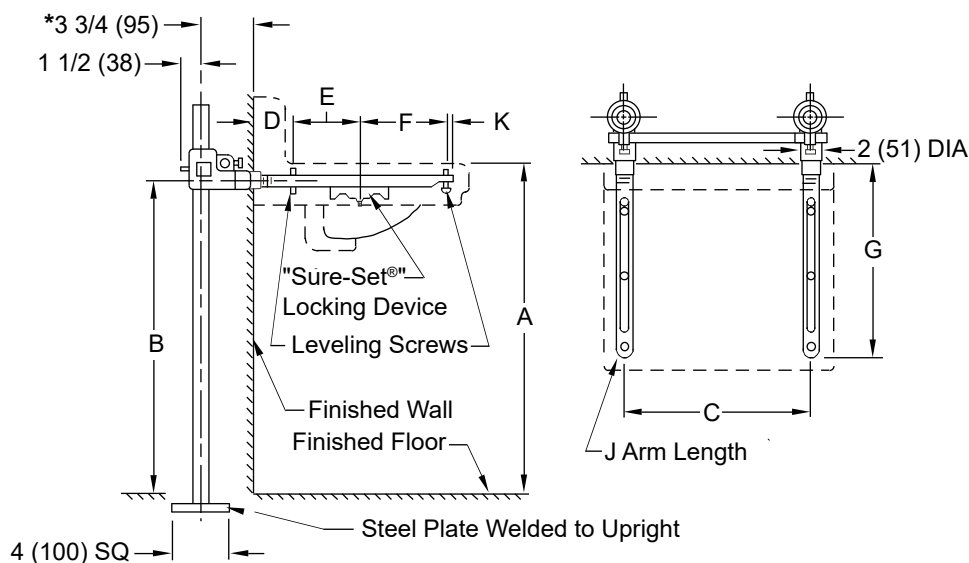
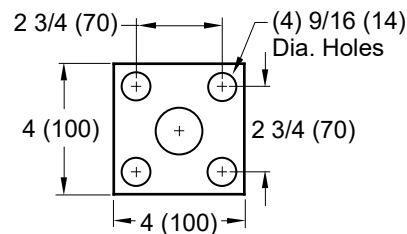


Fig. 0700 . . . FOR HIGH BACK LAVATORIES (shown)

NOTE: The rough-in dimensions listed on this drawing are based on current information available to Jay R. Smith by the fixture manufacturers. We recommend that you verify these dimensions with the fixture manufacturer. We will not accept responsibility for rough-in dimension changes by the fixture manufacturer.

NOTE: Dimensions shown in parentheses are in millimeters.



BASE DETAIL

Conforms to ASME A112.6.1M.

ROUGHING-IN TABLE

FIXTURE MFG.	NAME	NUMBER	SIZE	A	B	C	D	E	F	G	J	K
AM. STD.	LUCERNE	0355.012	20 (510) x 18 (455)	31 (785)	29 1/4 (760)	16 3/4 (425)	4 (100)	6 (150)	6 (150)	16 1/2 (420)	19 (485)	1/2 (13)
AM. STD.	LUCERNE	0356.015	20 (510) x 18 (455)	31 (785)	29 1/4 (760)	16 3/4 (425)	4 (100)	6 (150)	6 (150)	16 1/2 (420)	19 (485)	1/2 (13)
AM. STD.	LUCERNE	0356.421	20 (510) x 18 (455)	31 (785)	29 1/4 (760)	16 3/4 (425)	4 (100)	6 (150)	6 (150)	16 1/2 (420)	19 (485)	1/2 (13)
AM. STD.	MURRO	0954.000	20 (510) x 21 (455)	34 (865)	32 3/8 (823)	16 3/4 (425)	4 (100)	6 (150)	6 (150)	14 1/2 (370)	17M (430)	1/2 (13)
AM. STD.	MURRO	0955.000	22 (560) x 21 (535)	34 (865)	32 3/8 (823)	16 3/4 (425)	4 (100)	6 (150)	4 (100)	14 1/2 (370)	17M (430)	1/2 (13)
AM. STD.	MURRO	0958.000	22 (560) x 21 (535)	34 (865)	32 3/8 (823)	16 3/4 (425)	4 (100)	6 (150)	4 (100)	14 1/2 (370)	17M (430)	1/2 (13)
KOHLER	CHESAPEAKE	K-1722, 24, 28	19 (483) x 17 (432)	31 (785)	29 3/8 (746)	15 3/8 (391)	3 1/4 (83)	5 1/4 (135)	5 1/4 (135)	14 1/4 (360)	17M (430)	1/2 (13)
KOHLER	CHESAPEAKE	K-1721, 23, 29	20 (510) x 18 (455)	31 (785)	29 7/8 (759)	15 3/4 (400)	3 1/4 (83)	5 1/4 (135)	5 1/4 (135)	14 5/8 (371)	17 1/4 (430)	7/8 (22)
KOHLER	GREENWICH	K-2030, 1, 2, 4, 6	20 (510) x 18 (455)	31 (785)	29 (735)	17 1/4 (440)	3 1/4 (83)	5 1/4 (135)	5 1/4 (135)	14 1/4 (360)	16M (405)	1/2 (13)
KOHLER	KINGSTON	K-2005, 6, 7	20 (510) x 18 (455)	31 (785)	29 1/8 (740)	17 1/4 (440)	2 5/8 (67)	5 7/8 (149)	5 1/4 (135)	14 1/4 (360)	16M (430)	1/2 (13)
KOHLER	SOHO	K-2053, 2054, 2084	20 (510) x 18 (455)	31 (785)	29 1/4 (745)	16 1/2 (420)	3 1/4 (83)	5 1/4 (135)	5 1/4 (135)	14 1/4 (360)	17M (430)	1/2 (13)
TOTO	COMMERCIAL LAV.	LT 307	21 (533) x 18 (455)	34 (864)	33 1/4 (845)	16 3/4 (425)	3 3/4 (95)	8 1/8 (206)	4 1/2 (114)	16 7/8 (429)	19 (483)	1/2 (13)
ZURN	LAV	Z5361, 4, 8	20 (510) x 18 (455)	31 (785)	29 1/2 (749)	17 (432)	3 1/4 (83)	5 1/4 (135)	5 1/4 (135)	14 1/4 (360)	17M (430)	1/2 (13)
ZURN	LAV	Z5321, 4, 8, 9	23 (584) x 20 (510)	34 (864)	31 3/4 (806)	16 3/4 (425)	4 (100)	7 (178)	7 (178)	18 1/2 (470)	20 (510)	1/2 (13)

***NOTE:** This dimension can be increased when fixture support is to be installed behind the wall. (Specify when required and wall thickness).

P
N
M
L

4-18-24
9-27-23
9-14-23
4-19-23

Added ASME Note
Revised Table
Revised Table
Revised Table

HS
HS
HS
KK

JT
JT
JT
JT

WEIGHT
POUNDS

VOLUME
CUBIC FEET

FIGURE NUMBER

0700



WARNING: Cancer and Reproductive Harm - www.P65Warnings.ca.gov

DRAWING NUMBER
S0700SIZE
ASCALE:
NONEDATE:
2-20-85APPROVED BY:
SJMCHECKED BY:
SJMDRAWN BY:
CR

0700

FIGURE
NUMBER

WE CAN ASSUME NO RESPONSIBILITY FOR USE OF SUPERSEDED OR VOID DATA

DIMENSIONS ARE SUBJECT TO MANUFACTURERS TOLERANCE AND CHANGE WITHOUT NOTICE

RD-1



1380508

SMITH® JAY R. SMITH MFG. CO.®
MEMBER OF MORRIS GROUP INTERNATIONAL
POST OFFICE BOX 3237
MONTGOMERY, ALABAMA 36109-0237 (USA)
TEL: 334-277-8520 FAX: 334-272-7396 www.jrsmith.com

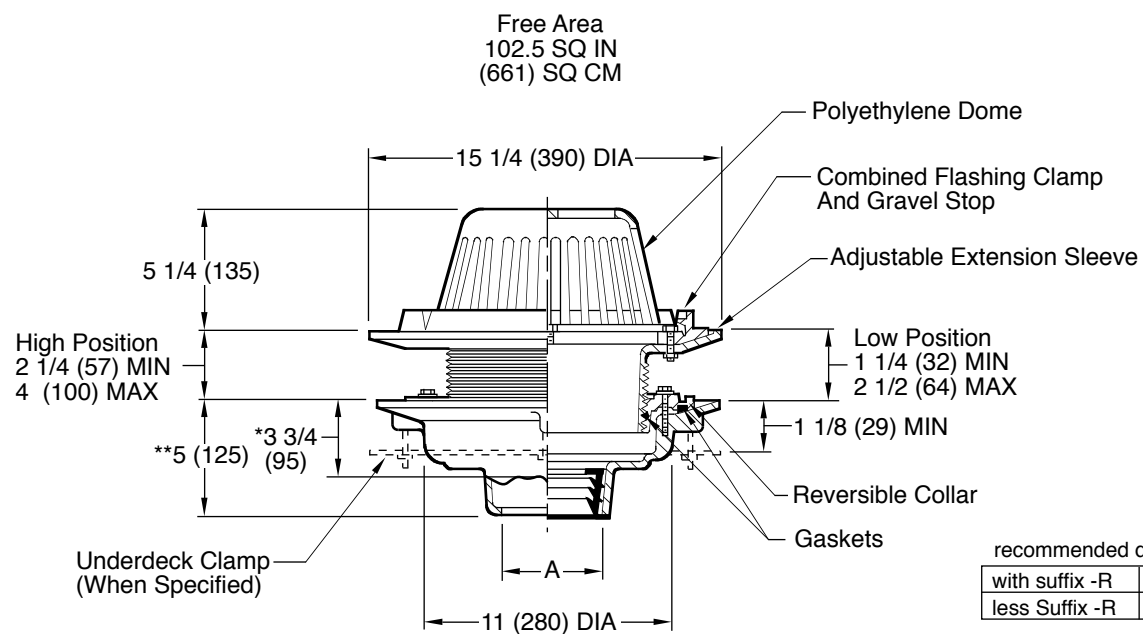


LOCATION

GENERAL PURPOSE ROOF DRAINS

ROOF DRAIN WITH ADJUSTABLE EXTENSION

FUNCTION: Used in flat roofs of any construction. The adjustable extension sleeve permits adjustment from a minimum of 1 1/4" (32) to a maximum of 4" (100), easily accommodating up to 2 3/4"(70) of varying insulation thicknesses. The positive threading allows smooth perimeter movement and circumferential support of the top at any adjusted elevation. Large low profile dome provides sufficient free area for quick drainage of rainwater and protection against debris.



recommended deck opening	
with suffix -R	17 (430) DIA
less Suffix -R	14 (355) DIA

*This dimension to internal stop of Speedi-Set Gasket
**Same for Caulk, NO-HUB and Speedi-Set, 3 3/4 (95) for Threaded

- ☐ Fig. 1015C CAULK OUTLET
- ☒ Fig. 1015Y NO-HUB OUTLET

NOTE: Stated flow rates for roof drains are based on particular test piping configurations and dimensions. Installed roof drain flow rates cannot be stated unless the attached piping configuration is known and the effects of the drainage calculated.

REGULARLY FURNISHED:
Duco Cast Iron Body with Adjustable Extension Sleeve, Reversible Collar, Combined Flashing Clamp and Gravel Stop, with Polyethylene Dome.

Note: See reverse side for assembly instructions.

NOTE: Dimensions shown in parentheses are in millimeters.

- VARIATIONS:**
- ☐ Expansion Joint (Specify Fig. 1710)
 - ☐ "L" Shaped Underdeck Clamp -CL
 - ☐ L Speedi-Set Service Weight 02(50), 03(75) & 04(100)" sizes only
 - ☐ LXH Speedi-Set Extra Heavy 02(50), 03(75) & 04(100)" sizes only
 - ☐ Secondary Flashing Clamp -C2
 - ☐ Sump Receiver -R
 - ☐ Underdeck Clamp -C
 - ☐ Vandal Proof Dome -U
 - ☐ Side Outlet Drain (Specify Fig. 1025)
 - ☐ T Threaded Outlet

- OPTIONAL MATERIALS:**
- ☐ Aluminum Dome -AD
 - ☐ Cast Iron Dome -CID
 - ☐ Galvanized Cast Iron -G
 - ☐ Rough Bronze Dome -RBD

▲ Optional domes' free area are as follows:

Aluminum	98 (632)
Cast Iron	80 (516)
Bronze	95 (613)

FIGURE NUMBER

1015

12-19-84

DATE:

12-19-84

APPROVED BY:

JD

CHECKED BY:

VGD

DRAWN BY:

CMB

SCALE:

NONE

SIZE:

A

DRAWING NUMBER

S1015

M

REV.	DATE	DESCRIPTION	BY	CKD. BY	WEIGHT POUNDS	VOLUME CUBIC FEET	FIGURE NUMBER
	8-18-15	Added Flow Rate Note	TBW	BW			1015
	06/01/09	Added Optional Dome Notes	JJ	CL			
	12-15-06	Added Note	RN	CL			



H

DRAWING NUMBER
S1015BS

SIZE
A

SCALE:
NONE

DATE:
6-29-88

APPROVED BY:
TD

CHECKED BY:
MCD

DRAWN BY:
EMB

FIGURE
NUMBER
1015BS

SMITH®

JAY R. SMITH MFG. CO.®

MEMBER OF MORRIS GROUP INTERNATIONAL
POST OFFICE BOX 3237
MONTGOMERY, ALABAMA 36109-0237 (USA)
TEL: 334-277-8520 FAX: 334-272-7396 www.jrsmith.com

ASPE

MEMBER OF:

PDI

MEMBER OF:

LOCATION

ADJUSTABLE ROOF DRAIN

1

Position flat gasket on drain body, align holes in gasket with holes in body.

2

Determine if reversible collar is to be installed in the low (1 1/4" to 2 1/2") or high (2 1/4" to 4") position.

2a

Place reversible collar on body and align holes in collar with holes in gasket and body.

3

Secure collar to body with 1 1/2" long bolts insuring neoprene backed washers are installed with neoprene side facing down at proper locations between top of reversible collar and bolt head. Tighten bolts in the following manner: 1, 3, 2 & 4. Tighten No. 1 bolt approximately one full turn then No. 3, No. 2 and No. 4 (same manner as for No. 1). Continue this method until reversible ring has been secured evenly all around.

4

Wipe "O" Ring recess in reversible collar clean. Insert dry "O" Ring in groove of threads at inside diameter of reversible collar. Lubricate the face of "O" Ring in it's entirety with (1) tube of the lubricant provided and remaining threads.

5

Lubricate liberally the entire lead-in chamfer and first 2 threads of the adjustable sleeve. Thread adjustable sleeve into reversible collar and adjust to required height. Do not exceed minimum/maximum adjustment dimensions.

6

Check bolts a final time to insure tightness.

7

Run waterproofing membrane over top of adjustable sleeve to inside of bolt circle.

8

Install flashing clamp and secure with (4) bolts, use pipe sealant on threads. Again tighten bolts in the 1, 3, 2, 4 manner. (See 3)

9

Install Dome

NOTE: Stated flow rates for roof drains are based on particular test piping configurations and dimensions. Installed roof drain flow rates cannot be stated unless the attached piping configuration is known and the effects of the drainage calculated.

NOTE: Dimensions shown in parentheses are in millimeters.

H G F E	8-18-15 1-13-15 12-15-06 6-14-95	Added Flow Rate Note Rev. Callouts, Text Rev. Dwg. Added Millimeters	TBW TBW RN EMB	BW CL CL BS	WEIGHT POUNDS	VOLUME CUBIC FEET	FIGURE NUMBER 1015BS
REV.	DATE	DESCRIPTION	BY	CKD. BY			

WARNING: Cancer and Reproductive Harm - www.P65Warnings.ca.gov

1380508

RH-1





Build With Purpose

P-RH4 Sanitary Roof Hydrant



full view



Find us in

MasterSpec®

a product of The American Institute of Architects



NSF 372

PRIER's P-RH4 Sanitary Roof Hydrant stores water in a lead free canister instead of requiring a drain line like traditional roof hydrants. While opening, the stored water re-enters the flow path and is first to exit the hydrant. The P-RH4 uses a piston system, rather than venturi, so it does not require winterization.

The Roof Mounting Bracket enables the hydrant to be attached to any structural member with rigid support, instead of to the skin of the building.

The bracket requires a single penetration point in the roof of 3 5/8", and can be attached to concrete, wood or steel, I-beams, trusses, perkins or other structures with standard construction materials (Unistrut, threaded rods, or other).

- Freezeless
- Lead Free
- No Drain Line Needed
- No Winterization Needed

FEATURES

- Single hole penetration in roof of just 3 5/8"
- No winterization required, drain water contained within hydrant
- Standard 3/4" Garden Hose Outlet
- 1" NPT inlet
- ASSE 1057 approved, pressure rated to 125 PSI, temperature to 140°
- Component maintenance can be completed at roof level
- Hydrant head marked for High "H" or Low "L" desired flow rates
- Heavy-duty ductile cast iron head and lockable lever handle
- Easy to open and close O-ring seals utilizing pressure balance design
- High flow rate of 10 gallons per minute at 25 PSI differential
- Bucket catch on spout
- Included roof mounting bracket attaches to any roof structural member (truss, I-beam, concrete roof, or other)
- Lead-Free NSF 372 listed & labeled

ORDERING

P-RH4

Type

H4- Sanitary Roof Hydrant

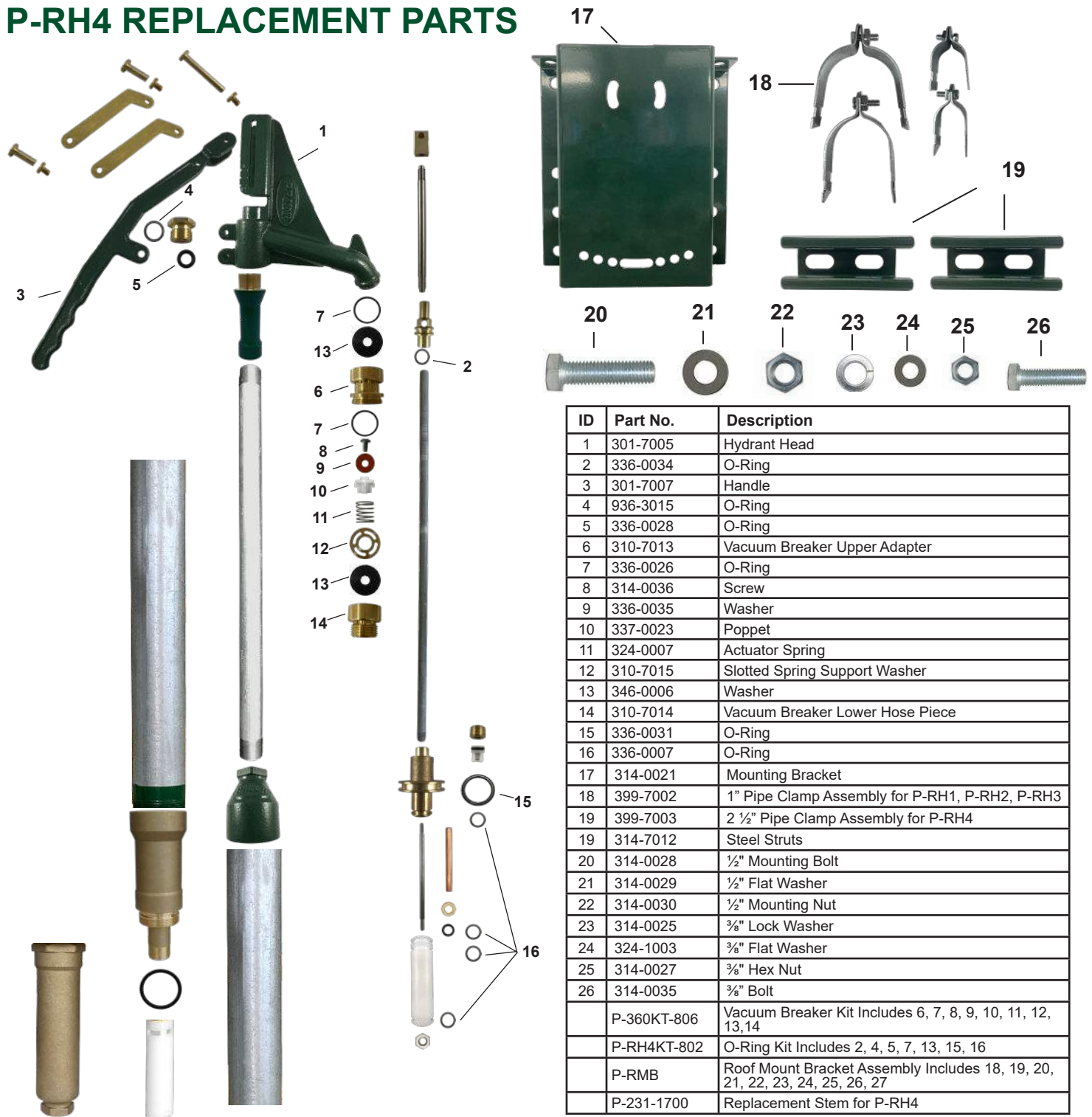


2693001



Build With Purpose

P-RH4 REPLACEMENT PARTS



ID	Part No.	Description
1	301-7005	Hydrant Head
2	336-0034	O-Ring
3	301-7007	Handle
4	936-3015	O-Ring
5	336-0028	O-Ring
6	310-7013	Vacuum Breaker Upper Adapter
7	336-0026	O-Ring
8	314-0036	Screw
9	336-0035	Washer
10	337-0023	Poppet
11	324-0007	Actuator Spring
12	310-7015	Slotted Spring Support Washer
13	346-0006	Washer
14	310-7014	Vacuum Breaker Lower Hose Piece
15	336-0031	O-Ring
16	336-0007	O-Ring
17	314-0021	Mounting Bracket
18	399-7002	1" Pipe Clamp Assembly for P-RH1, P-RH2, P-RH3
19	399-7003	2 1/2" Pipe Clamp Assembly for P-RH4
19	314-7012	Steel Struts
20	314-0028	1/2" Mounting Bolt
21	314-0029	1/2" Flat Washer
22	314-0030	1/2" Mounting Nut
23	314-0025	3/8" Lock Washer
24	324-1003	3/8" Flat Washer
25	314-0027	3/8" Hex Nut
26	314-0035	3/8" Bolt
	P-360KT-806	Vacuum Breaker Kit Includes 6, 7, 8, 9, 10, 11, 12, 13, 14
	P-RH4KT-802	O-Ring Kit Includes 2, 4, 5, 7, 13, 15, 16
	P-RMB	Roof Mount Bracket Assembly Includes 18, 19, 20, 21, 22, 23, 24, 25, 26, 27
	P-231-1700	Replacement Stem for P-RH4

SPECS

Furnish and install Model P-RH4 Sanitary Roof Hydrant manufactured by PRIER Products, Inc. It must be listed to ASSE 1057 and NSF 372 standards. This hydrant shall have freeze protection and a vacuum breaker/backflow preventer. Hydrant head shall be durable cast iron with freezess water flow and self-contained draining functions. Hydrant must be multi-piston design and completely evacuate drain canister upon use. The hydrant shall have a 1" NPT inlet and does not require winterization or a drain line. The roof penetration is to be sealed by the roofing contractor in accordance with the roofing codes and condition in their municipality. Hy shall be installed through a 3 5/8" single penetration in the roof and mounted to the interior structure with the Roof Hydrant Mounting Bracket (P-RMB) provided bracket must be manufactured of heavy gauge powder-coated steel and must allow for adjustment to meet roof angles up to 2/12 pitch. The mounting bracket be attached to concrete, a roofing truss, steel I-beam or interior wall with standard construction materials (threaded rods, Unistrut or other).

4515 E. 139th Street Grandview, MO 64030 | (800) 362-9055 | PRIER.com

P-RH



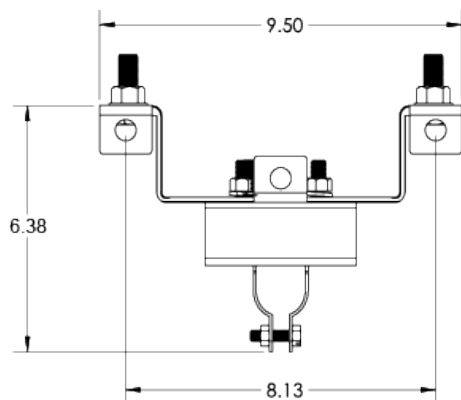
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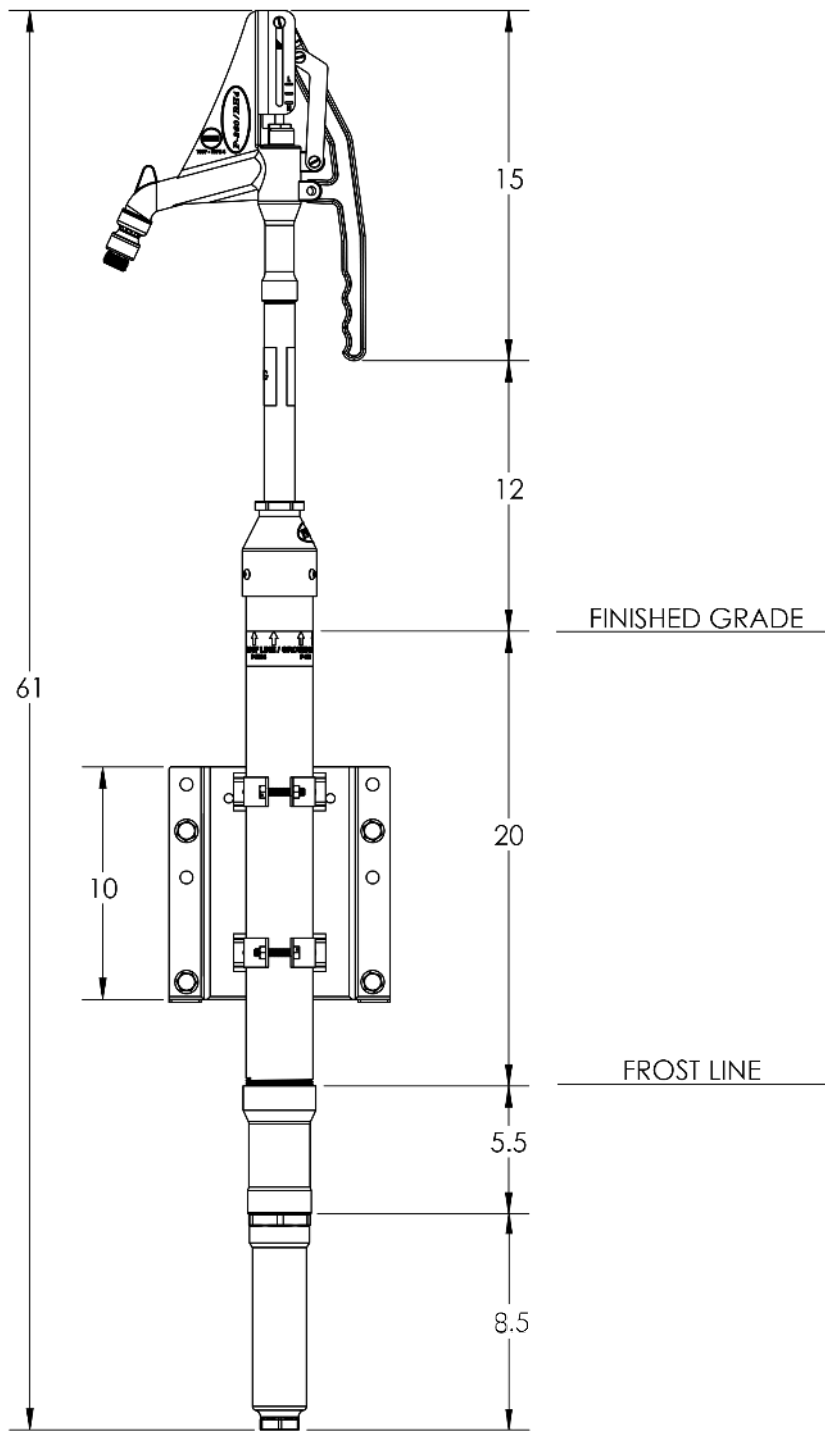
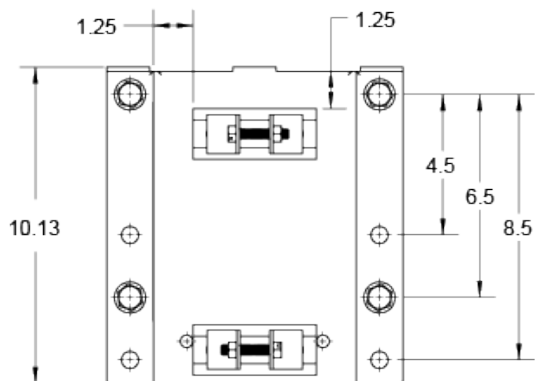
Build With Purpose

DIMENSIONS

Bottom Bracket View



Front Bracket View



2693001



SK-1





HAND SINKS

WALL MOUNT HAND SINKS

MODEL: _____ PROJECT: _____ ITEM #: _____ QTY: _____

PRODUCT IMAGES



STANDARD FEATURES

- **Fabrication**
20 gauge stainless steel. All seams tig welded and polished
- **Bowl**
Deep drawn with stamped rim to prevent spillage
- **Wall Mounting Bracket**
Offset design for added strength
- **Faucet**
4" on center wall mount faucet included on most models
- **Drain**
Stainless Steel
- **Drain with Overflow**
Stainless Steel with plastic overflow tube and inlet
- **Plumbing**
1/2" IPS hot and cold water. 1 1/2" IPS drain outlet. Install at 36" working height. 1/2" faucet supply 12" from floor. 1 1/2" drain line 23 1/4" from floor. (3 1/2" IPS drain on Model HS-20)
- **Low Lead Compliance**
Low Lead Compliant faucet options are available to meet California AB-1953 and Vermont S152 standards

OPTIONAL ACCESSORIES

- **H-100** Chrome Plated 1 1/2" IPS P-Trap
- **H-101** Deck Mount Soap Dispenser
- **H-102** Upgrade: Low Lead Wrist Handle Faucet
- **H-103** Wrist Handle Kit
- **H-104** Wall Mount Soap Dispenser
- **H-105** Wall Mount Towel Dispenser
- **H-106** One Side Splash (Specify Side)
- **H-107** Two Side Splashes
- **H-108** Stainless Steel Skirt
- **H-109** Upgrade: Low Lead Royal Series Faucet
- **H-110** Side Support Brackets
- **H-111** Soap & Towel Dispenser
- **H-200** Upgrade: Low Lead Commercial Series Faucet



CERTIFICATIONS:

Printed in the USA

APPROVED BY:

Due to our commitment to continued product improvement, specifications are subject to change with

Krowne Metal Corporation

100 Haul Rd. Wayne, NJ 07470 • Toll Free: (800) 631-0442 • Fax: (973) 872-1129
sales@krowne.com • www.krowne.com • www.facebook.com/KrowneMetal • www.twitter.com/KrowneMetal

KaTom Restaurant Supply, Inc.

18222786





HAND SINKS

WALL MOUNT HAND SINKS

MODEL: _____ PROJECT: _____ ITEM #: _____ QTY: _____

WALL MOUNT HAND SINKS

HS-9 	HS-2 	HS-4
HS-5 	HS-6 	HS-7
HS-8 	HS-10 	HS-20

18222786



CERTIFICATIONS:

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Krowne Metal Corporation

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KaTom Restaurant Supply, Inc.

Rev. 1/
Nc




Model: **10-400L** Item #: _____ Date: _____
 Project: _____ Qty: _____ Approved By: _____

4" Center Wall Mount Faucet w/ 3-1/2" Wide Gooseneck

Silver Series



DESCRIPTION

- 4" Center Wall Mount Faucet with 3-1/2" Wide Gooseneck, Chrome plated finish, and 0.5 GPM aerator installed, 0.5 GPM

STANDARD FEATURES

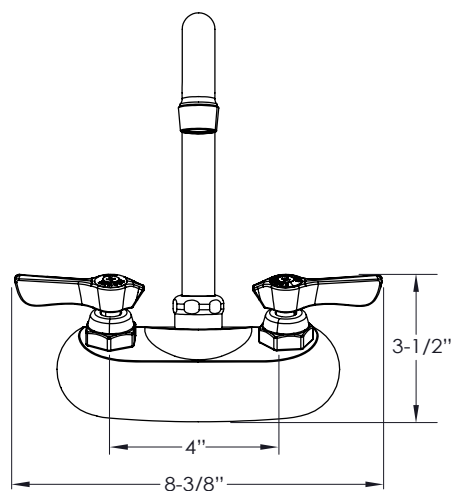
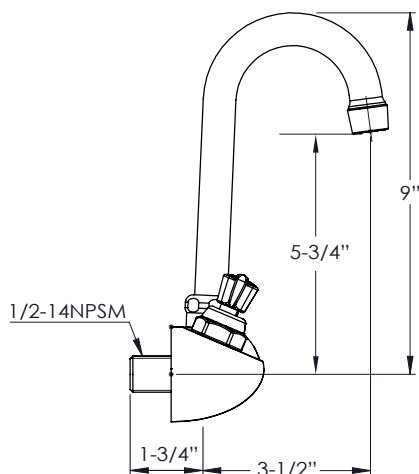
- 1/4 Turn ceramic valves
- 1/2" NPS male inlets on 4" centers
- Durable full range swing spout with double o-ring construction
- Chrome plated finish

SPECIFICATIONS

Water Inlet	1/2" NPT
Temp. Range	40°F - 180°F
Rough In	7/8" Round Holes on 4" Center
Flow Rate	0.5 GPM
Material	Chrome Plated Brass
Weight	2 lbs.

PRODUCT COMPLIANCE

- NSF-61 & NSF/ANSI 372
- ASME A112.18.1-2012/CSA BA125.1-12
- Massachusetts Approved
- CEC Approved



WARNING: Plumbing connections should be made by qualified personnel who will observe all applicable plumbing, sanitary and safety codes.

California Proposition 65

Krowne Metal Corporation • 100 Haul Rd. Wayne, NJ 07470 • P: (800) 631-0442 • customerservice@krowne.com • krowne.com

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Rev. 10/2020 • No. 10-400L



1225542

SK-2



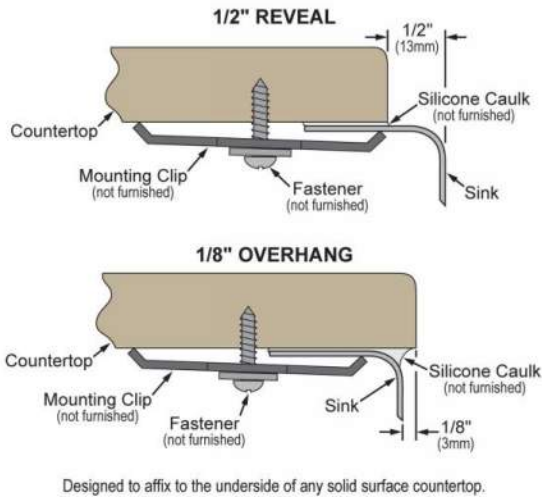
PRODUCT SPECIFICATIONS

Lustertone® Classic Stainless Steel, 30-1/2" x 18-1/2" x 5-3/8" Single Bowl Undermount ADA Sink. Sink is manufactured from 18 gauge 304 Stainless Steel with a Lustrous Satin finish, Rear Center drain placement, and Bottom only pads.

Installation Type:	Undermount
Material:	304 Stainless Steel
Finish:	Lustrous Satin
Gauge:	18
Sound Deadening:	Bottom only pads
Number of Bowls:	1
Sink Dimensions:	30-1/2" x 18-1/2" x 5-3/8"
Bowl 1 Dimensions:	28" x 16" x 5-3/8"
Drain Size:	3-1/2" (89mm)
Drain Location:	Rear Center
Minimum Cabinet Size:	36"
Mounting Hardware:	Undermount brackets sold separately
Template Included:	Yes
Cutout Template #:	1000001414

Template is available for download at [elkay.com](#). CAD software will be required to open the template.

Installation Profile:



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Product Compliance: ADA & ICC A117.1
ASME A112.19.3/CSA B45.4



Sinks are listed by IAPMO® as meeting the applicable requirements of the Uniform Plumbing Code®, International Plumbing Code®, and National Plumbing Code of Canada.

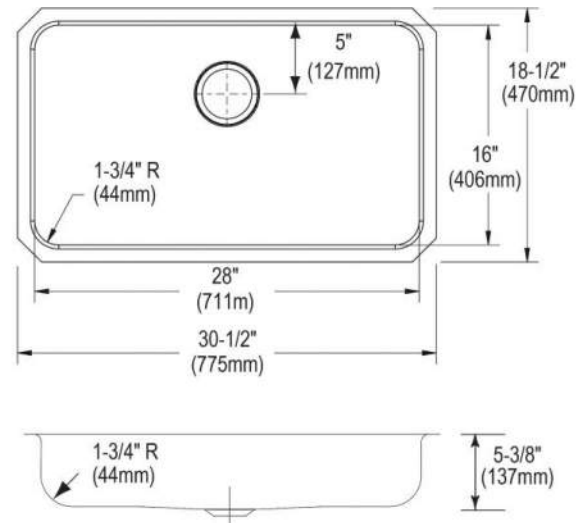


Complies with ADA & ICC A117.1 accessibility requirements when installed according to the requirements outlined in these standards.

[Clean and Care Manual \(PDF\)](#)

[Installation Instructions \(PDF\) - 1000005239](#)

[Warranty \(PDF\)](#)



PART: _____ QTY: _____
PROJECT: _____
CONTACT: _____
DATE: _____
NOTES: _____
APPROVAL: _____

In keeping with our policy of continuing product improvement, Elkay reserves the right to change product specifications without notice. Please visit [elkay.com](#) for the most current version of Elkay product specification sheets. This specification describes an Elkay product with design, quality, and functional benefits. When making a comparison of other producers' offerings, be certain these features are not overlooked.



970812

PRODUCT SPECIFICATIONS

Elkay Avado Single Hole Bar Faucet with Pull-down Spray and Lever Handle. Faucet has a flow rate of 1.8 GPM, and is made of Brass material, with a Ceramic Disc valve. Faucet requires 1 faucet holes.

Mounting Type:	Deck Mount
Special Features:	Solid Brass Construction
Spray Type:	Pull Down
Finish:	Black Stainless (BK), Chrome (CR), Lustrous Steel (LS), Matte Black (MB)
Handle Type:	Lever Handle
Deck Clearance:	7-1/2"
Spout Reach:	7"
Spout Height:	14-13/16"
Hole Drillings:	1
Material:	Brass
Valve Type:	Ceramic Disc
Valve Connection:	3/8" Female Compression Hose Assembly
Flow Rate:	1.8 GPM
Faucet Hole Size (min):	1-1/2"
Countertop Thickness:	3
Spout Swing Rotation:	360°
Spout Type:	Pull-down Spray
Spray Functions:	Aerated, Nylon Hand Spray Hose, Spray



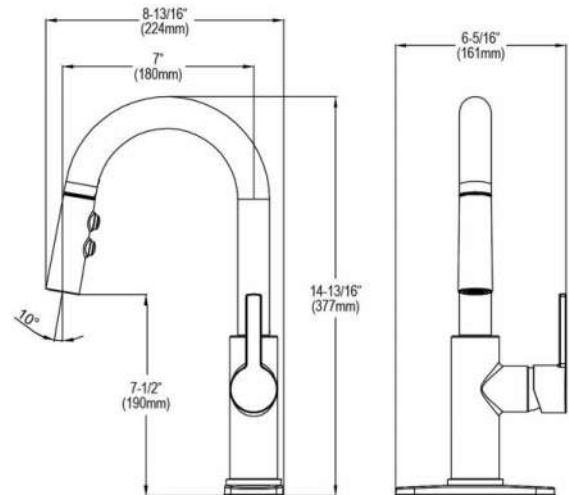
Included with Product: **Optional Deck Plate/Escutcheon included**

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Product Compliance: ADA & ICC A117.1
ASME A112.18.1/CSA B125.1
CEC
NSF 372 (lead free)
NSF 61



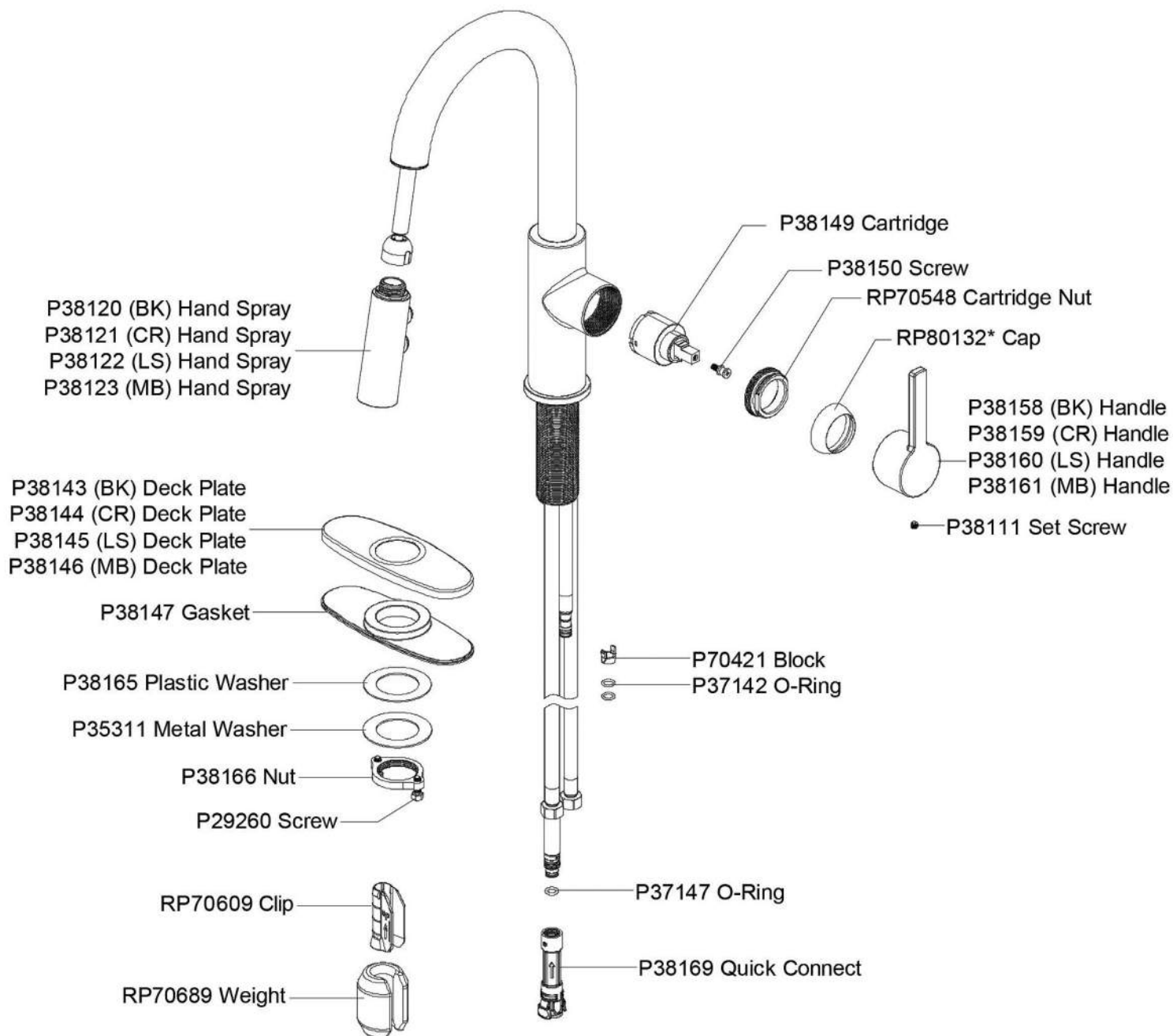
[Clean and Care Manual \(PDF\)](#)
[Installation Instructions \(PDF\) - LKAV3032 - Install Sheet](#)
[Warranty \(PDF\)](#)



PART: _____ QTY: _____
PROJECT: _____
CONTACT: _____
DATE: _____
NOTES: _____
APPROVAL: _____

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2715468

Food Waste Disposers



Submittal Sheet

This popular model offers the following features and benefits:

- 1/2 Horsepower Heavy Duty Motor (Quiet Dura-Drive® Induction Motor)
- We Come To You® 2-Year In-Home Limited Warranty
- Rugged Galvanized Steel Construction (For Disposer Durability)
- Space-Saving Compact Design
- Available with or without a factory installed power cord.

Sample Specification

Food Waste Disposer(s) shall be InSinkErator Badger® 5, continuous feed, with 1/2 H.P. motor, galvanized steel grinding elements with two stainless steel 360° swivel lugs. Self-service wrench.

Available with or without a factory installed power cord.

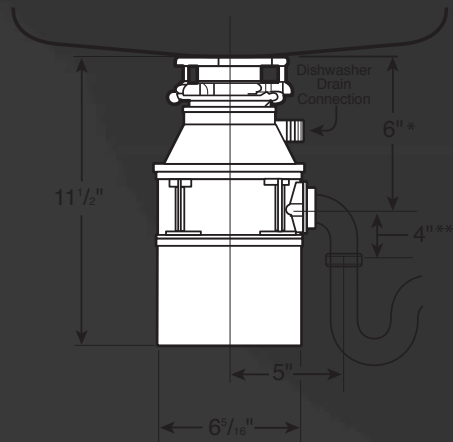
Warranty:
2-Year parts and in-home service.

* The complete InSinkErator warranty is included in the Care & Use Booklet packed with each unit.

Specifications

Type of Feed	Continuous
On/Off Control	Wall Switch
Motor	Single Phase
HP	1/2
Volts	120
HZ	60
RPM	1725
Amp. (Avg. Load)	6.3
Time Rating	Intermittent
Lubrication	Permanently Lubricated Upper & Lower Bearings
Shipping Weight (Approx.)	13 lbs. 10 oz. (without cord) 13 lbs. 14 oz. (with cord)
Unit Finish	Condor Grey
Overall Height	11-1/2"
Grind Chamber Capacity	26 oz.
Motor Protection	Manual Reset Overload
Average Water Usage	Approx. 1 Gallon per Person Per Day
Average Electrical Usage	3-4 KWh per Year
Drain Connection	1-1/2" Cushioned Slip Joint
Dishwasher Drain Connection	Yes
Factory Installed Power Cord	Available with or without

Dimensions



* Distance from bottom of sink to center line of disposer outlet. Add 1/2" when stainless steel sinks are used.
** Length of tailpipe from center line of disposer outlet to end of tailpipe.
NOTE: Plumb waste line to prevent standing water in disposer motor housing.

Job Specifications



Food waste disposers can provide an environmentally responsible alternative to transporting food waste to landfills. And they can help reduce greenhouse gas emissions. At capable wastewater treatment plants, food waste can be recycled to produce renewable energy. Additionally, capable wastewater treatment plants can process food waste into fertilizer. (Check the plant in your area.)



1-800-558-5700
www.insinkerator.com

The Emerson logo is a trademark and service mark of Emerson Electric Co. The mounting collar configuration is a trademark of Emerson Electric Co.

InSinkErator may make improvements and/or changes in the specifications at any time, in its sole discretion, without notice or obligation and further reserves the right to change or discontinue models.



SK-3



PRODUCT SPECIFICATIONS

Elkay Sturdibilt Stainless Steel 102" x 27-1/2" x 14" Floor Mount, Triple Compartment Scullery Sink with Drainboard. Sink is manufactured from 14 gauge 304 Stainless Steel with a Buffed Satin finish, Center drain placement. Sinks supported by (4) 16 gauge stainless steel, 1-5/8" O.D. tubular legs with bullet shaped feet adjustable up to 1".

Material:	304 Stainless Steel
Special Features:	Drainboard
Finish:	Buffed Satin
Gauge:	14
Number of Bowls:	3
Sink Dimensions:	102" x 27-1/2" x 44"
Bowl 1 Dimensions:	18" x 24" x 14"
Bowl 2 Dimensions:	18" x 24" x 14"



ELKAY
SPECIFICATIONS

Dependabilt™ Stainless Steel 72-1/2" x 25-13/16" x 43-3/4" 16 Gauge Three Compartment Sink w/ 18" Left Drainboard and Stainless Steel Legs
Model(s) 14-3C16X20-L-18X

PRODUCT SPECIFICATIONS

Dependabilt™ Stainless Steel 72-1/2" x 25-13/16" x 43-3/4" 16 Gauge Three Compartment Sink w/ 18" Left Drainboard and Stainless Steel Legs. Sink is manufactured from 16 gauge 300 series Stainless Steel with a #4 Finish, Center drain placement.

Material:	300 series Stainless Steel
Finish:	#4 Finish
Gauge:	16
Weight Without Packaging:	174
Number of Bowls:	3
Sink Dimensions:	72-1/2" x 25-13/16" x 43-3/4"
Bowl 1 Dimensions:	16" x 20" x 14"
Bowl 2 Dimensions:	16" x 20" x 14"
Bowl 3 Dimensions:	16" x 20" x 14"
Drainboard Location:	Left
Drainboard Size:	18
Backsplash Height:	9
Leg Type:	Stainless Steel
Drain Size:	3-1/2" (89mm)
Drain Location:	Center

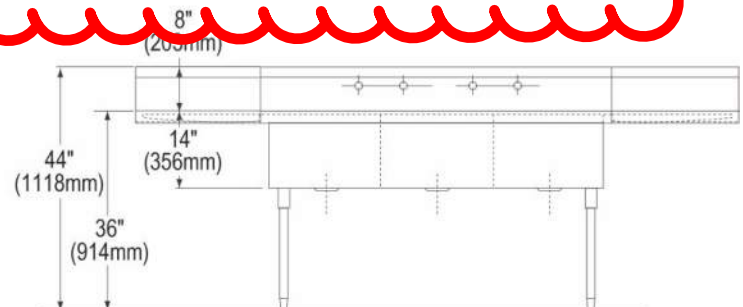
Special Note: Requires single splash mount faucet (not included).



Included with Product: Stainless steel legs and adjustable bullet feet.

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Product Compliance: ASME A112.19.3/CSA B45.4
NSF 2



**PROVIDE THIS
MODEL BASED ON
CCD 01 ISSUED ON
05/28/2026**

PART: _____ QTY: _____
PROJECT: _____
CONTACT: _____
DATE: _____
NOTES: _____
APPROVAL: _____

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Model: **17-109WL** Item #: _____ Date: _____
Project: _____ Qty: _____ Approved By: _____

8" Center Wall Mount Pre-Rinse with Add-On Faucet (12" Spout) Royal Series Plumbing



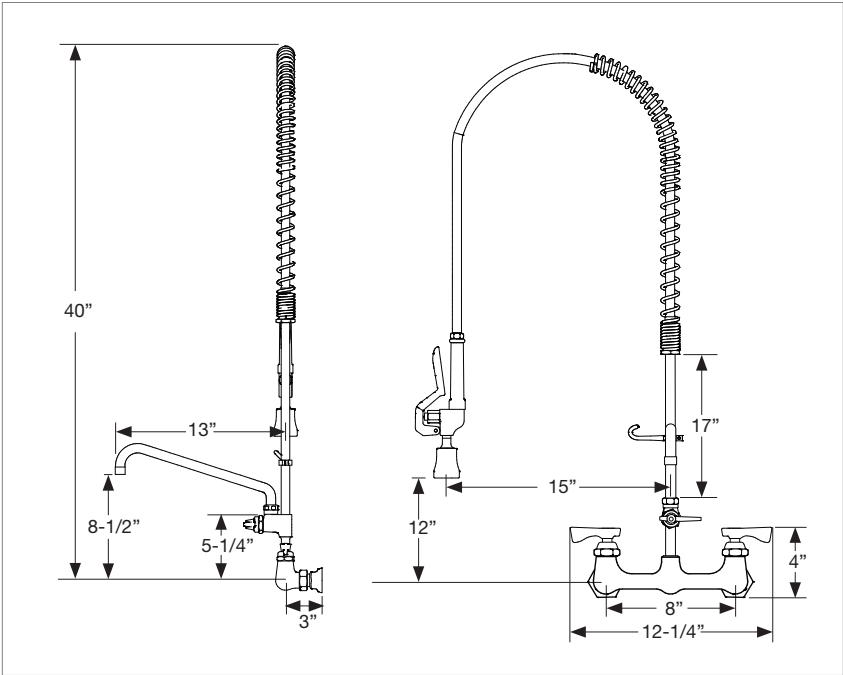
Standard Features

- Pre-assembled to cut installation time
- Includes add-on faucet with 12" spout
- Includes wall bracket
- 1/4 turn high performance ceramic cartridge valves with integral check
- 44" stainless steel hose with grip
- 38"H with 15" overhang
- 3/8" NPT x 18"L riser pipe with hook
- Mounting kit included
- Full replacement parts available

Specifications

Water Inlet	1/2" NPT Female
Temp. Range	40° - 180°F
Centers	8" Center
Mounting Style	Wall Mount
Rough In	Two 7/8" Round Holes
Add-On Faucet	12" Spout, 1.8 GPM
Spray Head Flow Rate	1.15 GPM @ 60 PSI
Material	Solid Brass
Finish	Polished Chrome
Weight	15 lbs.

Model	Description
17-109WL	Royal Series 8" Center Wall Mount Pre-Rinse with Add-On Faucet, 12" Spout



Product Compliance

- NSF-61 & NSF/ANSI 372
- ASME A112.18.1-2011/CSA B125.1-11 & NSF/ANSI 372
- City of Los Angeles
- CEC Listed
- Commonwealth of Massachusetts



WARNING:
Plumbing connections should be made by qualified personnel who will observe all applicable plumbing, sanitary and safety codes.

California Prop 65 Warning:
This product can expose you to chemicals including lead and lead compounds, which are known to the State of California to cause cancer and birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.



1130255



Model: **17-109WL** Item #: _____ Date: _____
Project: _____ Qty: _____ Approved By: _____

8" Center Wall Mount Pre-Rinse with Add-On Faucet (12" Spout) Royal Series Plumbing



Replacement Parts

No.	Model	Description
1	21-162	24" Spring
2	21-133L	44" Hose with Grip
3	21-129L	1.15 GPM Spray Head
4	21-165	Hook Assembly
5	21-161L	Spring Retainer
6	21-137	Wall Bracket
7	21-160L	18" Riser
8	21-139L	Add-On Faucet with 12" Spout
9	21-108L	8" Center Pre-Rinse Faucet Body

WARNING:
Plumbing connections should be made by qualified personnel who will observe all applicable plumbing, sanitary and safety codes.

California Prop 65 Warning:
This product can expose you to chemicals including lead and lead compounds, which are known to the State of California to cause cancer and birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.



1130255

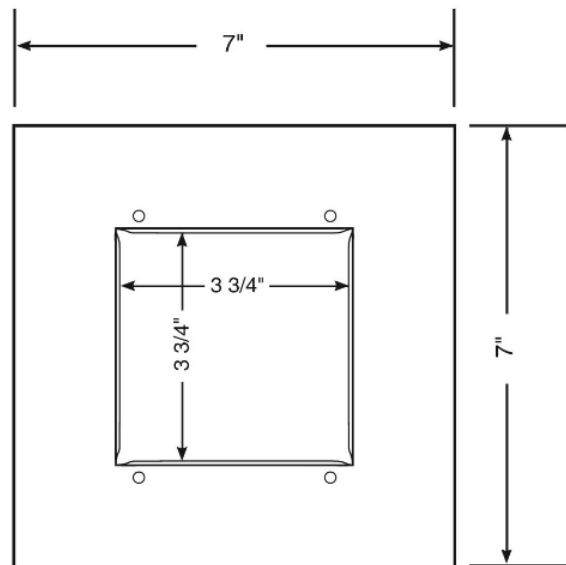
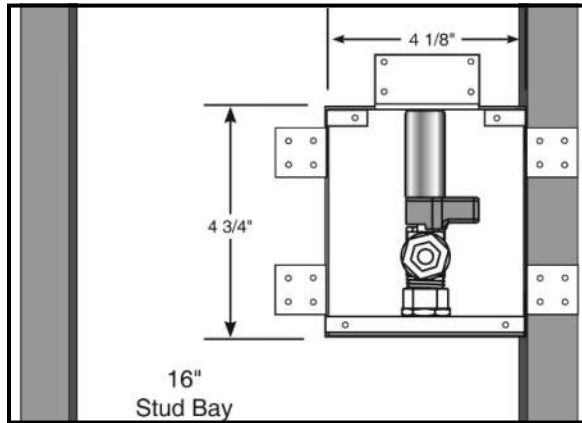
UB-1





White Powder Coated Ice Maker Outlet Box

With Lead Free Hammer Arrester Valves



1245652

SPECIFICATION

Furnish and install recessed, white powder coated ice maker outlet box. Ice maker outlet box shall have a brass plated, quarter turn hammer arrester valve meeting Lead-Free requirements. Valve shall feature either Mylar® tag and AB1953 stamping or white handle and check-mark logo for easy Lead-Free identification. Unit shall be Guy Gray™ product code checked below as manufactured by IPS Corporation.

Product Code #	Model Number	Product Description	Units/Case
88531	MIB1HAAB	Brass Qtr Turn Hammer Arrester Valve, Installed, 1/2" Sweat Conx.	5
88533	MIB2HAAB	Brass Qtr Turn Hammer Arrester Valve, Installed, 1/2" CPVC Conx.	5

IPS
CORPORATION

www.ipscorp.com watertite@ipscorp.com

500 Distribution Parkway, Collierville, TN 38017, USA ■ TEL: 901-853-5001 ■ WATS: 800-888-8312 ■ FAX: 901-853-5001



WC-1





Kingston™ Ultra

Wall-mount toilet bowl with top spud
K-84325

Features

- Elongated bowl offers added room and comfort
- Siphon jet
- 1-1/2" (38 mm) top spud
- 1.0 gpf (3.8 lpf), 1.1 gpf (4.2 lpf), 1.28 (4.8 lpf), or 1.6 gpf (6.0 lpf) depending on flushometer specified
- 11" x 8-1/4" (279 mm x 210 mm) water surface
- Wall-mounted vitreous china water closet bowls withstand a vertical force up to 1,000 lbs

Material

- Vitreous china

Technology

- Improved flush for toilet seat cover removal
- Re-engineered for strong flush performance in buildings with lower water pressure or flow
- Cast-on putty plate resists breakage
- New design eliminates gas leaks, splashing and paper shredding

Installation

- Wall-mount
- This product replaces K-4325
- This product is not designed or intended for bariatric use



ADA

CSA B651

OBC

Codes/Standards

ASME A112.19.2/CSA B45.1
DOE - Energy Policy Act 1992
EPA WaterSense®
California Energy Commission (CEC)
ADA
ICC/ANSI A117.1
CSA B651
OBC

KOHLER® One-Year Limited Warranty

See website for detailed warranty information.

Available Colors/Finishes

Color tiles intended for reference only.

Color	Code	Description
	0	White
	96	Biscuit
	7	Black Black™

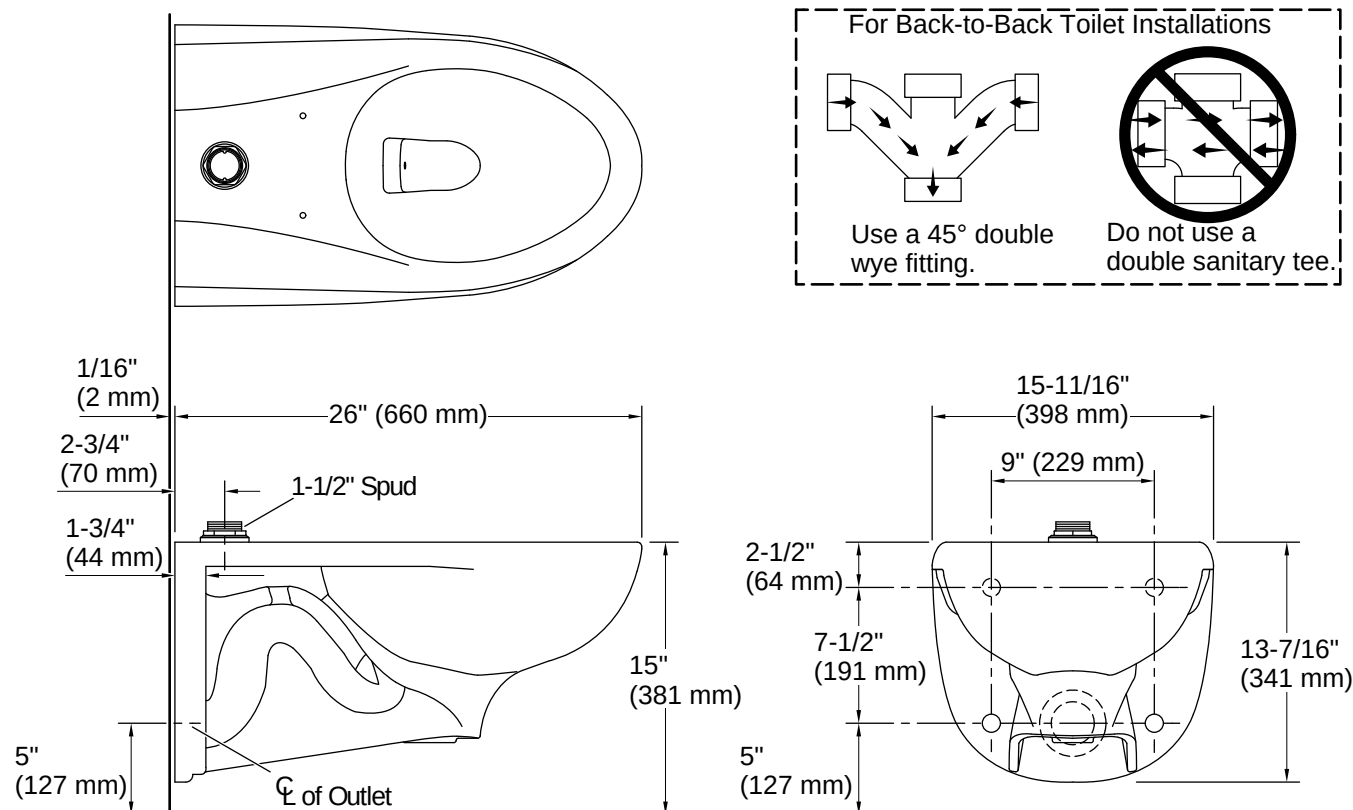
2002188



KOHLER®

Kingston™ Ultra

Wall-mount toilet bowl with top spud
K-84325



Technical Information

All product dimensions are nominal.

Toilet type:	Wall-hung
Waste Outlet:	Wall
Bowl shape:	Elongated
Flush type:	Siphon
Spud size:	1 1/2", Inlet, Top
Trap passageway:	2-1/8" (54 mm)
Water surface size:	11" x 8-1/4" (279 mm x 210 mm)
Rim to water surface:	5-13/16" (148 mm)
Seat-mounting holes:	5-1/2" (140 mm)

Fixture Supply Requirements

Min static pressure:	35 psi (241.3 kPa)
Max static pressure:	80 psi (551.6 kPa)
Min flowing pressure:	25 psi (172.4 kPa)
Min flow rate:	25 gpm (94.6 lpm)

Notes

Install this product according to the installation instructions.

For back-to-back toilet installations: Use only a 45° double wye fitting.

Refer to manufacturer's instructions and local codes for flushometer requirements.

Ensure valve is braced properly.

Ensure pressure and flow are correct.

Product dimensions are nominal and may vary per ASME A112.19.2 tolerances.

This product should be used with a WaterSense labeled counterpart with the same rated flush volume to ensure the complete system meets the requirements of this specification for water efficiency and performance.

ADA, OBC, CSA B651 compliant when installed to the specific requirements of these regulations.

Plumbing codes may require elongated toilets

1-800-4KOHLER (1-800-456-4537)

Kohler Co. reserves the right to make revisions without notice to product specifications.

For the most current Specification Sheet, go to www.kohler.com USA or www.kohler.ca Canada

2-11-2026 20:46 - US/CA

THE BOLD LOOK
OF **KOHLER®**



2002188



ZTR6200

Battery Powered Exposed Sensor Flush Valve
for Water Closets

TAG _____

Architectural/Engineering Specification:

Exposed, quiet piston-type, chrome plated flushometer valve with a polished exterior. Complete with chloramine resistant, filtered piston kit. The valve incorporates a battery powered solenoid actuator, automatic sensor with manual override push button, and robust vandal resistant metal cover with 10 degree angled sensor.

Product Features:

- Proprietary Dezincification resistant low lead brass alloy
- Control Stop has internal siphon-guard protection and vandal resistant stop cap, Sweat solder kit, wall flange with set screw.
- High back pressure Vacuum Breaker with one piece hex coupling nut
- Chloramine resistant Internal seals
- Adjustable tailpiece
- Spud coupling and flange for top spud connection

Compliance and Certification:

- ADA Compliant - Operational Controls



*This product should be used with a WaterSense labeled counterpart with a compatible flush volume to ensure that the entire system meets the requirements for water efficiency and performance.

WHEN INSTALLING IN A HANDICAP STALL:

Per the ADA Guidelines (section 604.9.4) it is recommended that the grab bars be split or shifted to the wide side of the stall. If grab bars must be present over the valve, use the alternate ADA installation as shown on the next page.



1208094

Flow Options:

	Flush Volume gpf [Lpf]	Dual Flush	WaterSense Labeled*
☐ -ONE	1.1 gpf [4.1 Lpf]		✓
☐ -EV	1.28 gpf [4.8 Lpf]		✓
☒ -WS1	1.6 gpf [6.1 Lpf]	1.6/1.1 Gpf [6.1/4.1 Lpf]	

Note: Dual Flush feature has 60 second threshold

Suffix Options:

- ☐ -DF Dual Flush Option (See Chart Above)
- ☐ -FM12 Floor Mount 12" Rough In
- ☐ -LL 10 Year Long Life Battery
- ☐ -YJ Split Ring Pipe Support
- ☐ -YK Solid Ring Pipe Support

Valve Height Options:

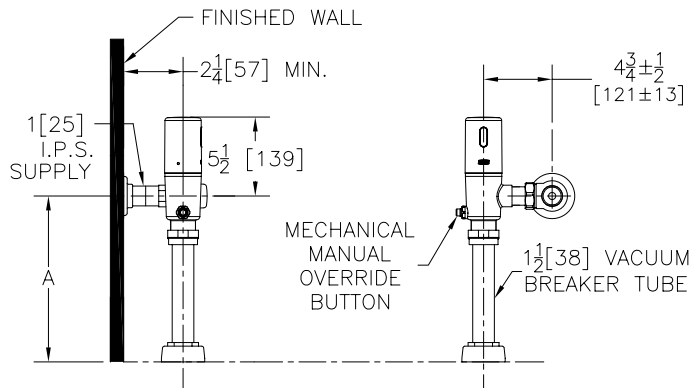
	A
☐ Standard	11 1/2" [292]
☐ -1	16" [406]

Architectural/Engineering Approval

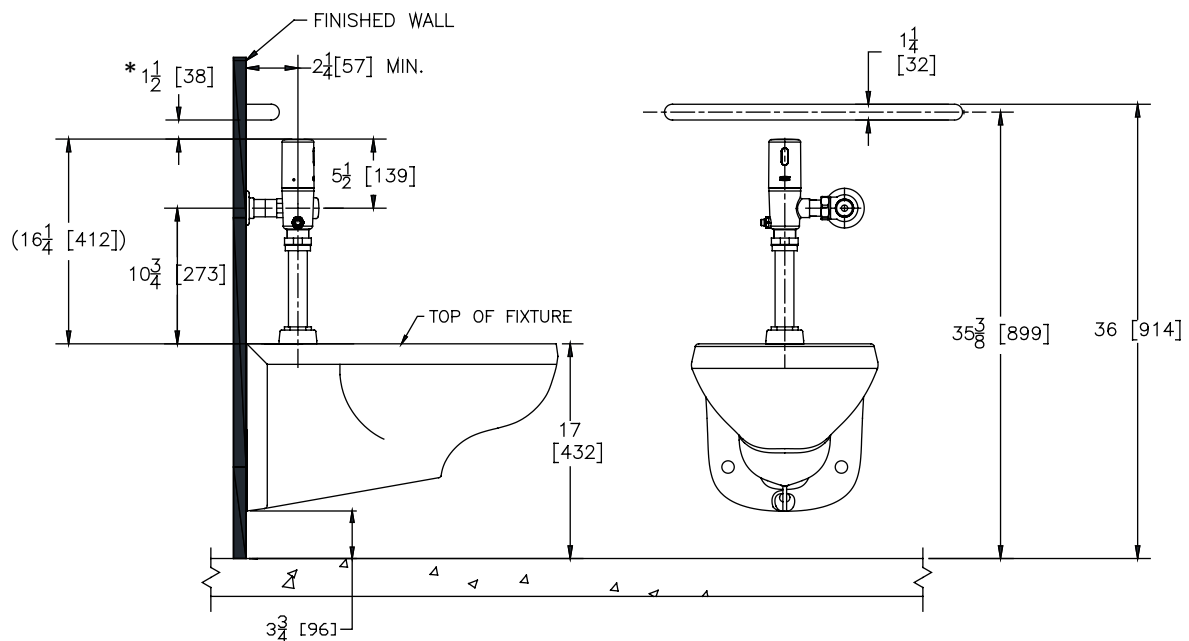
The information contained in this document is subject to change without notice.
Please contact Zurn for most up to date information.



Rough-in/Overview dimensions



Alternate ADA Installation

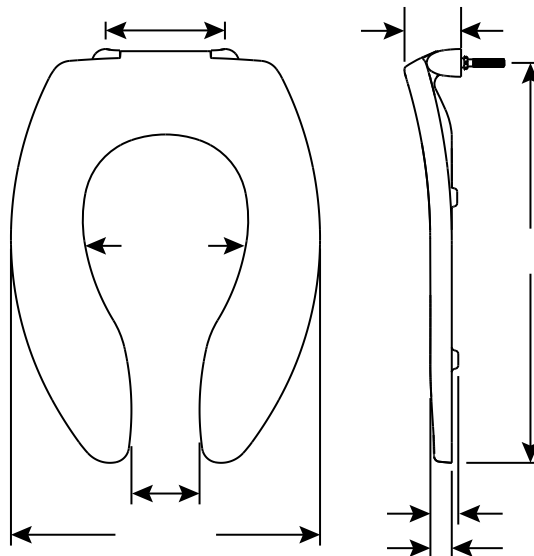


- Water supply rough-in to 10 $\frac{3}{4}$ " (minimum) with grab bar at 36".
- ADA compliance met using 1 $\frac{1}{4}$ " grab bar as shown.
- *ADA compliant minimum gap between wall and grab bar 1 $\frac{1}{2}$ ".
- Does not meet California ADA compliance for grab bar height (33" grab bar centerline required)



1655CT / 1655SSCT

COMMERCIAL EXTRA HEAVY-DUTY
PLASTIC TOILET SEAT



793988

FEATURES:

- **STA-TITE®**
Commercial Fastening System™
Eliminates callbacks for loosened seats

SPECIFICATIONS:

Size: Elongated

Material: Plastic

Style: Open Front less Cover

Ring Bumpers: Four

Hinges: Plastic Non Self-Sustaining Check Hinges (1655CT) or Self-Sustaining Check Hinges (1655SSCT) with Non-Corroding 300 Series Stainless Steel Posts and Pintles

Hardware: STA-TITE® Commercial Fastening System™

Codes & Standards:



Complies with
IAPMO/ANSI Z124.5
Performance Standards
Heavy-Duty Commercial



Proudly Made in the USA

Eco-Friendly



WC-1 CARRIER OPTIONS



1886448

G	SMITH® JAY R. SMITH MFG. CO.® MEMBER OF MORRIS GROUP INTERNATIONAL POST OFFICE BOX 3237 MONTGOMERY, ALABAMA 36109-0237 (USA) TEL: 334-277-8520 FAX: 888-377-7818 www.jrsmith.com MEMBER OF: MEMBER OF:				LOCATION																					
	ADJUSTABLE FIXTURE SUPPORTS FOR SIPHON JET WATER CLOSETS																									
DRAWING NUMBER s0211Y, 0221Y (-M54)	SIZE A																									
SCALE: NONE	DATE: 4-3-12																									
APPROVED BY: JM	CHECKED BY: AM																									
DRAWN BY: CR	FIGURE NUMBER																									
NOTES: - Base Support must be securely anchored to the floor (with one 1/2"(13) bolt per base support provided by installer.) - All single supports shall have the rear support lug anchored by contractor to assure a rigid installation. The rear support completes the 3 point anchoring system for ultimate stability of the installation. - Dimensions shown in parentheses are in millimeters. *Dimension to top of vent.																										
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LOCATION

DRAWING NUMBER
S0200BS

SIZE
A

SCALE:
NONE

DATE:
2-28-86

APPROVED BY:
JD

CHECKED BY:
CR

DRAWN BY:
VGD

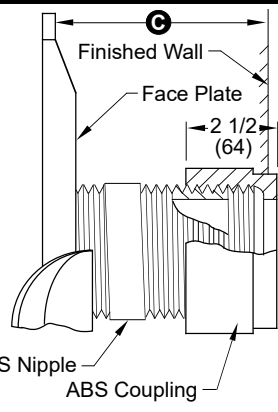
MODIFICATIONS & VARIATIONS

Ⓢ DIMENSION - ADJUS-TO-WALL TYPE OUTLET CONNECTION

The Smith Adjus-to-Wall outlet connection provides adjustment at the face of the wall at the time the fixture is set; thus, compensating for wall construction variations. The Ⓢ dimension is measured from the front of the face plate to the finished wall. To properly determine the length of nipple required, deduct 1" (25) from the Ⓢ dimension. When the Ⓢ dimension exceeds 7" (180), the -M40 Wide Pipe Chase Support is recommended. The Adjus-to-Wall coupling will provide 1/2" (13) horizontal adjustment in either direction with the coupling extended 5/16" (8) past face of the finished wall.

Ⓢ dimension = 4" (100) (regularly furnished) consisting of nipple and adjus-to-wall coupling permitting adjustment from 3" (76) min. to 4" (100) max.

*ABS (Regularly Furnished) other length PVC.



NIPPLE LENGTHS AVAILABLE

LENGTH	MIN	Ⓢ MAX
2 1/2 (64)	2 1/4 (57)	3 1/4 (83)
*3 1/4 (83)	3 (76)	4 (100)
*8 1/4 (210)	8 (205)	9 (230)
10 (255)	9 3/4 (250)	10 3/4 (275)
12 (305)	11 3/4 (300)	12 3/4 (325)
15 (380)	14 3/4 (376)	15 3/4 (400)
18 (455)	17 3/4 (450)	18 3/4 (475)

NOTE: When nipple lengths between 3 1/4" (83) and 8 1/4" (210) are specified, the 8 1/4" (210) nipple will be supplied. Nipple must be cut to specified length by others.

RECOMMENDED SETTING OF CLOSET OUTLET CONNECTION AND FIXTURE STUDS

Manufacturers have slight differences in the depth of the closet horn and flange thickness of their closets. Usually the fixture support closet connection should extend 5/16" (8) beyond face of finished wall (Dimension "R") and the fixture studs 2 1/4" (57) (Dimension "X"). For exact dimensions please use the formulas below.

DETAIL A Formula for "R" is: $S + 1/16" (2) - D = R$.

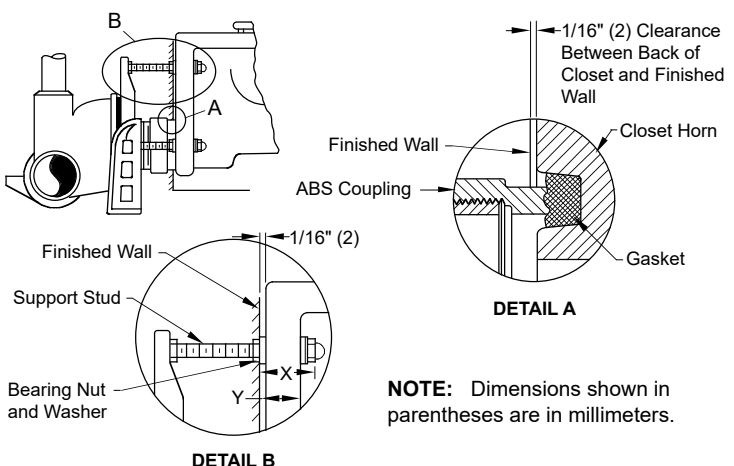
S = Depth of closet horn

1/16" (2) = Distance closet is to set away from finished wall.

D = 1/2" (13) for felt gasket and 3/8" (10) for neoprene gasket.

R = Distance coupling should extend beyond finished wall.

NOTE: When the fixture is installed, closet gasket must be compressed sufficiently to assure a gas and water-tight seal.



DETAIL B Formula for "X" is: $Y + 1/16" (2) + 9/16" (14) = X$.

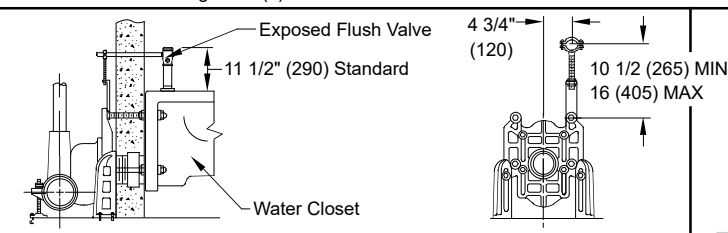
Y = Thickness of closet wall flange.

1/16" (2) = Distance closet is to set away from finished wall.

X = Distance supporting stud should extend beyond finished wall.

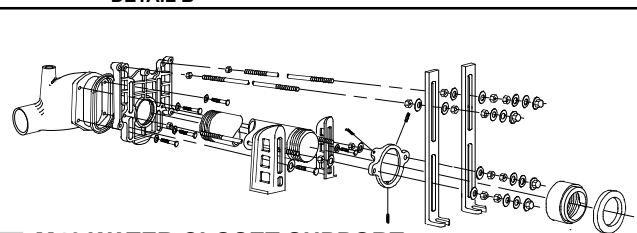
NOTE: Bearing nuts and washers must be set to take full loading from the fixture allowing 1/16" (2) clearance between fixture and wall.

NOTE: Dimensions shown in parentheses are in millimeters.

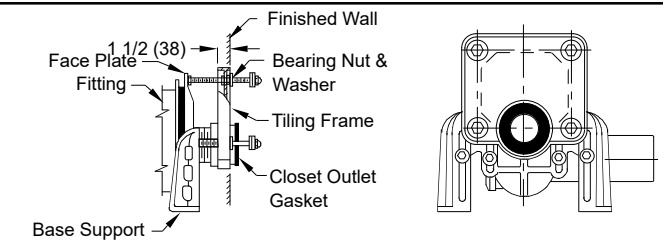


Provides support and alignment for the supply pipe to an exposed flush valve; must be specified for either 1" (25) copper tube or 1" (25) iron pipe.

☐ **SUFFIX -M12 EXPOSED FLUSH VALVE SUPPLY SUPPORT**



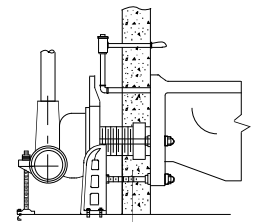
☐ **-M40 WATER CLOSET SUPPORT FOR WIDE PIPE CHASE (USE WHEN Ⓢ DIMENSION EXCEEDS 7" (180))**



☐ **SUFFIX -M13 TILING FRAME - POLYSTYRENE**

CONCEALED FLUSH VALVES

A concealed flush valve when used with a blowout water closet support will not clear the face plate. Therefore 4 1/2" (115) to 6 1/2" (165) is required from back of wall to face plate to permit the installation of the concealed flush valve piping.



0200/0300 Series Back Sheet

FIGURE NUMBER	C B A	12-10-03 6-1-95 7-1-93	Revised Notes Added Millimeters Submittal Update	RN EMB EMB	CL BS BS	WEIGHT POUNDS	VOLUME CUBIC FEET	FIGURE NUMBER 0200/0300 Ser Back Sheet
REV.	DATE	DESCRIPTION	BY	CKD. BY				



G						LOCATION 																	
	ADJUSTABLE FIXTURE SUPPORTS FOR SIPHON JET WATER CLOSETS																						
	<table border="1" style="width: 20%; margin-left: auto;"> <tr> <th>A</th> <th colspan="2">D</th> </tr> <tr> <th>SIZE</th> <th>MIN</th> <th>MAX</th> </tr> <tr> <td>4(100)</td> <td>3/4 (19)</td> <td>3 3/8(87)</td> </tr> </table>							A	D		SIZE	MIN	MAX	4(100)	3/4 (19)	3 3/8(87)							
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LOCATION

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CR

DRAWN BY:
VGD

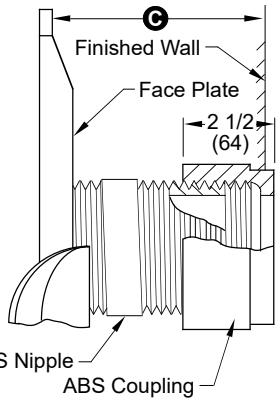
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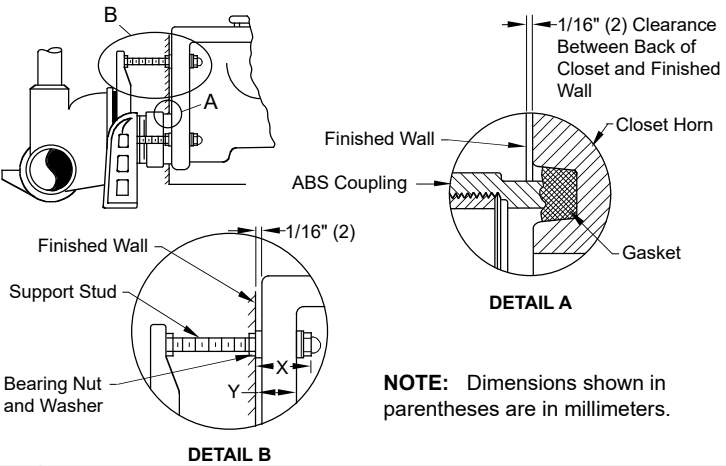
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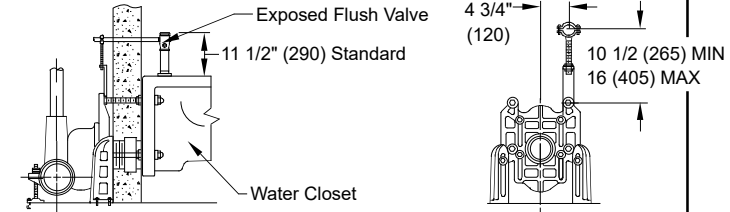
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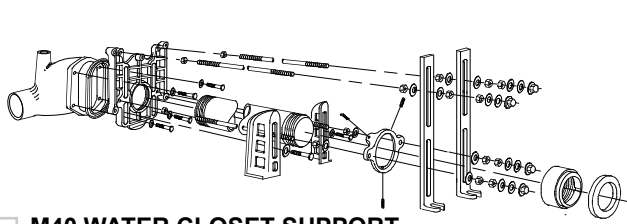
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Y = Thickness of closet wall flange.
1/16" (2) = Distance closet is to set away from finished wall.
X = Distance supporting stud should extend beyond finished wall.
NOTE: Bearing nuts and washers must be set to take full loading from the fixture allowing 1/16" (2) clearance between fixture and wall.

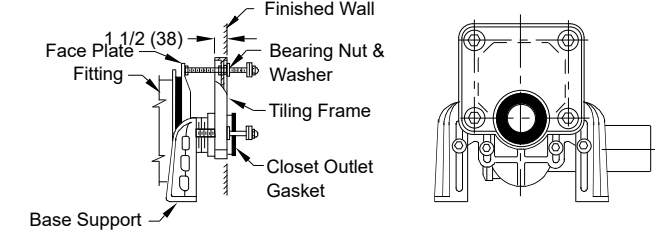


Provides support and alignment for the supply pipe to an exposed flush valve; must be specified for either 1" (25) copper tube or 1" (25) iron pipe.

☐ SUFFIX -M12 EXPOSED FLUSH VALVE SUPPLY SUPPORT



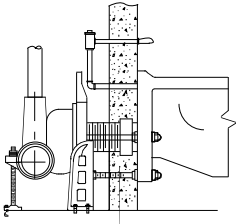
☐ -M40 WATER CLOSET SUPPORT FOR WIDE PIPE CHASE (USE WHEN Ⓒ DIMENSION EXCEEDS 7" (180))



☐ SUFFIX -M13 TILING FRAME - POLYSTYRENE

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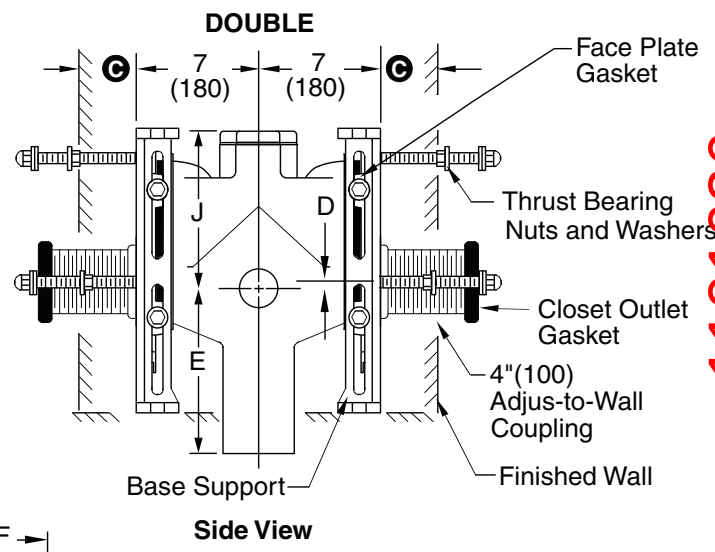
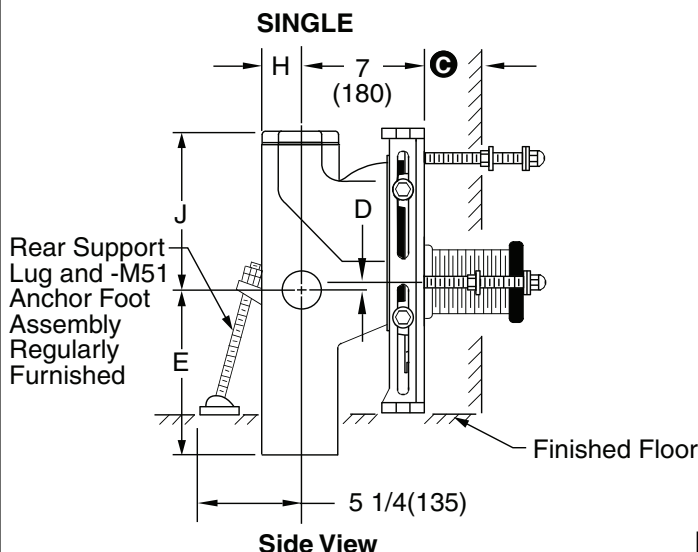


0200/0300 Series Back Sheet

FIGURE NUMBER	C B A	12-10-03 6-1-95 7-1-93	Revised Notes Added Millimeters Submittal Update	RN EMB EMB	CL BS BS	WEIGHT POUNDS	VOLUME CUBIC FEET	FIGURE NUMBER 0200/0300 Ser Back Sheet
REV.	DATE	DESCRIPTION	BY	CKD. BY				



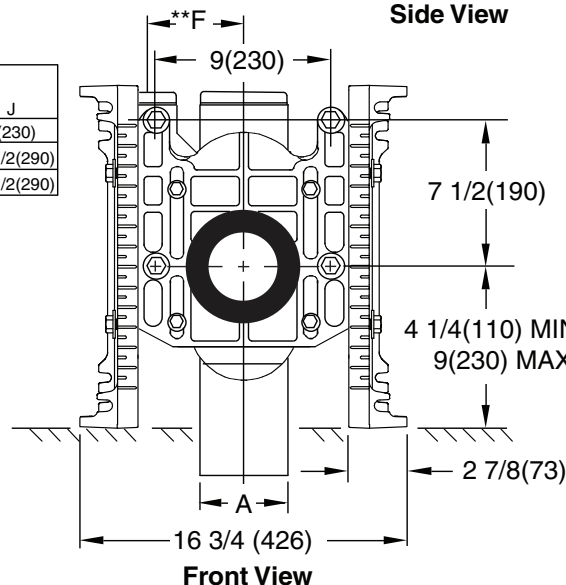
ADJUSTABLE FIXTURE SUPPORTS FOR SIPHON JET WATER CLOSETS



A SIZE	D		E	**F	H	J
	MIN	MAX				
4(100)	1 1/2(38)	4 1/4(110)	14(355)	5 1/4(135)	2 1/4(57)	9(230)
5(125)	1 1/2(38)	4 1/4(110)	11 1/2(290)	6 1/4(160)	2 3/4(70)	11 1/2(290)
6(150)	1 1/2(38)	4 1/4(110)	11 1/2(290)	6 1/4(160)	3 1/4(83)	11 1/2(290)

NOTE: Base support must be firmly anchored to floor with one 1/2"(13) bolt per base support provided by contractor.

NOTE: All single supports shall have the rear support lug anchored by the contractor to assure a rigid installation. The rear support completes the 3-point anchoring system for ultimate stability of the installation.



NOTE: When used with concealed flush valve locate face plate behind supply pipe.

NOTE: Shallower minimum rough-in dimension available to accommodate lower rough bowls. Specify dimension required.

NOTE: 1 1/2(38) or 2(51) NPT inlet available. Add suffix -N

*NOTE: 9"(230) core drilled hole is required for rough-in 4 1/4(110) thru 6 1/2(165)

**NOTE: "F" dimension is from centerline of vent to centerline of stack.

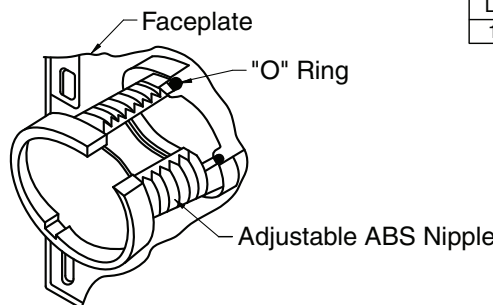
Fig. 0230Y (-M54-M58) . . . NO-HUB

Fig. 0230DY (-M54-M58) . . . NO-HUB DOUBLE

Conforms with the requirements of ASME A112.6.1M up to a 500 lbs (227kg) static load rating.

NOTE: Dimensions shown in parentheses are in millimeters.

NOTE: When nipple lengths between 3 1/4"(83) and 10 3/4"(275) are specified, the 10"(255) nipple will be supplied. The threaded end of nipple must be cut to specified length by others. Absolute minimum dimension is 1 1/2" (38).



LENGTH	MIN.	MAX.
10(255)	9 3/4"(250)	10 3/4"(275)

*ABS (Regularly Furnished).

F	6-16-26	Revised NOTE	TB	CL	WEIGHT	VOLUME	FIGURE NUMBER
E	12-7-22	Added Note	KK	CL	POUNDS	CUBIC FEET	0230Y (-M54
D	8-19-20	Rev. Draw./Dim: 16 7/16 16 3/4	MW	CL			
C	2-18-13	Rev. Dim., Table	TBW	CL			

REV.	DATE	DESCRIPTION	BY	CKD. BY
------	------	-------------	----	---------



WARNING: Cancer and Reproductive Harm - www.P65Warnings.ca.gov



1101626

0230Y (-M54-M58)

FIGURE NUMBER

DRAWING NUMBER
S0230Y (-M54-M58)

SIZE
A

SCALE:
NONE

DATE:
12/09/04

APPROVED BY:
AM

CHECKED BY:
AM

DRAWN BY:
JJ

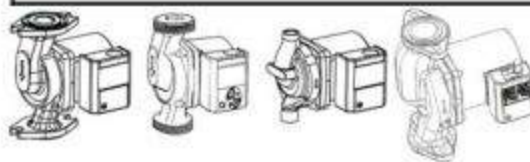
WE CAN ASSUME NO RESPONSIBILITY FOR USE OF SUPERSEDED OR VOID DATA

DIMENSIONS ARE SUBJECT TO MANUFACTURERS TOLERANCE AND CHANGE WITHOUT NOTICE

PP-1



JOB:		REPRESENTATIVE:	
UNIT TAG:		ORDER NO.	DATE:
ENGINEER:		SUBMITTED BY:	DATE:
CONTRACTOR:		APPROVED BY:	DATE:



Lead-Free[†] Bronze and Stainless Steel Bodied System Lubricated Circulators



DESCRIPTION

A series of in-line wet rotor circulation pumps designed specifically for quiet operation in open (potable) water systems. These pumps have lead free[†] bronze or stainless steel bodies.

Automatic Timer and Aquastat accessories are available. (See Submittal A-128A for details).

CONSTRUCTION MATERIALS

Pump Body:
NBF: Lead-Free[†] Bronze
SSF: Stainless Steel
Bearings: Carbon

Impeller: Noryl
Shaft: Ceramic

OPERATING DATA

Maximum Working Pressure: 150 psi (10.3 Bar)
Minimum Operating Temperature: 40°F (5°C)
Maximum Operating Temperature:
NBF-25, NBF-33, NBF-36, NBF-45: 225°F (107°C)
All Others: 230°F (110°C)

[†]Contains less than 0.25% Lead content by weight on wetted surfaces.

MODEL NUMBER	PART NUMBER	CONNECTION	APPROX. SHPG. WT. Lbs. (Kg)	STANDARD 60 CYCLE 115 VOLT SINGLE PHASE			TAGGING INFORMATION	QUANTITY
				WATTS	F.L. AMPS	RPM		
NBF-8S/LW	103257LF	1/2" Sweat	9.0 (4.1)	39	0.38	2800		
NBF-9U/LW	103258LF	Union (See Following Page)	9.3 (4.2)	41	0.40	2800		
SSF-9U/LW	103360LF	Union (See Following Page)	9.3 (4.2)	41	0.40	2800		
NBF-10S/LW	103259LF	1/2" Sweat	9.0 (4.1)	52	0.46	2800		
NBF-12U/LW	103261LF	Union (See Following Page)	9.3 (4.2)	55	0.48	2800		
SSF-12U/LW	103361LF	Union (See Following Page)	9.3 (4.2)	55	0.48	2800		
NBF-12F/LW	103260LF	Flange 3/4, 1, 1-1/4, 1-1/2	9.5 (4.3)	55	0.48	2800		
SSF-12F/LW	103358LF	Flange 3/4, 1, 1-1/4, 1-1/2	9.5 (4.3)	55	0.48	2800		
NBF-18S	103316LF	1/2" Sweat	9.0 (4.1)	90	0.74	3000		
NBF-22U	103255LF	Union (See Following Page)	9.3 (4.2)	92	0.80	2940		
SSF-22U	103362LF	Union (See Following Page)	9.3 (4.2)	92	0.80	2940		
NBF-22	103252LF	Flange 3/4, 1, 1-1/4, 1-1/2	9.5 (4.3)	92	0.80	2940		
SSF-22	103357LF	Flange 3/4, 1, 1-1/4, 1-1/2	9.5 (4.3)	92	0.80	2940		
NBF-25 [†]	103418LF	Flange 3/4, 1, 1-1/4, 1-1/2	10.4 (4.7)	125	1.10	2950		
NBF-33	103351LF	Flange 3/4, 1, 1-1/4, 1-1/2	10.4 (4.7)	125	1.10	2950		
NBF-36 [†]	103401LF	Flange 3/4, 1, 1-1/4, 1-1/2	13.1 (6.0)	270	2.30	3300		
NBF-45 [†]	103405LF	Flange 1, 1-1/4, 1-1/2	14.5 (6.6)	270	2.30	3300		

[†]3-speed circulators

TYPICAL SPECIFICATIONS

The contractor shall furnish and install in-line circulating pumps as illustrated on the plans and in accordance with the following specifications:

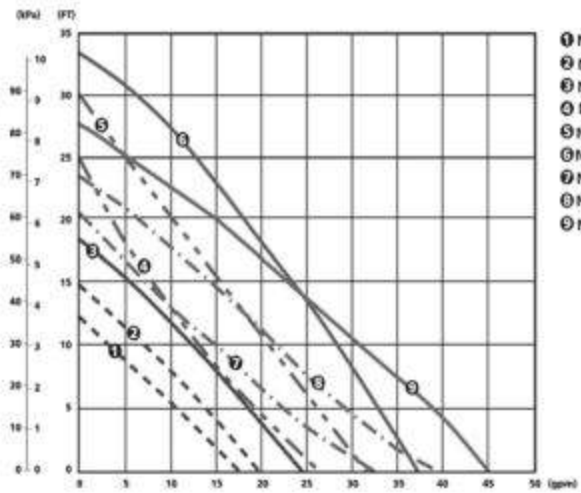
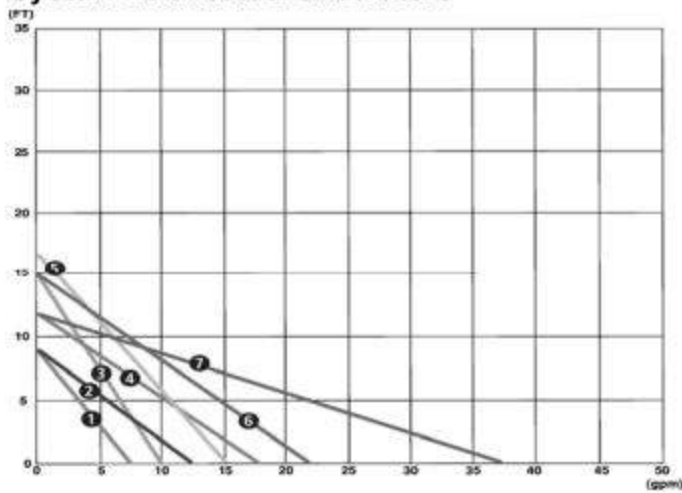
- The pumps shall be of the horizontal system lubricated type specifically designed and guaranteed for quiet operation.
- Pump to be suitable for _____°F (____°C) [choose one: 225°F (107°C) for NBF-25, NBF-33, NBF-36, NBF-45 or 230°F (110°C) for all other circulators] operation at 150 psig (10.3 Bar) working pressure.
- The pumps shall have a ceramic shaft supported by carbon bearings. Bearings are to be lubricated by the circulating fluid.

- Pump body shall be lead-free bronze for NBF circulators or stainless steel for SSF circulators.
- Motor stator to be isolated from circulating fluid through use of stainless steel can. Rotor to be sheathed in stainless steel.
- Motors shall be non-overloading at any point on the pump curve. NBF-36 & NBF-45 to have built-in thermal protection. All other motors to have built-in impedance protection.
- NBF-25 has an optional check valve.

Pumps to have a capacity of _____ GPM at _____ foot head when powered by 115 volt, 60 cycle single phase electrical supply.

All pumps are to be Xylem - Bell & Gossett Model _____.

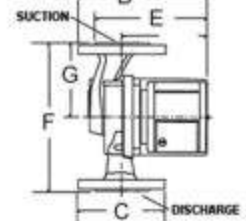
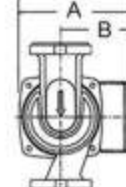
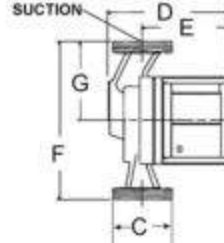
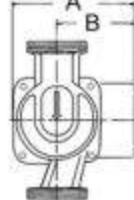
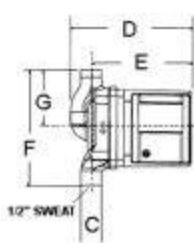
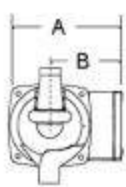
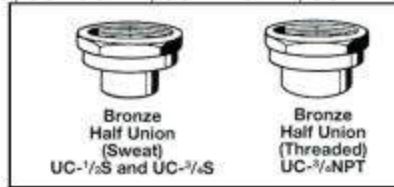
System Lubricated Circulators



HALF UNION CONNECTIONS

For NBF-9U/LW, NBF-12U/LW, NBF-22U, SSF-9U/LW, SSF-12U/LW, SSF-22U

MODEL NUMBER	PART NUMBER	DESCRIPTION (SETS OF 2)
UC-1/2S	113203LF	1/2" Bronze Union Sweat
UC-3/4S	113201LF	3/4" Bronze Union Sweat
UC-3/4NPT	113202LF	3/4" Bronze Union NPT Female



MODEL	PART NUMBER	A in (mm)	B in (mm)	C in (mm)	D in (mm)	E in (mm)	F in (mm)	G in (mm)
NBF-8S/LW	103257LF	4-7/8 (124)	3-3/16 (81)	1/2 (13)	5-7/32 (132)	4-9/32 (109)	5 (127)	2-1/2 (63)
NBF-9U/LW	103258LF	4-7/8 (124)	3-3/16 (81)	1-1/4 (32)	5-1/16 (129)	3-11/16 (93)	6-1/8 (156)	3-1/16 (78)
SSF-9U/LW	103360LF	4-7/8 (124)	3-3/16 (81)	1-1/4 (32)	5-1/16 (129)	3-11/16 (93)	6-1/8 (156)	3-1/16 (78)
NBF-10S/LW	103259LF	4-7/8 (124)	3-3/16 (81)	1/2 (13)	5-7/32 (132)	4-9/32 (109)	5 (127)	2-1/2 (63)
NBF-12U/LW	103261LF	4-7/8 (124)	3-3/16 (81)	1-1/4 (32)	5-1/16 (129)	3-11/16 (93)	6-1/8 (156)	3-1/16 (81)
SSF-12U/LW	103361LF	4-7/8 (124)	3-3/16 (81)	1-1/4 (32)	5-1/16 (129)	3-11/16 (93)	6-1/8 (156)	3-1/16 (78)
NBF-12F/LW	103260LF	4-7/8 (124)	3-3/16 (81)	3-3/16 (81)	5-9/16 (141)	3-11/16 (93)	6-3/8 (162)	3-3/16 (82)
SSF-12F/LW	103358LF	4-7/8 (124)	3-3/16 (81)	3-3/16 (81)	5-9/16 (141)	3-11/16 (93)	6-3/8 (162)	3-3/16 (82)
NBF-18S	103316LF	4-7/8 (124)	3-3/16 (81)	1/2 (13)	5-7/32 (132)	4-9/32 (109)	5 (127)	2-1/2 (63)
NBF-22U	103255LF	4-7/8 (124)	3-3/16 (81)	1-1/4 (32)	5-1/16 (129)	3-11/16 (93)	6-1/8 (156)	3-1/16 (78)
SSF-22U	103362LF	4-7/8 (124)	3-3/16 (81)	1-1/4 (32)	5-1/16 (129)	3-11/16 (93)	6-1/8 (156)	3-1/16 (78)
NBF-22	103252LF	4-7/8 (124)	3-3/16 (81)	3-3/16 (81)	5-9/16 (141)	3-11/16 (93)	6-3/8 (162)	3-3/16 (82)
NBF-25*	103418LF	5-1/8 (130)	3-3/16 (81)	3-3/16 (81)	6-3/16 (157)	4-7/8 (124)	6-3/8 (162)	2-1/2 (63)
NBF-33	103351LF	4-7/8 (124)	3-3/16 (81)	3-3/16 (81)	6-3/16 (157)	3-11/16 (94)	6-3/8 (162)	3-3/16 (82)
NBF-36*	103401LF	5-3/4 (146)	3-9/16 (91)	3-3/16 (81)	6-3/16 (157)	5-3/8 (137)	6-3/8 (162)	3-3/16 (82)
NBF-45*	103405LF	5-3/4 (146)	3-9/16 (91)	3-7/16 (87)	7-3/8 (187)	5-1/2 (140)	8-1/2 (216)	4-1/4 (108)

Dimensions are subject to change. Not to be used for construction purposes unless certified.

Companion Flanges Available in Sizes: 3/4", 1", 1-1/4", and 1-1/2"

* 3-speed circulators

Xylem Inc.
8200 N. Austin Avenue
Morton Grove, IL 60053
Phone: (847)966-3700
Fax: (847)965-8379
www.xyleminc.com/brands/bellgossett

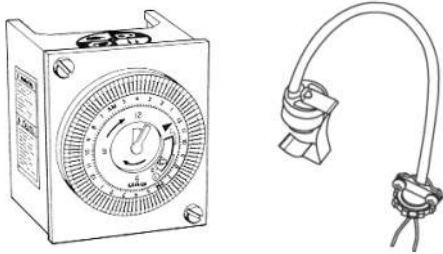
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997733

JOB: UNIT TAG: ENGINEER: CONTRACTOR:	REPRESENTATIVE: ORDER NO. SUBMITTED BY: APPROVED BY:	DATE: DATE: DATE:
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Automatic Timer Kit

Model TC-1

Aquastat

Models: AQS-1/2, AQS-3/4

532078

DESCRIPTION

TC-1 AUTOMATIC TIMER KIT - To increase the overall efficiency of a hot water recirculation system, the TC-1 timer control kit can be installed for use on any B&G NBF/NRF/SSF/PL and most 3-piece circulators. The TC-1 timer control is programmable to turn the circulator ON and OFF automatically at preset times. This permits the user to have the pump circulate hot water only during those times when high usage can be expected throughout the day.

AQS-1/2 and AQS-3/4 AQUASTAT - are designed to thermostatically turn any B&G NBF/NRF/SSF circulator ON and OFF. The AQS-1/2 or AQS-3/4 will switch the pump OFF at 120°F (48.9°C) and ON at 100°F (37.8°C). The aquastats are available in separate models that will sense the temperature for either 1/2" or 3/4" copper pipe.

AUTOMATIC TIMER KIT and AQUASTAT COMBINATION - The automatic timer and aquastat kits can be used in combination. When they are used together, the pump will only circulate water when the ON time conditions are met and when water temperature is low enough to cause the aquastat to switch ON.

OPERATIONAL LIMITS

TC-1 AUTOMATIC TIMER KIT:

Power Supply: 115/120 VAC, 60 HZ, 1Ø
 Minimum Switch Interval: 15 minutes
 Run Modes: ON (continuous run), OFF (at all times),
 TIMER (run at programmed intervals)
 Maximum Switch Current: 16 amps

AQS-1/2, AQS-3/4 AQUASTATS

Thermostatic Switch Modes: OFF (open) at 120°F (48.9°C) and ON (closed) at 100°F (37.8°C) operation of the aquastat is dependent on ambient settings. In setting where there is excessive heat loss or in installations where heat from the boiler or other components can affect the aquastat, the actual on and off temperatures may vary.

Thermostatic Element: Bimetal, senses surface temperature of outside diameter of pipe.

Pipe Size: AQS-1/2 clips onto 1/2" copper pipe or 3/8" steel pipe (OD of pipe 5/8"). AQS-3/4 clips onto 3/4" copper pipe or 1/2" steel pipe (OD of pipe 7/8").

Mounting: May be installed to sense temperature at the suction or discharge pipe of the pump.

CONSTRUCTION MATERIALS

TC-1 AUTOMATIC TIMER KIT:

Timer Body: Noryl Plastic

AQS-1/2, AQS-3/4 AQUASTATS:

Thermostat Covering: Epoxy (Environmentally Sealed)

Pipe Clip: Stainless Steel

Sensing Element: Bimetal

Wire Leads: Insulated 18" (457mm) #18 AWG

APPROX. SHIPPING WEIGHT

TC-1: 11oz. (.31Kg)

AQS-1/2: 5oz. (.14Kg)

AQS-3/4: 5oz. (.14Kg)

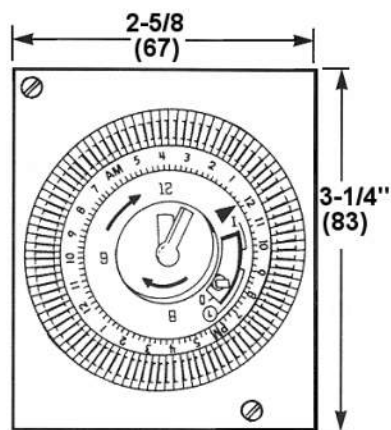
SCHEDULE

MODEL NUMBER	PART NUMBER	SIZE INCHES	VOLTAGE RATING 60 HZ 1Ø	TAGGING INFORMATION	QUANTITY
TC-1	113210	-	115-120		
AQS-1/2	113223	1/2	115-120		
AQS-3/4	113224	3/4	115-120		

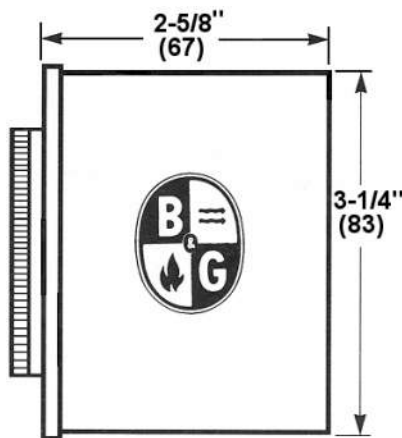
AUTOMATIC TIMER KIT AQUASTATS

TIMER KIT DIMENSION IN INCHES (MM)

Spec Seq#: 42
A-128D



FRONT VIEW



SIDE VIEW

TYPICAL SPECIFICATIONS

Furnish and install as shown on the circulator plans and in accordance with the following specifications:

AUTOMATIC TIMER KIT:

1. The timer kit shall be UL approved.
2. The timer kit shall be installed on the connection box of any Bell & Gossett NBF/NRF/SSF system lubricated circulator.
3. The timer kit will be suitable for 115/120V, 60 HZ operation.
4. The timer shall provide automatic ON-OFF control at minimum interval of every 15 minutes. It shall also have the option of providing manual ON-OFF control.

All units shall be Xylem - Bell & Gossett Model No.: TC-1

AQUASTATS:

1. The aquastat shall be UL approved.
2. The aquastat shall be connected to the lead wires in the connection box of any Bell & Gossett NBF/NRF/SSF system lubricated circulator.
3. The aquastat will be suitable for 115/120V, 60 HZ operation.
4. The aquastat shall provide thermostat control to the circulator. It will turn OFF (open) at 120°F (48.9°C) and ON (closed) at 100°F (37.8°C) the use of insulation in installation with excessive heat or cooling may be necessary. Use of fiberglass insulation with R-values greater than 1.6 is recommended.

All units shall be Xylem - Bell & Gossett Model No.:
(Choose: AQS-1/2 or AQS-3/4).

AUTOMATIC TIMER KIT AND AQUASTAT COMBINATION:

1. The automatic timer kit and either aquastat model can be combined to provide automatic time and temperature control to any Bell & Gossett NBF/NRF/SSF circulator.
2. When the automatic timer and the aquastat are used together, the pump will only circulate water when the ON time conditions are met and when the water temperature is low enough to cause the aquastat to switch ON.

All units shall be Xylem - Bell & Gossett Model No.: TC-1 and Bell & Gossett Model No.: (Choose: AQS-1/2 or AQS-3/4).

Xylem Inc.
8200 N. Austin Avenue
Morton Grove, IL 60053
Phone: (847)966-3700
Fax: (847)965-8379
www.xyleminc.com/brands/bellgossett

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532078

ISOLATION FLANGES

Description

The isolation flange is a combination of an isolation ball valve and a companion flange for circulators. The isolation flange allows easy service or replacement of the circulator without the need to drain the system. The isolation flange fits the Bell & Gossett NRF/NBF/SSF wet rotors, Series PL, Series 100, HV, PR and LR circulators.

Operating Data

Maximum Working Pressure: 150 PSI (10 bar)

Maximum Operating Temperature: 250°F (121°C)

Materials of Construction

Valve Body: Lead-Free* Brass

Flange: Chrome Plated Steel

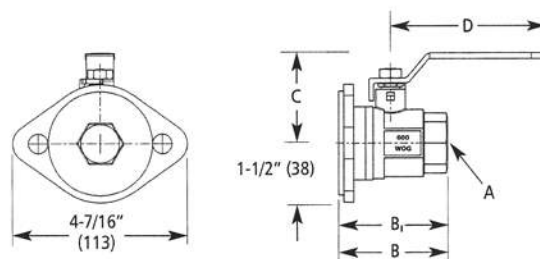
Ball: Chrome Plated Lead-Free* Brass

Packing: PTFE

Seat Ring: PTFE

Stem: Lead-Free* Brass

*CSA certified to NSF/ANSI 372 that product contains less than 0.25% lead content by weight on wetted surface.



Specifications

Model Number	Size Inches	Use with Following Circulators*	Dimensions - Inches (mm) Following Circulators				Approx. Shpg. Wt. lbs. (Kg)
			A	B	C	D	
101221LF	3/4" NPTF IF	ecocirc 20-18 and ecocirc+ 20-18 ecocirc XL 20-35, 36-45, 55-45 NRF/NBF/SSF wet rotors Series PL-30, PL-36, PL-55, PL-100 Series 100, PR and LR	3/4" NPT	2-27/64" (61.5)	2" (50.5)	4-47/64" (120)	3.2 (1.5)
101222LF	1" NPTF IF		1" NPT	2-57/64" (73.3)	2-5/32" (54.7)	4-47/64" (120)	4.1 (1.9)
101223LF	1-1/4" NPTF IF		1-1/4" NPT	3-19/64" (84)	3" (76.2)	6-7/32" (158)	5.8 (26)
101241LF	1-1/2" NPTF IF		1-1/2" NPT	3-23/64" (85.5)	3" (76.2)	6-7/32" (158)	6.1 (28)
101226LF	3/4" SWT IF		3/4" SWT	2-41/64" (67)	2" (50.5)	4-23/32" (120)	3.2 (1.5)
101227LF	1" SWT IF		1" SWT	3" (75.3)	2-5/32" (54.7)	4-23/32" (120)	3.9 (1.8)
101228LF	1-1/4" SWT IF		1-1/4" SWT	3-25/64" (86)	3" (76.2)	6-7/32" (158)	5.4 (25)
101243LF	1-1/2" SWT IF		1-1/2" SWT	3-35/64" (90)	3" (76.2)	6-7/32" (158)	6 (27)

"IF" = "Isolation Flange"

Note: Dimensions and weights are approximate and subject to change. Contact factory for certified dimensions. The part numbers and shipping weights are for two isolation flanges, capscrews and nuts.

* Not for use with NRF/NBF-45.

COMPANION FLANGES

Flanges for Cast Iron Circulators

	Size (NPT)	Master Carton of 12 Part No.	Set of 2 Part No.
Series 100, PR NRF-22, NRF-9F/LW, NRF-33, NRF-36 PL-30, PL-36, PL-55, ecocirc XL	3/4"	101001	101201
	1"	101002	101202
	1-1/4"	101003	101203
	1-1/2"	101004	101204
Series HV, PL-45 PL-50, PL-100, NRF-45 ecocirc XL	1"	101005	101205
	1-1/4"	101006	101206
	1-1/2"	101007	101207

	Size (NPT)	Set of 2 Part No.*
PL-75, PL-130/2" ecocirc XL 15-75	2"	101215
PL-130/3" ecocirc XL 40-275	2-1/2"	101219
	3"	101217

*Includes Fasteners

Union Connection for NBF Circulators

	Union Connection	Set of Two	
		Model No.	Part No.
NBF-22U, NBF-12U/LW NBF-9U/LW	1/2" sweat	UC-1/2S	113203LF
	3/4" sweat	UC-3/4S	113201LF
	3/4" NPT	UC-3/4NPT	113202LF

Flanges for Bronze Circulators

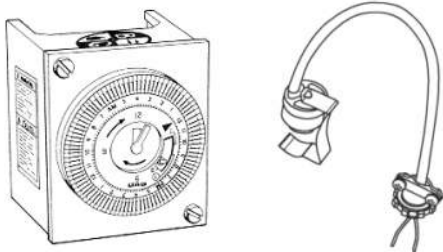
	Size (NPT)	Master Carton of 12 Part No.	Set of 2 Part No.
Series 100B, PRAB, NBF-22, NBF-12F/LW, NBF-33, NBF-36 PL-30B, PL-36B, ecocirc XLB	3/4"	101511LF	101501LF
	1"	101512LF	101502LF
	1-1/4"	101513LF	101503LF
	1-1/2"	1101514LF	101504LF
Series HV, PL-45B PL-50B, PL-100B, NBF-45 ecocirc XLB	1"	101515LF	101505LF
	1-1/4"	101516LF	101506LF
	1-1/2"	101517LF	101507LF

	Size (NPT)	Set of 2 Part No.*
PL-75B, PL-130B/2" ecocirc XLB 15-75	2"	101508LF
PL-130B/3" ecocirc XLB 40-275	2-1/2"	101510LF
	3"	101509LF

*Includes Fasteners



JOB: UNIT TAG: ENGINEER: CONTRACTOR:	REPRESENTATIVE: ORDER NO. SUBMITTED BY: APPROVED BY:	DATE: DATE: DATE:
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Automatic Timer Kit

Model TC-1

Aquastat

Models: AQS-1/2, AQS-3/4

684757

DESCRIPTION

TC-1 AUTOMATIC TIMER KIT - To increase the overall efficiency of a hot water recirculation system, the TC-1 timer control kit can be installed for use on any B&G NBF/NRF/SSF/PL and most 3-piece circulators. The TC-1 timer control is programmable to turn the circulator ON and OFF automatically at preset times. This permits the user to have the pump circulate hot water only during those times when high usage can be expected throughout the day.

AQS-1/2 and AQS-3/4 AQUASTAT - are designed to thermostatically turn any B&G NBF/NRF/SSF circulator ON and OFF. The AQS-1/2 or AQS-3/4 will switch the pump OFF at 120°F (48.9°C) and ON at 100°F (37.8°C). The aquastats are available in separate models that will sense the temperature for either 1/2" or 3/4" copper pipe.

AUTOMATIC TIMER KIT and AQUASTAT COMBINATION - The automatic timer and aquastat kits can be used in combination. When they are used together, the pump will only circulate water when the ON time conditions are met and when water temperature is low enough to cause the aquastat to switch ON.

OPERATIONAL LIMITS

TC-1 AUTOMATIC TIMER KIT:

Power Supply: 115/120 VAC, 60 HZ, 1Ø
 Minimum Switch Interval: 15 minutes
 Run Modes: ON (continuous run), OFF (at all times),
 TIMER (run at programmed intervals)
 Maximum Switch Current: 16 amps

AQS-1/2, AQS-3/4 AQUASTATS

Thermostatic Switch Modes: OFF (open) at 120°F (48.9°C) and ON (closed) at 100°F (37.8°C) operation of the aquastat is dependent on ambient settings. In setting where there is excessive heat loss or in installations where heat from the boiler or other components can affect the aquastat, the actual on and off temperatures may vary.

Thermostatic Element: Bimetal, senses surface temperature of outside diameter of pipe.

Pipe Size: AQS-1/2 clips onto 1/2" copper pipe or 3/8" steel pipe (OD of pipe 5/8"). AQS-3/4 clips onto 3/4" copper pipe or 1/2" steel pipe (OD of pipe 7/8").

Mounting: May be installed to sense temperature at the suction or discharge pipe of the pump.

CONSTRUCTION MATERIALS

TC-1 AUTOMATIC TIMER KIT:

Timer Body: Noryl Plastic

AQS-1/2, AQS-3/4 AQUASTATS:

Thermostat Covering: Epoxy (Environmentally Sealed)

Pipe Clip: Stainless Steel

Sensing Element: Bimetal

Wire Leads: Insulated 18" (457mm) #18 AWG

APPROX. SHIPPING WEIGHT

TC-1: 11oz. (.31Kg)

AQS-1/2: 5oz. (.14Kg)

AQS-3/4: 5oz. (.14Kg)

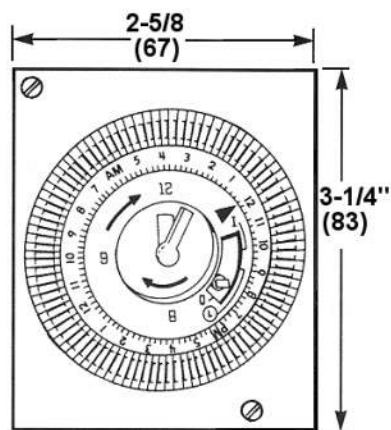
SCHEDULE

MODEL NUMBER	PART NUMBER	SIZE INCHES	VOLTAGE RATING 60 HZ 1Ø	TAGGING INFORMATION	QUANTITY
TC-1	113210	-	115-120		
AQS-1/2	113223	1/2	115-120		
AQS-3/4	113224	3/4	115-120		

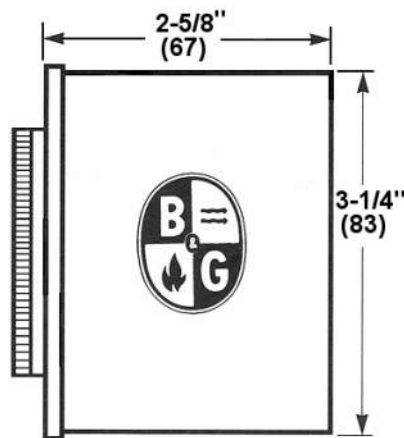
AUTOMATIC TIMER KIT AQUASTATS

TIMER KIT DIMENSION IN INCHES (MM)

Spec Seq#: 44
A-128D



FRONT VIEW



SIDE VIEW

TYPICAL SPECIFICATIONS

Furnish and install as shown on the circulator plans and in accordance with the following specifications:

AUTOMATIC TIMER KIT:

1. The timer kit shall be UL approved.
2. The timer kit shall be installed on the connection box of any Bell & Gossett NBF/NRF/SSF system lubricated circulator.
3. The timer kit will be suitable for 115/120V, 60 HZ operation.
4. The timer shall provide automatic ON-OFF control at minimum interval of every 15 minutes. It shall also have the option of providing manual ON-OFF control.

All units shall be Xylem - Bell & Gossett Model No.: TC-1

AQUASTATS:

1. The aquastat shall be UL approved.
2. The aquastat shall be connected to the lead wires in the connection box of any Bell & Gossett NBF/NRF/SSF system lubricated circulator.
3. The aquastat will be suitable for 115/120V, 60 HZ operation.
4. The aquastat shall provide thermostat control to the circulator. It will turn OFF (open) at 120°F (48.9°C) and ON (closed) at 100°F (37.8°C) the use of insulation in installation with excessive heat or cooling may be necessary. Use of fiberglass insulation with R-values greater than 1.6 is recommended.

All units shall be Xylem - Bell & Gossett Model No.:
(Choose: AQS-1/2 or AQS-3/4).

AUTOMATIC TIMER KIT AND AQUASTAT COMBINATION:

1. The automatic timer kit and either aquastat model can be combined to provide automatic time and temperature control to any Bell & Gossett NBF/NRF/SSF circulator.
2. When the automatic timer and the aquastat are used together, the pump will only circulate water when the ON time conditions are met and when the water temperature is low enough to cause the aquastat to switch ON.

All units shall be Xylem - Bell & Gossett Model No.: TC-1 and Bell & Gossett Model No.: (Choose: AQS-1/2 or AQS-3/4).

Xylem Inc.
8200 N. Austin Avenue
Morton Grove, IL 60053
Phone: (847)966-3700
Fax: (847)965-8379
www.xyleminc.com/brands/bellgossett

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xylem
Let's Solve



684757

FD-1



Location: _____

**TS****TOP SET HEELPROOF STRAINER**

Specification: MIFAB® Series TS(*)-(**) (*specify strainer diameter, ** specify strainer material) MIFAB Top Set heelproof strainer that can be adjusted vertically and side to side after the pour with a 1/2" thick composite strainer protective cover and indicating whisker. Complete with securing and adjustable hardware and allen key vandal resistant stainless steel grate screws.

Function: Used in showers, kitchens, restrooms, and other non-membrane finished areas. Round adjustable heelproof strainer is used for all types of poured finished floors. The concrete is poured to the same level as the orange cap. After the pour, the cap is removed with channel locks and the three Phillips head screws underneath the grate can be engaged to raise the strainer vertically and to also adjust it side to side

VANDAL PROOF
ALLEN KEY
STAINLESS STEEL
SCREWS

PROTECTIVE CAP
WITH WHISKER

TOP SET
HEELPROOF
STRAINER

SCREWS THAT
PROVIDE VERTICAL
SIDE TO SIDE
ADJUSTMENT
AFTER THE
CONCRETE POUR

TS5-1	5" Round Nickel Bronze Top Set Strainer
TS5-3	5" Round Stainless Steel Top Set Strainer
TS6-1	6" Round Nickel Bronze Top Set Strainer
TS6-3	6" Round Stainless Steel Top Set Strainer

STRAINER DIAMETER	FREE AREA	ANSI / ASME RATING LOAD RATING Lbs.		WT. Lbs.	STRAINER ADJUSTMENT	
		N.B. (-1)	S.S. (-3)		MIN	MAX
● 5" (127)	8 in ² (5200 ²)	Up to 2,000	5,000 to 7,499	1.25	1/2" (13)	2 1/2" (64)
○ 6" (152)	9 in ² (5800 ²)	Up to 2,000	5,000 to 7,499	1.75	11/16" (17)	2 1/2" (64)

SUFFIX	OPTIONAL VARIATIONS
☒ -1	Nickel Bronze
☐ -3	Stainless Steel (Type 304)
☐ -6	Security Screws (TR25 Torx and pin - H-1039A)
☐ -49	Polished Chrome

CALIFORNIA PROPOSITION 65 WARNING: This product contains chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

Job Name: _____

Page No: _____

Section No: _____

Contractor: _____

Schedule No: _____

Purchase Order No: _____



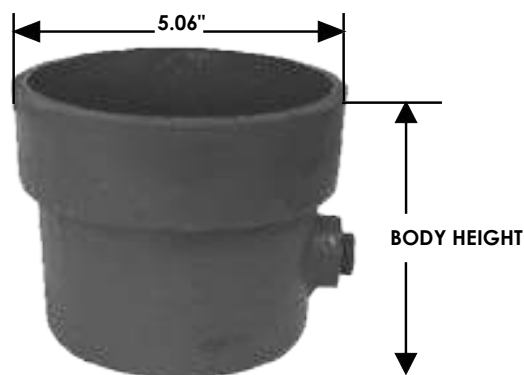
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Location: _____

**A4 BODY****A4 BODY**

Specification: MIFAB's A4 body Series, 5" diameter body with ½" trap seal primer tapping and plug with secondary tapping to receive an optional secondary composite plug. Available in either cast iron, PVC and / or ABS material in various outlet connection sizes (see chart below).

Function: Universal body that is used with all MIFAB F1000, F1100, F1300, F1320, F1420, C1000, C1100, Series of floor and area drains and cleanouts that do not require a water-proofing membrane. The most common application for the A4 bodies is with the C1000 and C1100 cleanouts that require a secondary plug.



2726031

Note: No hub outlet connections are the standard outlet connection provided by MIFAB. No suffix is required to indicate the no hub outlet connection.

PIPE SIZE	NO HUB (STANDARD)	PUSH ON (P)	PVC (-30) ABS (-31)
2"	4"	4 3/8"	3 5/8"
3"	4"	4 3/8"	4 5/16"
4"	4"	4 3/8"	4 11/16"

SUFFIX	OPTIONAL VARIATIONS
-11	Acid Resistant Coated Epoxy Body
-13	Galvanized Body
-30	PVC Connection Body (2", 3" or 4")
-31	ABS Connection Body (2", 3" or 4")
-34	Close plug (C1440 Series)
-90	Threaded side outlet connection (2", 3" or 4"). Not available in ABS / PVC

SUFFIX	OPTIONAL VARIATIONS
-90NH	No hub side outlet connection (2", 3" or 4"). Not available in ABS / PVC
P	Push On Connection Body (2", 3" or 4")
T	Threaded Connection Body (2", 3" or 4")
X	Inside Caulk Connection Body (2", 3" or 4")

CALIFORNIA PROPOSITION 65 WARNING. This product contains chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

Job Name: _____

Page No: _____

Section No: _____

Contractor: _____

Schedule No: _____

Purchase Order No: _____



FD-2



Location: _____

**TSS****TOP SET SQUARE
HEELPROOF STRAINER**

Specification: MIFAB® Series TSS(*)-(**) (*specify strainer size, ** specify strainer material) MIFAB Top Set heelproof strainer that can be adjusted vertically and side to side after the pour with a ½" thick composite strainer protective cover and indicating whisker. Complete with securing and adjustable hardware and allen key vandal resistant stainless steel grate screws.

Function: Used in showers, kitchens, restrooms, and other non-membrane finished areas. Square adjustable heelproof strainer is used with floors that are tiled or have straight line patterns. The concrete is poured to the same level as the orange cap. After the pour, the cap is removed with channel locks and the three Phillips head screws underneath the grate can be engaged to raise the strainer vertically and to also adjust it side to side

U.S.
PATENT
#9,453,331

VANDAL PROOF ALLEN KEY
STAINLESS STEEL SCREWS

PROTECTIVE CAP
WITH WHISKER

TOP SET HEELPROOF
STRAINER

SCREWS THAT
PROVIDE VERTICAL
SIDE TO SIDE
ADJUSTMENT AFTER
THE CONCRETE POUR

TSS5-1	5" x 5" Square Nickel Bronze Top Set Strainer
TSS5-3	5" x 5" Square Stainless Steel Top Set Strainer
TSS6-1	6" x 6" Square Nickel Bronze Top Set Strainer
TSS6-3	6" x 6" Square Stainless Steel Top Set Strainer

STRAINER SIZE	FREE AREA	ANSI / ASME RATING LOAD RATING Lbs.		WT. Lbs.	STRAINER ADJUSTMENT	
		N.B. (-1)	S.S. (-3)		MIN	MAX
● 5" x 5" (127)	9 in ² (5200 ²)	Up to 2,000	5,000 to 7,499	2.00	1/2" (13)	2 9/16" (65)
○ 6" x 6" (152)	21 in ² (7700 ²)	Up to 2,000	5,000 to 7,499	3.00	11/16" (17)	2 9/16" (65)

SUFFIX	OPTIONAL VARIATIONS
☒ -1	Nickel Bronze
☐ -3	Stainless Steel (Type 304)
☐ -6	Security Screws (TR25 Torx and pin - H-1039A)
☐ -49	Polished Chrome

CALIFORNIA PROPOSITION 65 WARNING: This product contains chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

Job Name: _____

Page No: _____

Section No: _____

Contractor: _____

Schedule No: _____

Purchase Order No: _____



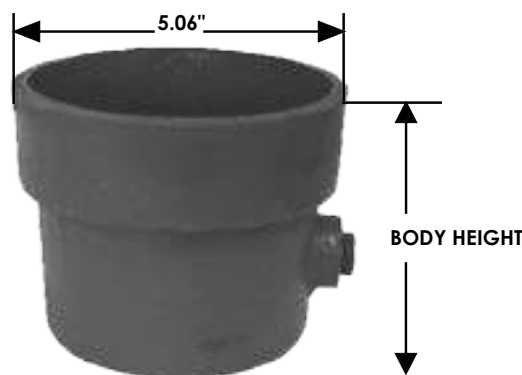
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Location: _____

**A4 BODY****A4 BODY**

Specification: MIFAB's A4 body Series, 5" diameter body with ½" trap seal primer tapping and plug with secondary tapping to receive an optional secondary composite plug. Available in either cast iron, PVC and / or ABS material in various outlet connection sizes (see chart below).

Function: Universal body that is used with all MIFAB F1000, F1100, F1300, F1320, F1420, C1000, C1100, Series of floor and area drains and cleanouts that do not require a water-proofing membrane. The most common application for the A4 bodies is with the C1000 and C1100 cleanouts that require a secondary plug.



2726031

Note: No hub outlet connections are the standard outlet connection provided by MIFAB. No suffix is required to indicate the no hub outlet connection.

PIPE SIZE	NO HUB (STANDARD)	PUSH ON (P)	PVC (-30) ABS (-31)
2"	4"	4 3/8"	3 5/8"
3"	4"	4 3/8"	4 5/16"
4"	4"	4 3/8"	4 11/16"

SUFFIX	OPTIONAL VARIATIONS
-11	Acid Resistant Coated Epoxy Body
-13	Galvanized Body
-30	PVC Connection Body (2", 3" or 4")
-31	ABS Connection Body (2", 3" or 4")
-34	Close plug (C1440 Series)
-90	Threaded side outlet connection (2", 3" or 4"). Not available in ABS / PVC

SUFFIX	OPTIONAL VARIATIONS
-90NH	No hub side outlet connection (2", 3" or 4"). Not available in ABS / PVC
P	Push On Connection Body (2", 3" or 4")
T	Threaded Connection Body (2", 3" or 4")
X	Inside Caulk Connection Body (2", 3" or 4")

CALIFORNIA PROPOSITION 65 WARNING. This product contains chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

Job Name: _____

Page No: _____

Section No: _____

Contractor: _____

Schedule No: _____

Purchase Order No: _____



FD-3



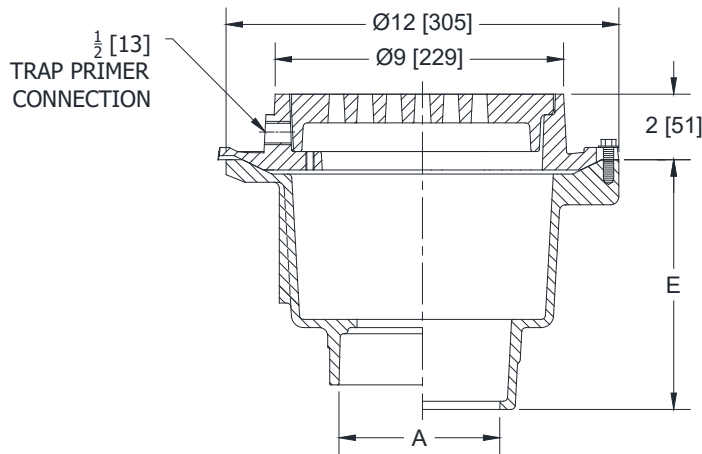
**Z511****9 [229] DIAMETER HEAVY-DUTY DRAIN W/
INTEGRAL TRAP PRIMER CONNECTION**

Spec Seq#: 52

SPECIFICATION SHEET

TAG _____

Design and Dimensional Data (inches and [mm]) are Subject to Manufacturing Tolerances and Change Without Notice



2102082

'A' Pipe Size In. [mm]	Approx. Wt. Lbs. [kg]	Grate Open Area Sq. In. [cm ²]
2, 3, 4, [51, 76, 102]	40 [18]	18 [116]

ENGINEERING SPECIFICATION: ZURN Z511

9 [229mm] Diameter top drain, Dura-Coated cast iron deep sump body with bottom outlet, seepage pan, and combination membrane flashing clamp and frame with integral trap primer and plug for heavy-duty ductile iron deep flange slotted grate.

OPTIONS (Check/specify appropriate options)**PIPE SIZE**

3, 4 [76, 102]
3, 4 [76, 102]
3, 4 [76, 102]
2, 3, 4 [51, 76, 102]

(specify size/type) **OUTLET**

___ IC Inside Caulk
___ IG Inside Gasket
___ IP Threaded
___ NL Neo-Loc

'E' BODY HT. DIM.

7-5/8 [194]
7-5/8 [194]
7-5/8 [194]
6-7/8 [175]

PREFIXES

___ Z D.C.C.I. Body and Top*
✓ ZN D.C.C.I. Body with Polished Nickel Bronze Top (Add 3/16 [5]) to 2 [51] Dim. and 3/4 [19] to 9 [229] Dim.)

SUFFIXES

___ -AR	Acid Resistant Epoxy Coated Cast Iron	___ -TC	Neo-Loc Test Cap Gasket (2, 3, 4 [51, 76, 102] NL Bottom Outlet Only)
___ -BT	Bell Trap	___ -TS	Top Secured with Slotted Screws
___ -F	Extension Frame (Specify Total Height Req'd.)	___ -V	Backwater Valve (For 3 & 4 [76 & 102] IC & IP or 3 [76] IG Connections only)
___ -G	Galvanized Cast Iron	___ -VP	Vandal-Proof Secured Top
___ -HP	Heel-Proof Grate	___ -W	Winter Closure Plug (For 3 & 4 [76 & 102] IC & IP or 3 [76] IG Connections only)
___ -J	Auxiliary Inlet Connection (Specify 1-1/2 [38] or 2 [51] IP, Number & Location)	✓ -Y	Sediment Bucket
✓ -S	Secondary Strainer		
___ -SC	Solid Cover w/ Lift Handle		
___ -SG	Solid Gasketed Cover		

* Regularly furnished unless otherwise specified

⚠ **WARNING:** Cancer and Reproductive Harm - www.P65Warnings.ca.gov⚠ **ADVERTENCIA:** Cáncer y daño reproductivo - www.P65Warnings.ca.gov⚠ **AVERTISSEMENT:** Cancer et effets néfastes sur la reproduction - www.P65Warnings.ca.gov

Zurn Industries, LLC | Specification Drainage Operation
1801 Pittsburgh Avenue, Erie, PA 16502, Ph. 855.663.9876

In Canada | Zurn Industries Limited
7900 Goreway Drive, Unit 10, Brampton, Ontario L6T 5W6, Ph. 877.892.5216

www.zurn.com

Rev. L
Date: 05/15/2024
C.N. No. 146062
Patent zurn.com/patents
Prod. | Dwg. No. Z511



FS-1



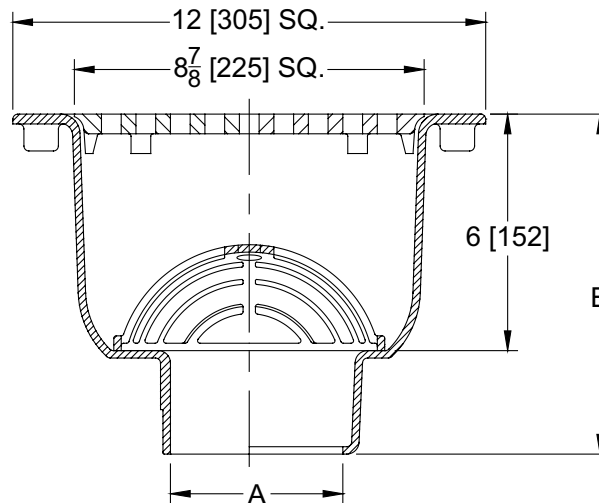
**Z1900****12 x 12 [305 x 305] A.R.E. SANI-FLOR RECEPTOR**
6 [152] SUMP DEPTH

Spec Seq#: 54

SPECIFICATION SHEET

TAG _____

Design and Dimensional Data (inches and [mm]) are Subject to Manufacturing Tolerances and Change Without Notice



A Pipe size in [mm]	Approx. Weight lbs [kg]	Grate Open Area in ² [cm ²]
2, 3, 4 [51, 76, 102]	32 [15]	31 [200]

ENGINEERING SPECIFICATION: ZURN Z1900

Sani-Flor Receptor 12" x 12" x 6" [305mm x 305mm x 152mm] deep cast iron body and square, light-duty grate with 1/2" [13mm] slotted openings, white acid resisting porcelain enamel interior and top, complete with white ABS anti-splash interior bottom dome strainer.

Options (Check/specify appropriate options)**PIPE SIZE**

3, 4 [76, 102]

2, 3, 4 [51, 76, 102]

2, 3, 4 [51, 76, 102]

(Specify Size/Type) **OUTLET**

___ IC Inside Caulk

___ NH No-Hub

___ NL Neo-Loc

'E' BODY HEIGHT DIMENSION**Z**

8-5/8 [219]

8-5/8 [219]

8 [203]

ZN/ZS

9-1/8 [232]

9-1/8 [232]

8-1/2 [216]

PREFIXES___ ☒ Z Cast Iron Body with White A.R.E. Interior*

___ ZN Cast Iron Body with White A.R.E. Interior, 12-1/2 [317] Square N.B. Frame and Full Grate with 1/2 [13] Openings (Add 1/2 [13] to 12 [305] Dim.)

___ ZS Cast Iron Body with White A.R.E. Interior, 12-1/2 [317] Square Stainless Steel Frame and Full Grate with 1/2 [13] Openings (Add 1/2 [13] to 12 [305] Dim.)

SUFFIXES

___ -DX Dex-O-Tex Flange (ZN Only)

___ -HD Stainless Steel Frame w/ X-Heavy Duty Stainless Steel Grate (ZS Only)

___ -HP Heel-Proof Grate (ZN Only)

___ -K Anchor Flange with Seepage Holes

___ -KC Anchor Flange with Seepage Holes and Clamp Collar

___ -LD (Less) Bottom Dome Strainer

___ -P 1/2 [13] Trap Primer Connection (See Z1023)

___ -SA Stabilizer Assembly (See Z1903)

___ -TC Neo-Loc Test Cap Gasket
(2, 3, 4 [51, 76, 102] NL Bottom Outlet Only)

___ -VPS Vandal-Proof Strainer

___ -1 (Less) Grate

___ ☒ -2 1/2 Grate

___ -3 3/4 Grate

___ -4 Full Grate w/ Center Opening (2 [51] Dia. for Z) and (3-1/32 [77] Square for ZN & ZS)

___ -6 Grate w/ 6 [152] Dia. x 6 [152] High Funnel

___ -8 Grate w/ 8-7/8 x 3-5/8 x 3-3/4 [225 x 92 x 95] High Oval Funnel

___ -11 Vandal-Proof Secured Top (ZN & ZS Only)

___ -15 Solid Loose Set Cover (Z & ZN Only)

___ -19 Full Hinged Grate (ZN Only)

___ -23 Aluminum Bucket

___ -25 White A.R.C. Bucket

___ -31 Stainless Steel Mesh Liner for Bucket

___ -32 Aluminum Dome Strainer

___ -33 White A.R.C. Anti-Splash Bottom Dome Strainer

⚠ WARNING: Cancer and Reproductive Harm - www.P65Warnings.ca.gov**⚠ ADVERTENCIA: Cáncer y daño reproductivo - www.P65Warnings.ca.gov****⚠ AVERTISSEMENT: Cancer et effets néfastes sur la reproduction - www.P65Warnings.ca.gov****Zurn Industries, LLC | Specification Drainage Operation**
1801 Pittsburgh Avenue, Erie, PA 16502, Ph. 855.663.9876**In Canada | Zurn Industries Limited**
7900 Goreway Drive, Unit 10, Brampton, Ontario L6T 5W6, Ph. 877.892.5216www.zurn.comRev. N
Date: 01/11/23
C.N. No. 144899
Patent zurn.com/patents
Prod. | Dwg. No. Z1900

1401756

FS-2



63M Mop Service Basin – 24" x 24" x 10"



The 63M is engineered and built to provide years of service. A complete range of accessories are available to outfit your Model 63M Mop Basin.

Features

- High impact resistant, one piece molded structural fiberglass.
- Generous 10" high side walls to contain splashes.
- Removable strainer.
- Integral molded-in-drain with seal for installing to 3" ABS, PVC (Sch. 80) or cast iron.
- Available drain seals for connecting 2" pipe or 3" no-hub soil pipe.
- Convenient anchor brackets for securing to wall.

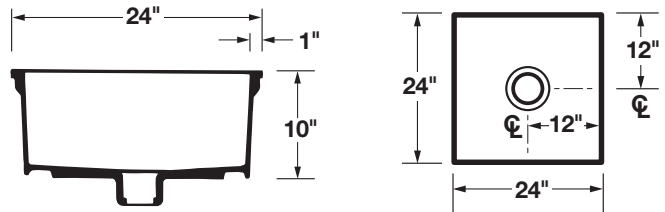
Code Accepted

- International Association of Plumbing and Mechanical Officials (IAPMO)® Listed
- Warnock Hersey® (Canada) Listed—CSA Standards: Mop Basins – #B45.0
- American National Standards Institute (ANSI) – Specification ANSI Z 124-2022
- American Society of Testing and Materials (ASTM) – E-84
- CAN/ULC S102-07 Burn and Smoke Test

Specifications

General: Mop Service Basin Model 63M, as manufactured by E.L. Mustee & Sons, Inc. One-piece, molded fiberglass made with matched metal molds. Height shall be 10" with not less than 1" wide shoulder. Size 24" x 24". Drain shall be integrally molded with drain seal for installation of 3" ABS, PVC (Sch. #80) and iron pipe. Removable stainless steel strainer. Shall meet ANSI Specifications Z 124-2022. Color: White.

Model	Color	Length	Width
63M	White	24"	24"



Accessories – for complete details and specifications visit www.mustee.com



- 1 63.600A SERVICE FAUCET** – Chrome plated brass service sink faucet, 1/2" pipe eccentric inlets on 8" centers. Vacuum breaker integral stops, chrome plated (H&C) handles, 3/4" hose end spout with pail hook, top reinforcing bar and mounting bracket. Shall meet ANSI Specifications A112.1.1, Section 2.9 and A112.18.1M, CSA Listed.

- 2 65.700 HOSE and HOSE HOLDER** – Heavy-duty 5/8" diameter, reinforced 31" rubber hose with brass couplings on one end. Spring loaded, molded rubber hose holder mounts on stainless steel wall plate.

- 3 65.600 MOP HANGER** – Three spring-loaded rubber holders mounted to 3" x 22" stainless steel wall plate. Wall anchors and mounting screws included.

- 4 BUMPER GUARDS** – Protect rims, snap-on

High impact resistant vinyl

63.401 – 20 3/4" long white, fits Models 63M and 65M
65.403 – 32 3/4" long white, fits Model 65M

Stainless steel

63.403 – 20 3/4" long, fits Models 63M and 65M
65.404 – 32 3/4" long, fits Model 65M

- 5 DURAGUARD™ WALL GUARDS** – Protect wall surfaces, 12" height, 20 gauge, #304 stainless steel.
67.2424 – two panels and bracket for 24" x 24" corner
67.2436 – two panels and bracket for 24" x 36" corner

- 6 OPTIONAL SEALS** – Order separately.

65.308 – for 2" PVC, ABS and iron
65.309 – for 3" PVC, ABS and iron, include mop basin
65.311 – for 3" soil pipe (no hub)

65M Mop Service Basin – 24" x 36" x 10"



The 65M is engineered and built to provide years of service. A complete range of accessories are available to outfit your Model 65M Mop Basin.

Features

- High impact resistant, one piece molded structural fiberglass.
- Elevated, self-draining shelf and removable strainer.
- Generous 10" high side walls to contain splashes.
- Integral molded-in-drain with seal for installing to 3" ABS, PVC (Sch. 80) or cast iron.
- Available drain seals for connecting 2" pipe or 3" no-hub soil pipe.
- Convenient anchor brackets for securing to wall.

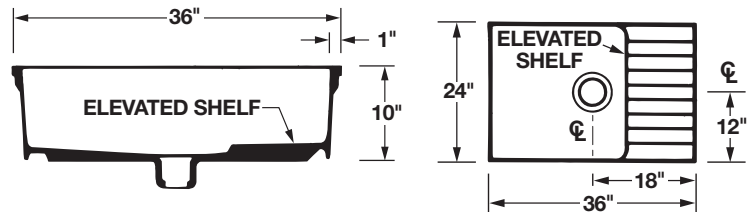
Code Accepted

- International Association of Plumbing and Mechanical Officials (IAPMO)[®] Listed 
- Warnock Hersey[®] (Canada) Listed—CSA Standards: Mop Basins – #B45.0 
- American National Standards Institute (ANSI) – Specification ANSI Z 124-2022 
- American Society of Testing and Materials (ASTM) – E-84 
- CAN/ULC S102-07 Burn and Smoke Test

Specifications

General: Mop Service Basin Model 65M, as manufactured by E.L. Mustee & Sons, Inc. One-piece, molded fiberglass made with matched metal molds. Height shall be 10" with not less than 1" wide shoulder. Size 24" x 36" with integrally molded, self-draining shelf and drain with seal for installation of 3" ABS, PVC (Sch. #80) and iron pipe. Removable stainless steel strainer. Shall meet ANSI Specifications Z 124-2022. Color: White.

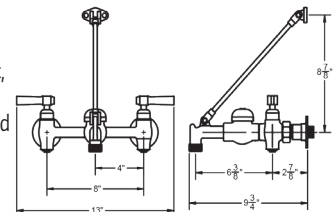
Model	Color	Length	Width
65M	White	24"	36"



Accessories – for complete details and specifications visit www.mustee.com



- 1 63.600A SERVICE FAUCET** – Chrome plated brass service sink faucet, 1/2" pipe eccentric inlets on 8" centers. Vacuum breaker integral stops, chrome plated (H&C) handles, 3/4" hose end spout with pail hook, top reinforcing bar and mounting bracket. Shall meet ANSI Specifications A112.1.1, Section 2.9 and A112.18.1M, CSA Listed. 



- 2 65.700 HOSE and HOSE HOLDER** – Heavy-duty 5/8" diameter, reinforced 31" rubber hose with brass couplings on one end. Spring loaded, molded rubber hose holder mounts on stainless steel wall plate.



- 3 65.600 MOP HANGER** – Three spring-loaded rubber holders mounted to 3" x 22" stainless steel wall plate. Wall anchors and mounting screws included.

- 4 BUMPER GUARDS** – Protect rims, snap-on

High impact resistant vinyl

63.401 – 20 3/4" long white, fits Models 63M and 65M
65.403 – 32 3/4" long white, fits Model 65M

Stainless steel

63.403 – 20 3/4" long, fits Models 63M and 65M
65.404 – 32 3/4" long, fits Model 65M

- 5 DURAGUARD[™] WALL GUARDS** – Protect wall surfaces, 12" height, 20 gauge, #304 stainless steel.

67.2424 – two panels and bracket for 24" x 24" corner
67.2436 – two panels and bracket for 24" x 36" corner

- 6 OPTIONAL SEALS** – Order separately.

65.308 – for 2" PVC, ABS and iron
65.309 – for 3" PVC, ABS and iron, included with mop basin
65.311 – for 3" soil pipe (no hub)

PRV



Job Name _____

Contractor _____

Job Location _____

Approval _____

Engineer _____

Contractor's P.O. No. _____

Approval _____

Representative _____

LEAD FREE*

Series LFN223B, LFN223BS

Super Capacity Water Pressure Reducing Valves**

Sizes: 2½" – 3"

Series LFN223B and LFN223BS Super Capacity Water Pressure Reducing Valves are designed to reduce incoming water pressure to a sensible level to protect plumbing system components and reduce water consumption. The LFN223B/LFN223BS features Lead Free* construction to comply with Lead Free* installation requirements. This series is suitable for water supply pressures up to 300psi (20.7 bar) and may be adjusted from 25 – 75psi (172 – 517 kPa). The standard setting is 50psi (345 kPa). Series LFN223B features an enlarged diaphragm, spring cage and seat orifice for super capacity performance. Series LFN223BS has the same options as the LFN223B, except it is furnished with a Lead Free* strainer. All parts are quickly and easily serviceable without removing the valve from the line. The standard bypass feature permits the flow of water back through the valve into the main when pressures, due to thermal expansion on the outlet side of the valve, exceed the pressure in the main supply.

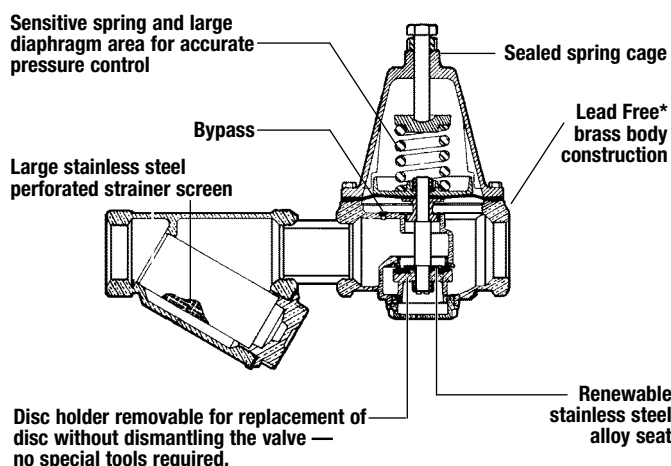
Features

- Enlarged diaphragm, spring cage and seat orifice for super capacity performance
- Lead Free* cast copper silicon alloy body construction
- Serviceable in line
- Series LFN223BS furnished with separate Lead Free* strainer
- Standard bypass feature controls thermal expansion pressure***
- Sealed spring cage on all models for accessible outdoor or pit installations

Specifications

A Lead Free* Water Pressure Reducing Valve shall be installed on the water service pipe near its entrance to the building where supply main pressure exceeds 60psi (413 kPa) to reduce it to 50psi (345 kPa) or lower. The water pressure reducing valve shall be constructed using Lead Free* materials. Lead Free* regulators shall comply with state codes and standards, where applicable, requiring reduced lead content. Provision shall be made to permit the bypass flow of water back through the valve into the main when pressures, due to thermal expansion on the outlet side of the valve, exceed the pressure in the main supply. Pressure reducing valves with built-in bypass check valves and strainer will be acceptable. Approved valves shall comply with ASSE 1003. Valve shall be a Watts Series LFN223B or LFN223BS (with strainer).

Series LFN223B

Available with or without
separate strainer

*The wetted surface of this product contacted by consumable water contains less than 0.25% of lead by weight.

**A water saving test program concluded that reducing the supply pressure from 80-50psi (551-345 kPa) resulted in a water savings of 30%.

***The bypass feature will not prevent the pressure relief valve from opening on the hot water supply system with pressure above 150psi (10.3 bar).

NOTICE

The information contained herein is not intended to replace the full product installation and safety information available or the experience of a trained product installer. You are required to thoroughly read all installation instructions and product safety information before beginning the installation of this product.

NOTICE

Inquire with governing authorities for local installation requirements

Watts product specifications in U.S. customary units and metric are approximate and are provided for reference only. For precise measurements, please contact Watts Technical Service. Watts reserves the right to change or modify product design, construction, specifications, or materials without prior notice and without incurring any obligation to make such changes and modifications on Watts products previously or subsequently sold.



1240786

Models

2½" LFN223M2-B	NPT threaded female inlet x NPT threaded female outlet
3" LFN223M1-B	NPT threaded female inlet x NPT threaded female outlet
2½" LFN223M2-BS	NPT threaded female inlet with strainer x NPT threaded female outlet
3" LFN223M1-BS	NPT threaded female inlet with strainer x NPT threaded female outlet

For ½" – 2½" threaded connections, refer to literature ES-LF223.

For 3" flanged connections, refer to literature ES-LF223F.

Materials

Body:	Lead Free* cast copper silicon alloy
Seat:	Replaceable stainless steel alloy
Strainer Screen:	Stainless steel (model LFN223S)
Diaphragm:	Reinforced Buna-N
Valve Disc:	EPDM

Pressure – Temperature

Temperature Range: 33°F – 160°F (0.5°C – 71°C)

Maximum Working Pressure: 300psi (20.7 bar)

Adjustable Reduced Pressure Range: 25 – 75psi (172 – 517 kPa)

Standard Reduced Pressure Setting: 50psi (345 kPa)

Standards



Both 2½" and 3" are listed to ASSE 1003 through IAPMO (UPC).

The 2½" is also listed to ASSE 1003 through ASSE.

Both 2½" and 3" are listed to NSF 61-G

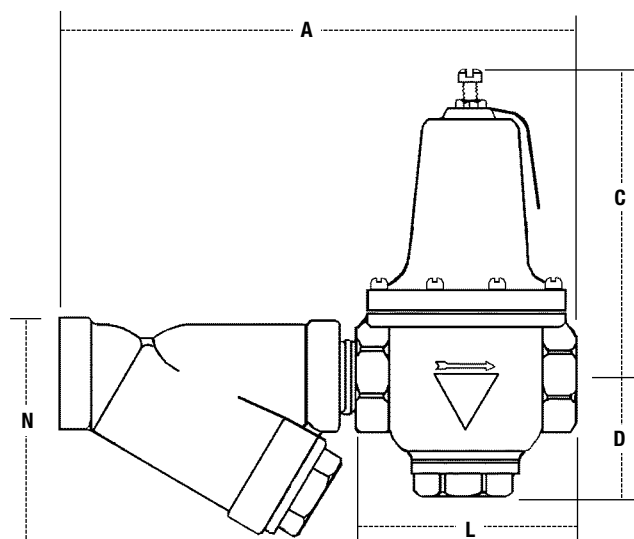
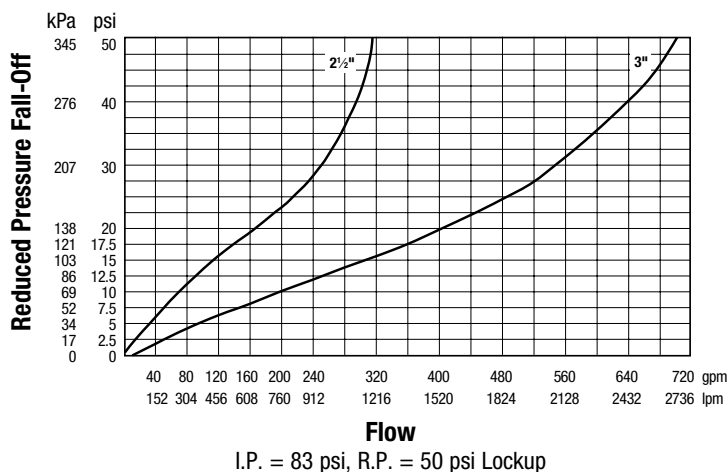
Options

HP High pressure range 75 – 125psi (172 – 862 kPa)

2½" HP, LP, G and GC options

3" HP option

Capacity



Dimensions – Weights

SIZE	DIMENSIONS						WEIGHT			
	A (LFN223BS)		C		D		L		N (LFN223BS)	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
2½	17	432	10¾	273	2⅞	73	7⅞	200	5	127
3	20¾	527	12¾	324	4⅞	105	10½	267	6¾	172



EWH-1





The new degree of comfort.®



Spec Seq#: 62
Commercial Electric
Light-Duty Water Heaters

Light-Duty Commercial Electric Water Heaters

With easy field-level conversion for multiple light duty applications

Flexible design for a variety of light duty applications

- 30 - 119.9-Gallon models
- 208/240/277/480 Voltages available on all models
- 30 to 50-Gallon models available in 3kW-12 kW, tall and short
- 66, 80 and 119.9-Gallon models only available in 12.1 kW
- Factory shipped in simultaneous configuration

FEATURES & BENEFITS	
Terminal Block	Easy to convert standard models from simultaneous to non-simultaneous, single phase to three phase
	Field convertible to meet any light-duty application need
	Available on 30 - 119.9-gallon models
Wiring Options	Custom field level configuration based on application requirements
	Simultaneous and non-simultaneous wiring
	Single & three phase available
Energy Efficient Design	Reduces energy consumption & standby heat loss with 2-1/2" of rigid polyurethane foam insulation
Designed for Long Life & Top Performance	Proprietary protective steel formulation with a unique coat of high temperature porcelain enamel
	Design maximizes corrosion resistance & life of anode rod
	Patented corrosion-resistant elements include double layer of magnesium oxide and copper to resist corrosion
	Durable factory-installed full flow brass drain valve
Automatic Temperature Control	Automatic Temperature Control - Surface mounted thermostat automatically cycles on and off to maintain desired water temperature
	170°F Maximum temperature setting
Warranty	3-year limited tank, upgradable to 5-years and a 1-year limited parts warranty*

*See Commercial Warranty Certificate for complete information.

All models come with a terminal block. Models with power requirements over 48 Amps come with a fused terminal block. See Electrical Characteristics Chart on Back.



Rheem Light-Duty
30 to 119.9-Gallon Capacities
208/240/277/480 Voltages
3kW - 12.1 kW
Electric



Consult factory for certification listing.

Safety and Construction | These products are design certified by Underwriters Laboratories (UL) to meet UL safety standards as electric storage tank water heaters. All models are North Carolina and Massachusetts Code compliant.
Certified for 150 PSI maximum working pressure.

See specifications chart on back.



INTEGRATED AIR & WATER

12/21 FORM NO. RR102



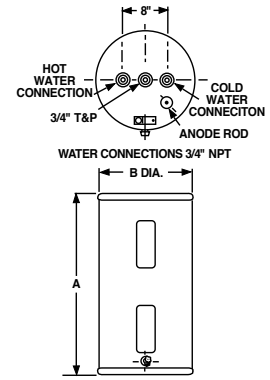


The new degree of comfort.®



Spec Seq#: 62
Commercial Electric
Light-Duty Water Heaters

DIMENSIONAL INFORMATION (All dimensions shown in English and Metric)														FIRST HOUR DEL. G.P.H.	UNIFORM ENERGY FACTOR (UEF)	STL
MODEL NUMBER	MIN. WATTS	MAX. WATTS	TANK CAPACITY (LISTED)		TANK CAPACITY (MEASURED)		A		B		APPROX. SHIPPING WT.					
			GAL.	LTR	GAL.	LTR	INCHES	MM	INCHES	MM	lbs.	kgs.				
ELD30-TB	3,000	12,000	30	114	27	102	47-1/2	1207	19	483	125	57	46	0.92	N/A	
ELD40-TB	3,000	12,000	40	151	36	136	48-1/4	1226	20-1/4	514	140	64	55	0.93	N/A	
ELD52-TB	3,000	12,000	50	189	45	170	58-5/8	1489	20-1/4	514	155	70	63	0.93	N/A	
ELD66-TB	12,100	12,100	65	246	58.5	221	59-3/8	1508	22-1/4	565	185	84	N/A	N/A	0.76	
ELD80-TB	12,100	12,100	80	303	72	273	59-5/8	1514	23-3/4	603	220	100	N/A	N/A	0.68	
ELD120-TB	12,100	12,100	119.9	454	108	409	62-7/8	1597	28-1/4	718	335	152	N/A	N/A	0.55	
ELDS30-TB	3,000	12,000	28	106	25	95	30	762	23	584	130	59	43	0.92	N/A	
ELDS40-TB	3,000	12,000	36	136	33	125	31-1/2	800	24-1/4	616	150	68	46	0.92	N/A	
ELDS52-TB	3,000	12,000	47	178	43	163	32	813	26-1/4	667	180	82	57	0.93	N/A	



PRODUCT AVAILABILITY		Electric Light Duty models are available with terminal block in both fused and non-fused configurations. Fused configuration is Standard in all water heaters with power requirements over 48 Amps.																			
SIM OR NON-SIM	SIZE	208V					240V					277V					480V				
		3/3	4/4	4.5/4.5	5/5	6/6	3/3	4/4	4.5/4.5	5/5	6/6	3/3	4/4	4.5/4.5	5/5	6/6	3/3	4/4	4.5/4.5	5/5	6/6
	30	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	40	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	50	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	66					✓					✓					✓					✓
	80					✓					✓					✓					✓
	120					✓					✓					✓					✓

- Fused or Non-Fused, Fused if > 48 Amps • Terminal Block allows for conversions from Simultaneous element usage to non-Simultaneous for maximum energy savings.
- ELD-C Models 66/80/120 still available without terminal block. • All models are factory shipped with sim configuration.

ELECTRICAL CHARACTERISTICS																	
ELEMENT WATTAGE	ELEMENT WATTAGE	NON-SIMULTANEOUS WIRING				SIMULTANEOUS WIRING											
		SINGLE/THREE PHASE OPERATION FULL LOAD CURRENT IN AMPERES				2-WIRE SINGLE PHASE OPERATION FULL LOAD CURRENT IN AMPERES				4-WIRE THREE PHASE OPERATION FULL LOAD CURRENT IN AMPERES			4-WIRE SINGLE PHASE OPERATION FULL LOAD CURRENT IN AMPERES (DOUBLE BRANCH CIRCUIT)				
UPPER	LOWER	TERMINALS L1 AND L2 FOR SINGLE PHASE, L1, L2, AND T2 FOR 3 PHASE				TERMINALS L1 AND L2				THERE'S NO 4 WIRE 3 PHASE THIS IS 3 PHASE SIM TERMINALS L1, L2, AND T2			TERMINALS L1 AND L2 FIRST CIRCUIT TERMINALS T1 AND T2 SECOND CIRCUIT				
		208	240	277	480	208	240	277	480	208	240	480	208	240	277	480	
3000	3000	14	13	11	6	29	25	22	13	25	22	11	SAME AS NON SIM, BUT THERE WILL BE TWO SETS OF INCOMING POWER. THIS IS THE FACTORY WIRING				
4000	4000	19	17	14	8	38	33	29	17	33	29	14					
4500	4500	22	19	16	9	43	38	32	19	37	32	16					
5000	5000	24	21	18	10	48	42	36	21	42	36	18					
6000	6000	29	25	22	13	58	50	43	25	50	43	22					
6050	6050					58	50	44	25	50	44	22					

RECOVERY CAPACITIES (Recovery in U.S. Gallons/Hr. (GPH) and Liters/Hr. (LPH) at various temperature rises)																				
ELEMENT WATTAGE UPPER/LOWER	NON-SIMULTANEOUS WIRING										SIMULTANEOUS WIRING									
	40°F / (22°C)		60°F / (33°C)		80°F / (45°C)		100°F / (56°C)		120°F / (67°C)		40°F / (22°C)		60°F / (33°C)		80°F / (45°C)		100°F / (56°C)		120°F / (67°C)	
	GPH	LPH	GPH	LPH	GPH	LPH	GPH	LPH	GPH	LPH	GPH	LPH	GPH	LPH	GPH	LPH	GPH	LPH	GPH	LPH
3000/3000	30	115	20	77	15	58	12	46	10	38	61	230	41	153	30	115	24	92	20	77
4000/4000	41	153	27	102	20	77	16	61	14	51	81	307	54	205	41	153	32	123	27	102
4500/4500	46	173	30	115	23	86	18	69	15	58	91	345	61	230	46	173	36	138	30	115
5000/5000	51	192	34	128	25	96	20	77	17	64	101	384	68	256	51	192	41	153	34	128
6000/6000	61	230	41	153	30	115	24	92	20	77	122	460	81	307	61	230	49	184	41	153
6050/6050											122	460	81	307	61	230	49	184	41	153

Consult factory for certification requirements.

Recommended Specifications (for trade reference only)

Water heater(s) shall be model _____, manufactured by Rheem, having electrical input of _____ kW and a recovery rate of _____ GPH at a 100°F temperature rise. Water heater(s) shall have a storage capacity of _____ gallons. Water heater(s) shall have the UL seal of certification and be factory equipped with an CSA/ ASME rated temperature and pressure relief valve. Tank(s) interior shall be coated with a high temperature porcelain enamel and furnished with

a magnesium anode rod rigidly supported. Water heater(s) shall meet or exceed the energy factor requirements of ASHRAE. Tanks shall have a working pressure rating of 150 psi, and shall be completely assembled. Water heater(s) shall be equipped with copper, resistored, "screw-in" type elements. Tank shall be insulated with 2-1/2" of rigid polyurethane foam insulation. Water heater(s) shall be equipped with surface mounted thermostats each with an integral, manual reset, high limit control. Water heater(s) shall be covered by a three year limited warranty against tank leaks.

In keeping with its policy of continuous progress and product improvement, Rheem reserves the right to make changes without notice.

Rheem Water Heating • 1115 Northmeadow Parkway, Suite 100
Roswell, Georgia 30076 • www.rheem.com



INTEGRATED AIR & WATER

12/21 FORM NO. RR102



1918115

GAS METER



ONICON

Flow and Energy Measurement

F-5500

THERMAL MASS FLOW METER

ONICON's
F-5500 Series
Thermal Mass
Flow Meters
provide
accurate and
reliable flow
measurement
for natural gas,
compressed
air, and other
industrial gas
applications.



2496658

• Natural Gas • Compressed Air • Industrial Gases •

F-5500

THERMAL MASS FLOW METER



2496658



DESCRIPTION

ONICON's F-5000 Series Thermal Mass Flow Meters provide accurate mass flow measurement of natural gas, compressed air and other industrial gases. The proprietary sensor design measures mass flow directly and does not require additional pressure or temperature compensation to deliver accurate flow rate and total data.

The F-5500 is available as an inline or an insertion style meter and includes an easy to operate user interface/display. The standard version of the meter is provided with a 4-20 mA analog output and an RS485 interface that is field configurable for BACnet® MS/TP or Modbus® RTU. A second output signal configuration is available that includes a 4-20 mA analog output and a programmable pulse output. The pulse output model is also available with HART.

APPLICATIONS

Accurate sub-metering of natural gas & propane for:

- Tenant space usage
- Boiler efficiency
- Campus monitoring

Also ideal for monitoring:

- Compressed Air
- Medical gases
- Other industrial gases

CALIBRATION

Every ONICON flow meter is wet calibrated in a flow laboratory against standards that are directly traceable to NIST*. A certificate of calibration accompanies every meter.

* National Institute of Standards and Technology

** Installations must comply with federal, state and municipal building codes. Review all proposed combustible gas installations with your local code enforcement officials before attempting to install.

FEATURES

BACnet MS/TP or MODBUS RTU - The standard F-5500 includes an RS485 output that provides BACnet MS/TP or Modbus RTU. Data reported to the network includes flow rate and total, temperature and elapsed time since reset.

User Friendly Interface / Display - The bright, easy-to-read, backlit display and intuitive menu structure simplify page navigation and allow for field programming. Free utility software is also available for programming and data logging.

Provides for Field Validation of Calibration - F-5500 internal diagnostic functions include a zero flow calibration check. This fast, easy to perform test allows for field validation of the factory zero flow calibration. The utility software provided with the meter allows you to print a certificate validating the test results.

Insertion Meters Can Be Installed Without Interrupting Gas Service** - ONICON's hot tap design allows for installation without interruption to the gas service. The meter can also be removed for service without disrupting flow.

Highly Accurate Over a Wide Operating Range - Our proprietary direct digital control sensing circuitry is very stable yet highly responsive to changes in flow. This design allows for accurate flow measurement over a very wide operating range (over 1000:1 for the inline version). It also makes the meter ideal for measuring low flow rates.

Excellent Value - ONICON insertion style meters are accurate, easy-to-use and reliable. They are also priced independently of pipe size. This makes them an excellent value, particularly in larger diameter pipes.



F-5500

THERMAL MASS FLOW METER



SPECIFICATIONS*

F-5500 THERMAL MASS FLOW METER	
FLOW ACCURACY	NATURAL GAS/PROPANE GAS ±1.0% of reading from 500 – 7,000 SFPM (14:1 turndown) ±2.0% of reading from 100 – 500 SFPM (5:1 turndown) COMPRESSED AIR & OTHER HIGH VELOCITY CALIBRATIONS ±1.0% of reading ± 0.5% of full scale over a 100:1 turndown
TEMPERATURE ACCURACY	±1.0°F over the range of -40°F to 250°F
OVERALL FLOW RANGE	15 - 35,000 SFPM
SENSING METHOD	Thermal mass flow utilizing direct digital control sensing circuitry
PIPE SIZE RANGE	INSERTION STYLE 1½ - 24" nominal diameter INLINE STYLE ¾ - 6" nominal diameter Note: Inline meters larger than 2.5" are shipped disassembled in two separate boxes.
INPUT POWER	12 - 28 VDC, 6W minimum power
TEMPERATURE RANGE	FLUID -40°F to 250°F AMBIENT -40°F to 158°F
MAXIMUM OPERATING PRESSURE	INSERTION STYLE Process adapter fitting - 60 psig (4.1 barg) max High pressure adapter fitting - 150 psig (10.3 barg) max INLINE STYLE ANSI Class 150 flanges - 230 psig at 100°F (16 barg) NPT - 300 psig (20.7 barg) All stainless steel ferrules
PRESSURE DROP (at 2500 SFPM, 70°F and 2 PSIG)	INSERTION STYLE Less than ½" W.C. (H2O) in 1½" diameter pipes, decreasing in larger pipes INLINE STYLE (with built-in flow conditioner) Less than ½" W.C (H2O) in 2" and larger diameter meters Less than 0.9" W.C (H2O) in 1" and 1½" diameter meters
PROGRAMMING/MEMORY	Factory programmed for specific application. Field programming available through mini-USB interface and utility program. Non-volatile memory retains all program parameters and totalized values in the event of power loss.
OUTPUT SIGNALS PROVIDED	Analog output: 4-20 mA Select from one of the following options as the second output: • RS485 interface: BACnet MS/TP or Modbus RTU (field selectable) • Programmable pulse output: Field selectable as scaled pulse or alarm (Isolated open collector output) • 4-20 mA with HART FSK (Only available with programmable pulse output)
ELECTRONICS ENCLOSURE	NEMA 4X Weathertight aluminum enclosure
ELECTRICAL CONNECTIONS	Enclosed terminal blocks, cable access through two ¾" NPT conduit fittings
MATERIAL	Wetted metal components: 316 stainless steel
APPROVALS	FM (USA) FMc (CAN): Approved Class 1, Div 1, Groups B, C, D; Class 2, Div 1, Groups E, F, G; Class 3, Div 1; T4, Ta = -40°C to 70°C; Class 1, Zone 1, AEx/Ex db IIB = H2 T4; Gb Ta = -40°C to 70°C; Type 4X, IP66/67 EMC Directive; 2014/30/EU Emissions and Immunity Testing: EN61326-1:2013 Massachusetts Board of State Examiners of Plumbers and Gasfitters Canadian Registration Number (CRN) according to CSA BS1

* SPECIFICATIONS subject to change without notice.

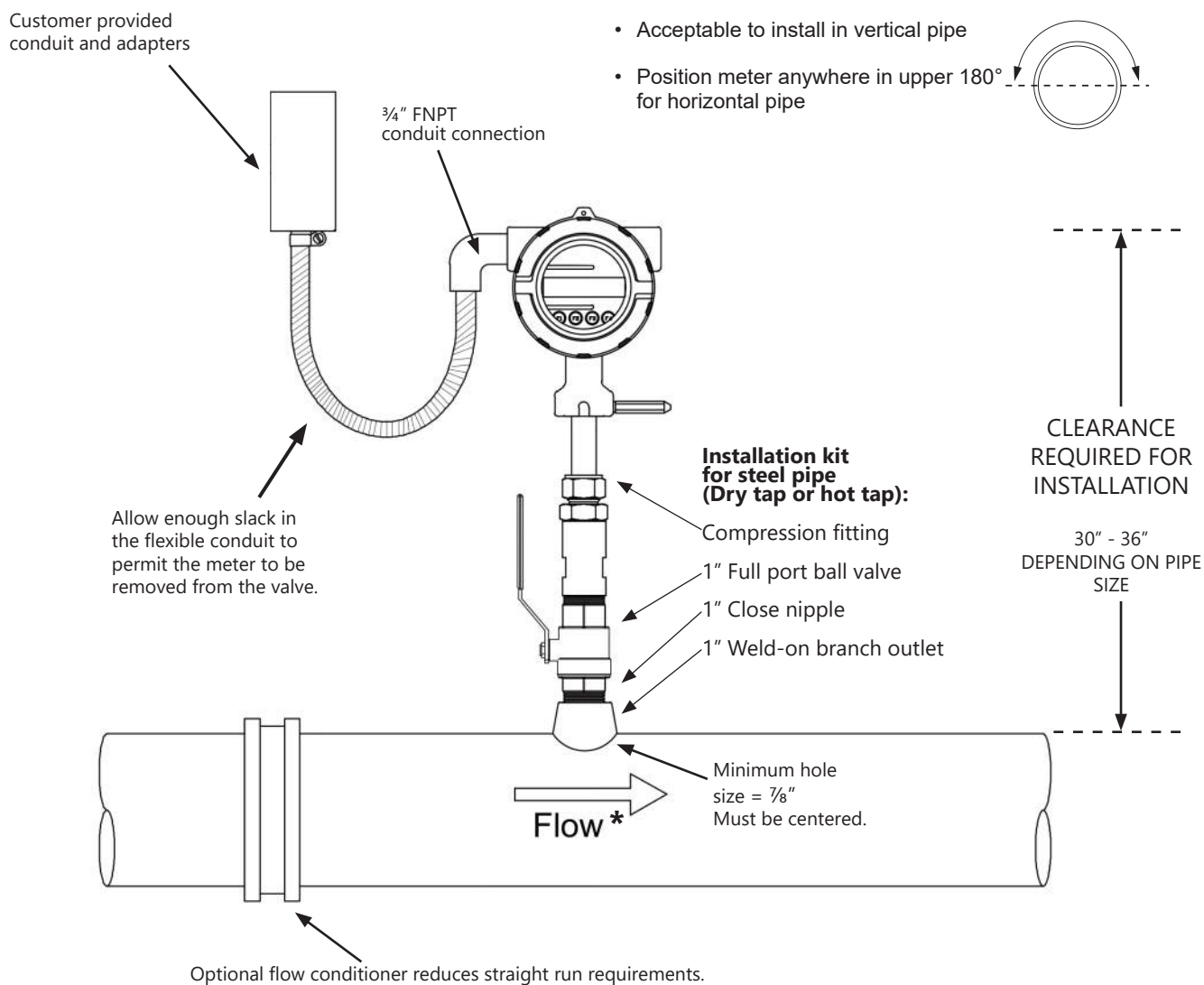


F-5500

THERMAL MASS FLOW METER



TYPICAL INSERTION METER INSTALLATION



2496658

*Standard orientation. Contact ONICON for other options.

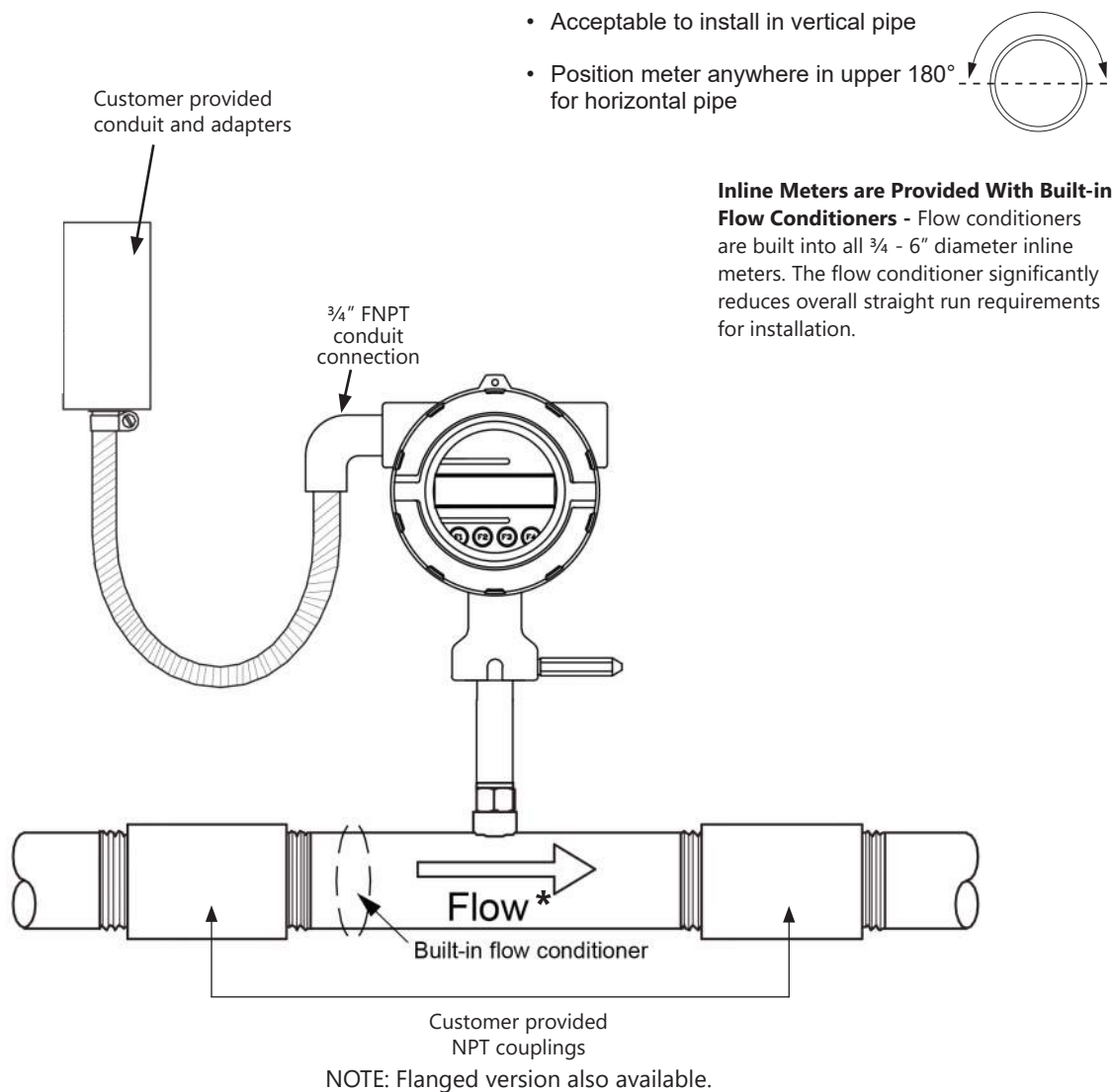


F-5500

THERMAL MASS FLOW METER



TYPICAL INLINE METER INSTALLATION



2496658

*Standard orientation. Contact ONICON for other options.



F-5500

THERMAL MASS FLOW METER



METER ORDERING INFORMATION

Meter Model Number Coding = F-55AA-BCDE-FGGH

F-55 = Thermal Mass Flow Meter with Display

AA = Pipe Diameter

00 = Insertion	02 = 2"
34 = 3/4"	25 = 2 1/2"
01 = 1"	03 = 3"
13 = 1 1/4"	04 = 4"
15 = 1 1/2"	06 = 6"

B = Output Signals

- 2 = Loop powered 4-20 mA & scaled pulse
- 3 = Loop powered 4-20 mA & RS485
- 4 = Loop powered 4-20 mA with HART & scaled pulse

C = Input Power

- 1 = 12 - 28 VDC

D = Electronics Enclosure Mounting Configuration

- 1 = Integral enclosure with LCD

E = Process Connection

- 4 = Insertion
- 5 = Threaded MNPT (3/4 - 3" pipe sizes only)
- 6 = ANSI Class 150 flanges

F = Flow Conditioner

- 1 = Insertion without flow conditioner
- 2 = Insertion with flow conditioner
- 3 = Inline with flow conditioner

GG = Pipe Size Range

- 00 = Inline Meter
- 15 = 1 1/2 - 6" nominal diameter
- 18 = >6" nominal diameter

H = Process Adapter Fitting

- 0 = Standard (0-60 psig)
- 1 = High pressure (0-150 psig)
- 9 = Inline meter

GAS TYPE

NG = Natural Gas	HE = Helium Gas
ME = Methane Gas	NI = Nitrogen Gas
PG = Propane Gas	AR = Argon Gas
AI = Air	CD = Carbon Dioxide
O2 = Oxygen Gas	BU = Butane
HY = Hydrogen	

ACCESSORY ORDERING INFORMATION

Install Kits for Carbon Steel Piping Systems

Model Number	Description
INSTL0094-FMH	Insertion flow meter hot tap installation kit, wetted materials are bronze, brass and steel

OPERATING RANGE FOR COMMON NATURAL GAS PIPE SIZES 15 to 7,000 SFPM in schedule 40 pipe

Pipe Size (Inches)	Flow Rate (SCFH)	
	Min	Max
3/4	3.3	1,560
1	5.4	2,521
1 1/4	9.3	4,362
1 1/2	13	5,938
2	21	9,740
2 1/2	30	13,964
3	46	21,562
4	80	37,130
5	125	58,350
6	181	84,263
8	313	145,912

OPERATING RANGE FOR COMMON COMPRESSED AIR PIPE SIZES 15 to 35,000 SFPM in schedule 40 pipe

Pipe Size (Inches)	Flow Rate (SCFM)	
	Min	Max
3/4	0.06	130
1	0.09	210
1 1/4	0.16	364
1 1/2	0.21	395
2	0.35	816
2 1/2	0.5	1,160
3	0.77	1,800
4	1.33	3,090
5	2.08	4,860
6	3	7,020
8	5.2	12,200
10	8.2	19,200



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ONICON INSTALLATION HARDWARE SELECTION GUIDE

ONICON installation kits are specifically designed for use with ONICON insertion flow and BTU (energy) meters. Their use is recommended with ONICON products.



F-1000 Series Turbine Flow Meter with INST0005-FMD Dry Ta

2052116



INSTALLATION HARDWARE SELECTION GUIDE

How to use this selection guide?

1. Select the type of installation hardware required, either Dry Tap or Hot Tap.
2. Choose the pipe material.
3. Choose the Installation Kit based on material preference and/or pipe size.

What does Dry Tap or Hot Tap mean?

The terms "Dry Tap" and "Hot Tap" refer to the installation method of the isolation valve kit only.

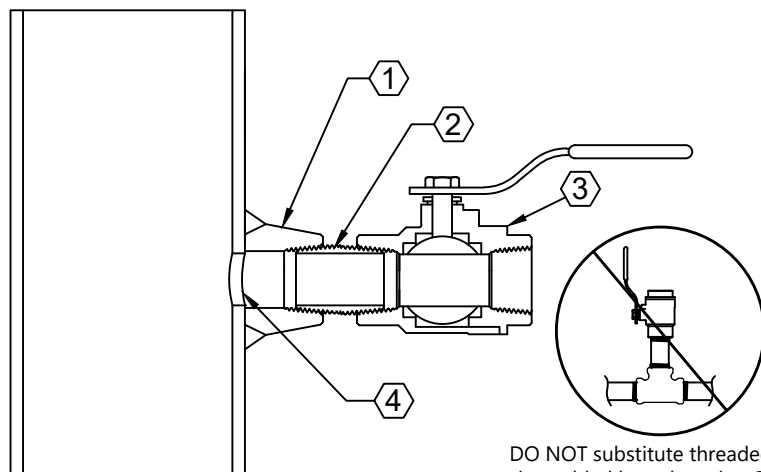
Dry Tap Installation Hardware

For new construction or scheduled shut down, the installation kit is installed into a dry, or unfilled system.

Hot Tap Installation Hardware

For installations which require the access hole in the pipe to be drilled through the installation valve, via a hot tap drilling machine, while the hydronic system is full and pressurized.

Below is an example of Dry Tap Installation Kit for Welded Steel Pipe:



1. 1" NPT welded branch outlet, typical for steel piping system installation.
2. 1" threaded close nipple
3. 1" full port ball valve
4. 1" minimum dia. access hole

DO NOT substitute threaded tees for the welded branch outlet. Contact ONICON if you need installation hardware for threaded pipe.

Important Note:

Installation hardware materials vary based on pipe material, pipe size and dry tap vs hot tap versions.

See next pages for the complete list of installation kits offered by ONICON. If you have any questions or need assistance, please contact ONICON at (727) 447-6 or sales@onicon.com.



INSTALLATION HARDWARE SELECTION GUIDE

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INSTALLATION HARDWARE SELECTION GUIDE

DRY TAP INSTALLATION HARDWARE



For F-1000 Series, FSM-3 and F-3500 Flow Meters

Pipe Material	Model Number	Description	Items Included with Kit	Notes
Carbon Steel/ Black Iron (welded pipe)	INSTL0001-FMD	Insertion flow meter installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.	1" Full port bronze ball valve 1" Brass close nipple 1" Weld-on carbon steel branch outlet	Designed for use in hydronic systems.
	INSTL01DW-FMD	Insertion flow meter installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, CU and low lead brass, for use in carbon steel piping systems.	1" Full port bronze ball valve 1" Brass close nipple 1" Weld-on carbon steel branch outlet	NSF certified for use in domestic water applications.
	INSTL0005-FMD	Insertion flow meter installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in carbon steel piping systems.	1" Full port SS ball valve 1" SS close nipple 1" Weld-on carbon steel branch outlet	Designed for use in hydronic systems.
Carbon Steel/ Black Iron (threaded pipe)	INSTL0007-FMD	Insertion flow meter installation kit for pipe size range from 1.25" to 2.5". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.	1" Full port bronze ball valve 1" Brass close nipple (Pipe dia x 12" long) threaded nipple 1" NPT outlet	Designed for use in hydronic systems.
	INSTL0008-FMD	Insertion flow meter installation kit for pipe size 1.25" to 2.5". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.	1" Full port SS ball valve 1" SS close nipple (Pipe dia x 12" long) threaded nipple 1" NPT outlet	Designed for use in hydronic systems.
Copper Tube	INSTL0003-FMD	Insertion flow meter installation kit for pipe size range from 1.25" to 2". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems.	1" Full port bronze ball valve 1" Copper street adapter with MNPT threads - Copper tee with 1" outlet	NSF certified for use in domestic water applications.
	INSTL0004-FMD	Insertion flow meter installation kit for pipe size range from 2.5" to 3". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems.	1" Full port bronze ball valve 1" Copper street adapter with MNPT threads - Copper tee with 1" outlet	NSF certified for use in domestic water applications.
	INSTL0009-FMD	Insertion flow meter installation kit for pipe size 4". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems.	1" Full port bronze ball valve 1" Copper street adapter with MNPT threads - Copper tee with 1" outlet	NSF certified for use in domestic water applications.
Stainless Steel	INSTL0018-FMD	Insertion flow meter installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in SS piping systems.	1" Full port SS ball valve 1" SS close nipple 1" SS weld-on branch outlet	Designed for use in process water applications.

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INSTALLATION HARDWARE SELECTION GUIDE

DRY TAP INSTALLATION HARDWARE



For System-10, System-20 and F-4300 BTU Meters (For use with current (mA) based temperature sensors)

Pipe Material	Model Number	Description	Items Included with Kit	Notes
Steel Pipe Welded	INSTL0032-TSD	Temperature sensor (pair) installation kit for pipe size range from 0.75" to 5". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.	• (2) Brass thermowells • (2) Weld-on carbon steel branch outlets, ½" NPT • (2) Transmitter enclosures	Designed for use in hydronic systems.
	INSTL0034-TSD	Temperature sensor (pair) installation kit for pipe size range from 0.75" to 42". Wetted materials are SS, for use in carbon steel piping systems.	• (2) SS thermowells • (2) Weld-on carbon steel branch outlets, ½" • (2) Transmitter enclosures	Designed for use in hydronic systems.
	INSTL0066-TSD	Temperature sensor (pair) installation kit for pipe size range from 1.5" to 42". Wetted materials are SS, for use in carbon steel piping systems.	• (2) SS thermowells • (2) Weld-on carbon steel branch outlets, 1" • (2) High temperature transmitter enclosure assemblies	Designed for use in hydronic systems, and in HTHW systems (T ≥ 300°F)
Steel Pipe Threaded	INSTL0052-TSD	Temperature sensor (pair) installation kit for pipe size range from 0.75" to 2.5". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.	• (2) Brass thermowells • (2) Brass reducer bushings, ½" • (2) Transmitter enclosures	Designed for use in hydronic systems.
Steel /PVC Pipe with Saddles	INSTL0045-TSD	Temperature sensor (pair) installation kit for pipe size range from 1.5" to 8". Wetted materials are SS, for use in PVC piping systems.	• (2) SS thermowells • (2) Brass reducer bushings, 1" x ½" • (2) Ductile iron saddles for Sch std./40/80 pipe with 1" NPT outlets • (2) Transmitter enclosures	Designed for use in hydronic systems.

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INSTALLATION HARDWARE SELECTION GUIDE

DRY TAP INSTALLATION HARDWARE



For System-10, System-20 and F-4300 BTU Meters (For use with current (mA) based temperature sensors)

Pipe Material	Model Number	Description	Items Included with Kit	Notes
Copper Tees (brass wells)	INSTL0036-TSD	Temperature sensor (pair) installation kit for pipe size range from 0.75" to 2". Wetted materials are bronze, brass and steel, for use in copper piping systems.	• (2) Copper tees with 1/2" outlet • (2) Brass thermowells • (2) Copper street adapter, 1/2" • (2) Transmitter enclosures	Designed for use in hydronic systems.
	INSTL0037-TSD	Temperature sensor (pair) installation kit for pipe size range from 2.5" to 3". Wetted materials are bronze, brass and steel, for use in copper piping systems.	• (2) Copper tees with 1/2" outlet • (2) Brass thermowells • (2) Copper street adapter, 1/2" • (2) Transmitter enclosures	Designed for use in hydronic systems.
	INSTL0038-TSD	Temperature sensor (pair) installation kit for pipe size 4". Wetted materials are bronze, brass and steel, for use in copper piping systems.	• (2) Copper tees with 1/2" outlet • (2) Brass thermowells • (2) Copper street adapter, 1/2" • (2) Transmitter enclosures	Designed for use in hydronic systems.
Copper Tees (SS wells)	INSTL0083-TSD	Temperature sensor (pair) installation kit for pipe size range from 0.75" to 2". Wetted materials are SS, for use in copper piping systems.	• (2) Copper tees with 1/2" outlet • (2) SS thermowells • (2) Copper street adapter, 1/2" • (2) Transmitter enclosures	Designed for use in hydronic systems.
	INSTL0086-TSD	Temperature sensor (pair) installation kit for pipe size range from 2.5" to 3". Wetted materials are SS, for use in copper piping systems.	• (2) Copper tees with 1/2" outlet • (2) SS thermowells • (2) Copper street adapter, 1/2" • (2) Transmitter enclosures	Designed for use in hydronic systems.
Stainless Steel Pipe Welded	INSTL0061-TSD	Temperature sensor (pair) installation kit for pipe size range from 0.75" to 42". Wetted materials are SS, for use in SS piping systems.	• (2) SS thermowells • (2) Weld-on stainless steel branch outlets, 1/2" • (2) Transmitter enclosures	Designed for use in process water applications.

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INSTALLATION HARDWARE SELECTION GUIDE

DRY TAP INSTALLATION HARDWARE



For System-20, System-40 and F-4300 BTU Meters (For use with RTD temperature sensor pair)

Pipe Material	Model Number	Description	Items Included with Kit	Notes
Steel Pipe Threaded	INSTL4001-TSD	Temperature sensor (pair) installation kit for pipe size range from 0.5" to 1". Wetted material is low lead brass, for use in carbon steel piping systems. NSF certified for use in domestic water applications.	(2) Lead free brass NPT thermowells	NPT threads sizes will match meter size. For use with customer supplied threaded steel tees. Suitable for use with SYS-40 where G=1 only. Refer to System-40 Selection and Ordering Guide for the meter ordering information.
	INSTL4002-TSI	Temperature sensor installation kit for pipe size range from 0.5" to 1". Wetted material is low lead brass, for use in carbon steel piping systems. Designed for use in hydronic systems.	(1) Lead free brass M10 x NPT reducer bushing	NPT thread size will match meter size. For use with customer supplied threaded steel tee. Suitable for use with SYS-40 where temperature sensor option G=0 only. Refer to System-40 Selection and Ordering Guide for the meter ordering information.
Copper Tube	INSTL4003-TSD	Temperature sensor (pair) installation kit for pipe size range from 0.5" to 1". Wetted material is low lead brass, for use in copper piping systems. NSF certified for use in domestic water applications.	(2) Lead free brass flush thermowells	Bushing sizes will match meter size. For use with customer supplied copper tees. Suitable for use with SYS-40 where G=1 only. Refer to System-40 Selection and Ordering Guide for the meter ordering information.
	INSTL4004-TSI	Temperature sensor installation kit for pipe size range from 0.5" to 1". Wetted material is low lead brass, for use in copper piping systems. Designed for use in hydronic systems.	(1) Lead free brass M10 x flush reducer bushing	Bushing size will match meter size. For use with customer supplied copper tee. Suitable for use with SYS-40 where temperature sensor option G=0 only. Refer to System-40 Selection and Ordering Guide for the meter ordering information.
Steel / PVC / Copper	INSTL4005-TSD	Temperature sensor (pair) installation kit for pipe size range from 0.5" to 2.5". Wetted material is low lead brass. NSF certified for use in domestic water applications.	(2) 5.0 mm dia., lead free, brass, 1/2" NPT thermowells (2) Set screws	Suitable for use with with F-4300, SYS-20, and SYS-40 (where G =2) BTU meters. Refer to SYS-40 or F-4300 Ordering Guides for the meter ordering information.

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INSTALLATION HARDWARE SELECTION GUIDE

DRY TAP INSTALLATION HARDWARE



For System-20, System-40 and F-4300 BTU Meters (For use with RTD temperature sensor pair)

Pipe Material	Model Number	Description	Items Included with Kit	Notes
Copper Tube	INSTL4007-TSD	Temperature sensor (pair) installation kit for pipe size range from 1.25" to 2.5". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems. NSF certified for use in domestic water applications.	• (2) 5.0 mm dia., lead free, brass, 1/2" NPT thermowells • (2) 1"x 1/2" street adapters • (2) Set screws	For use with customer supplied tees with 1" outlets. Suitable for use with F-4300, SYS-20 and SYS-40 BTU meters.
Kits with Copper Tees	INSTL4008-TSD	Temperature sensor (pair) installation kit for pipe size range from 1.25" to 2". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems. NSF certified for use in domestic water applications.	• (2) 5.0 mm dia., lead free, brass, 1/2" NPT thermowells • (2) 1"x 1/2" street adapters • (2) Copper tees with 1" outlets • (2) Set screws	Suitable for use with F-4300, SYS-20 and SYS-40 BTU meters.
	INSTL4009-TSD	Temperature sensor (pair) installation kit for pipe size 2.5". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems. NSF certified for use in domestic water applications.	• (2) 5.0 mm dia., lead free, brass, 1/2" NPT thermowells • (2) 1"x 1/2" street adapters • (2) Copper tees with 1" outlets • (2) Set screws	Suitable for use with F-4300, SYS-20 and SYS-40 BTU meters.
Steel	INSTL4021-TSD	Temperature sensor (pair) installation kit for pipe size range from 3" to 48". Wetted materials are SS, for use in carbon steel piping systems. Designed for use in hydronic systems.	• (2) 6.0 mm dia., SS, 1/2" NPT thermowells • (2) Carbon steel 1/2" NPT branch outlets	Suitable for use with F-4300 and SYS-20 BTU meters.

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INSTALLATION HARDWARE SELECTION GUIDE

HOT TAP INSTALLATION HARDWARE



For F-1000 Series, FSM-3 and F-3500 Flow Meters

Pipe Material	Model Number	Description	Items Included with Kit	Notes
Carbon Steel/ Black Iron (welded pipe)	INSTL0002-FMH	Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.	• 1 1/4" Full port bronze ball valve • 1 1/4" Brass close nipple • 1 1/4" x 1" Brass reducing bushing • 1 1/4" Weld-on carbon steel branch outlet	Designed for use in hydronic systems.
	INSTL02DW-FMH	Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, CU and low lead brass, for use in carbon steel piping systems.	• 1 1/4" Full port bronze ball valve • 1 1/4" SS close nipple • 1 1/4" x 1" SS reducing bushing • 1 1/4" Weld-on carbon steel branch outlet	NSF certified for use in domestic water applications.
	INSTL0006-FMH	Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in carbon steel piping systems.	• 1 1/4" Full port SS ball valve • 1 1/4" SS close nipple • 1 1/4" x 1" SS reducing bushing • 1 1/4" Weld-on carbon steel branch outlet	Designed for use in hydronic systems.
Carbon Steel or PVC Pipe w/Saddles	INSTL019A-FMH	Insertion flow meter hot tap installation kit for pipe size range from 1.5" to 6". Wetted materials are bronze, CU and low lead brass, for use in PVC piping systems.	• 1 1/4" Full port bronze ball valve • 1 1/4" Brass close nipple • 1 1/4" x 1" Brass reducing bushing • Ductile iron saddle with 1 1/4" outlet	
	INSTL019B-FMH	Insertion flow meter hot tap installation kit for pipe size range from 8" to 14". Wetted materials are bronze, CU and low lead brass, for use in PVC piping systems.	• 1 1/4" Full port bronze ball valve • 1 1/4" Brass close nipple • 1 1/4" x 1" Brass reducing bushing • Ductile iron saddle with 1 1/4" outlet	
Copper Tube w/Saddles	INSTL0021-FMH	Insertion flow meter hot tap installation kit for pipe size 2". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems.	• 1 1/4" Full port bronze ball valve • 1 1/4" Brass close nipple • 1 1/4" x 1" Brass reducing bushing • Ductile iron or SS saddle with 1 1/4" outlet	NSF certified for use in domestic water applications.
	INSTL0022-FMH	Insertion flow meter hot tap installation kit for pipe size range from 2.5" to 6". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems.	• 1 1/4" Full port bronze ball valve • 1 1/4" Brass close nipple • 1 1/4" x 1" Brass reducing bushing • SS saddle with 1 1/4" outlet	NSF certified for use in domestic water applications.

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INSTALLATION HARDWARE SELECTION GUIDE

HOT TAP INSTALLATION HARDWARE



For F-1000 Series, FSM-3 and F-3500 Flow Meters

Pipe Material	Model Number	Description	Items Included with Kit	Notes
Stainless Steel	INSTL0031-FMH	Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in SS piping systems.	• 1 1/4" Full port 316 SS ball valve • 1 1/4" 316 SS close nipple • 1 1/4" x 1" 316 SS reducing bushing • 1 1/4" Weld-on 316 SS branch outlet	Designed for use in process water applications.
HDPE/PPR	INSTL5001-FMH	Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, brass and steel, for use in HDPE/PPR piping systems.	• 1 1/4" Full port bronze ball valve • 1 1/4" Brass close nipple • 1 1/4" x 1" Brass reducing bushing	Designed for use in hydronic systems. For use with customer supplied branch/fusion outlet, total stack height not to exceed 7".
	INSTL5002-FMH	Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, CU and low lead brass, for use in HDPE/PPR piping systems.	• 1 1/4" Full port bronze ball valve • 1 1/4" SS close nipple • 1 1/4" x 1" SS reducing bushing	NSF certified for use in domestic water applications. For use with customer supplied branch/fusion outlet, total stack height not to exceed 7".
	INSTL5003-FMH	Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in HDPE/PPR piping systems.	• 1 1/4" Full port SS ball valve • 1 1/4" SS close nipple • 1 1/4" x 1" SS reducing bushing	Designed for use in hydronic systems. For use with customer supplied branch/fusion outlet, total stack height not to exceed 7".

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INSTALLATION HARDWARE SELECTION GUIDE

HOT TAP INSTALLATION HARDWARE



For F-2000 Series and F-1500 Flow Meters

Pipe Material	Model Number	Description	Items Included with Kit	Notes
Carbon Steel/ Black Iron (welded pipe)	INSTL201S-FMH	Insertion flow meter hot tap installation kit for pipe size range from 2" to 42". Wetted materials are carbon steel, for use in carbon steel piping systems.	• 2" Full port carbon steel class 150 flanged gate valve • 2" Carbon steel class 150 weld-neck flange • 2" Weld-on carbon steel branch outlet	Designed for use in steam systems.
	INSTL203S-FMH	Insertion flow meter hot tap installation kit for pipe size range from 2" to 42". Wetted materials are carbon steel, for use in carbon steel piping systems.	• 2" Full port carbon steel class 300 flanged gate valve • 2" Carbon steel class 300 weld-neck flange • 2" Weld-on carbon steel branch outlet	Designed for use in steam systems.

For F-5000 Series Flow Meters

Model Number	Description	Notes
INSTL0094-FMH	Insertion flow meter hot tap installation kit, wetted materials are bronze, brass and steel, for use in carbon steel piping systems.	Designed for use in NG and compressed air systems.
INSTL0096-FMH	Insertion flow meter hot tap installation kit, wetted materials are SS, brass and carbon steel, for use in carbon steel piping systems.	Designed for use in NG and compressed air systems.

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INSTALLATION HARDWARE SELECTION GUIDE

HOT TAP INSTALLATION HARDWARE



For System-10, System-20 and F-4300 BTU Meters (For use with current (mA) based temperature sensors)

Pipe Material	Model Number	Description	Items Included with Kit	Notes
Steel Pipe Welded	INSTL0033-TSH	Temperature sensor (pair) hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.	• (2) SS hot tap thermowells with brass hot tap adapters • (2) Full port bronze ball valves, 1" • (2) Brass close nipples • (2) Weld-on carbon steel branch outlets, 1" NPT • (2) Transmitter enclosures	Designed for use in hydronic systems.
	INSTL0035-TSH	Temperature sensor (pair) hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in carbon steel piping systems.	• (2) SS hot tap thermowells with stainless steel hot tap adapters • (2) Full port stainless steel ball valves, 1" • (2) SS close nipples • (2) Weld-on carbon steel branch outlets, 1" • (2) Transmitter enclosures	Designed for use in hydronic systems.
Steel or PVC Pipe with Saddles	INSTL0060-TSH	Temperature sensor (pair) hot tap installation kit for pipe size range from 1.5" to 6". Wetted materials are SS, for use in PVC piping systems.	• (2) SS hot tap thermowells with brass hot tap adapters • (2) Full port bronze ball valves, 1" • (2) Brass close nipples • (2) Ductile iron saddles for Sch std./40/80 pipe with 1" NPT outlets • (2) Transmitter enclosures	Designed for use in hydronic systems.
	INSTL060A-TSH	Temperature sensor (pair) hot tap installation kit for pipe size range from 8" to 14". Wetted materials are SS, for use in PVC piping systems.	• (2) SS hot tap thermowells with brass hot tap adapters • (2) Full port bronze ball valves, 1" • (2) Brass close nipples • (2) Ductile iron saddles for Sch std./40/80 pipe with 1" NPT outlets • (2) Transmitter enclosures	Designed for use in hydronic systems.
	INSTL0065-TSH	Temperature sensor (pair) hot tap installation kit for pipe size range from 6" to 12". Wetted materials are SS, for use in PVC piping systems.	• (2) SS hot tap thermowells with stainless steel hot tap adapters • (2) Full port stainless steel ball valves, 1" • (2) SS close nipples • (2) Ductile iron saddles for Sch std./40/80 pipe with 1" NPT outlets • (2) Transmitter enclosures	Designed for use in hydronic systems.

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INSTALLATION HARDWARE SELECTION GUIDE

HOT TAP AND CLAMP-ON INSTALLATION HARDWARE

For System-10, System-20 and F-4300 BTU Meters (For use with current (mA) based temperature sensors)

Pipe Material	Model Number	Description	Items Included with Kit	Notes
Copper Tube with Saddles	INSTL0040-TSH	Temperature sensor (pair) hot tap installation kit for pipe size range from 2" to 6". Wetted materials are SS, for use in copper piping systems.	• (2) SS hot tap thermowells with brass hot tap adapters • (2) Full port bronze ball valves, 1" • (2) Brass close nipples • (2) SS saddles for copper tube with 1" NPT outlets • (2) Transmitter enclosures	Designed for use in hydronic systems.
Steel Pipe	INSTL0501-TSC	Installation kit for pipe size range from 0.5" to 48". Materials consist of aluminum and brass, for use in carbon steel piping systems.	• (2) Clamp-on thermowell blocks w/clamps • (2) Transmitter enclosures	Designed for use in hydronic systems.
Copper Tube	INSTL0502-TSC	Installation kit for pipe size range from 0.5" to 6". Materials consist of aluminum and brass, for use in copper piping systems.	• (2) Clamp-on thermowell blocks w/clamps • (2) Transmitter enclosures	Designed for use in hydronic systems.

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